2 Radiographic Interpretation

MADHU K. NAIR, MARTIN D. LEVIN, and UMADEVI P. NAIR

CHAPTER OUTLINE

Radiographic Interpretation
Imaging Modalities
Image Characteristics and Processing
Digital Imaging and Communications in Medicine
Diagnostic Tasks in Endodontics
Diagnosis and Healing
Three-Dimensional Imaging
Principles of Cone Beam Computed Tomography
Voxels and Voxel Sizes
Field of View
Diagnostic Task
Type of Patient
Spatial Resolution Requirements
Imaging Tasks Improved or Simplified by Cone Beam Volumetric Computed Tomography
Differential Diagnosis
Lesions of Endodontic Origin
Lesions of Nonendodontic Origin
Evaluation of Anatomy and Complex Morphology
Dental Anomalies
Root Canal System Morphology
Diagnosis of Endodontic Treatment Failures
Intraoperative or Postoperative Assessment of Endodontic Treatment Complications
Vertical Root Fractures
Overextended Root Canal Obturation Material
Separated Endodontic Instruments
Calcified Canals
Perforations
Presurgical Treatment Planning
Dentoalveolar Trauma
Internal and External Root Resorption
Endodontic Treatment Outcomes Assessment
Dental Implant Case Planning
3D-Guided Endodontics
Image Perception and Viewing Environment
Future of Cone Beam Computed Tomography
Magnetic Resonance Imaging
Conclusions

Radiographic Interpretation

Interpretation of information captured by radiographic imaging modalities is central to the diagnostic process. It is very important to capture a diagnostically useful image using appropriate exposure parameters and view it with interactive manipulation of brightness and contrast or window/level (for cone-beam computed tomography [CBCT] studies) in an optimal environment to adequately evaluate anatomy and diagnose pathoses. Accurate interpretation of root and canal morphology, determination of radiographic canal length, diagnosis of radicular and periradicular disease (Fig. 2.1), and postsurgical and long-term evaluation of the outcome of endodontic treatment are some of the routine diagnostic imaging tasks in endodontics.[218] Systematic and methodical interpretation processes must be followed for all images. Recognition of anatomy, anatomic variants, and pathologic conditions or deviations from normal is important. Various imaging modalities exist in radiology. Some use ionizing radiation, whereas others use ultrasonic waves (ultrasonography [US]) or powerful external magnetic fields (magnetic resonance imaging [MRI]). Interventional and noninterventional imaging modalities are also available. Imaging modalities using ionizing radiation are most frequently used in endodontic diagnoses. The different image capture modalities include digital receptors using different technologies.

IMAGING MODALITIES

Digital radiography using electronic sensors or photostimulable phosphor (PSP) plates is widely used in endodontics. Digital imaging modalities in endodontics use different image capture technologies, which include a charge-coupled device (CCD), a complementary metal oxide semiconductor (CMOS), or a PSP (also sometimes referred to as an indirect acquisition modality).

CCD-based solid-state sensors were used extensively in endodontics initially. However, the earlier generation sensors had a smaller active area and limited x-ray absorption and conversion efficiency, in addition to being bulky. Sensors use an array of radiation-sensitive elements that generate electric charges proportionate to the amount of incident radiation. To reduce the amount of radiation

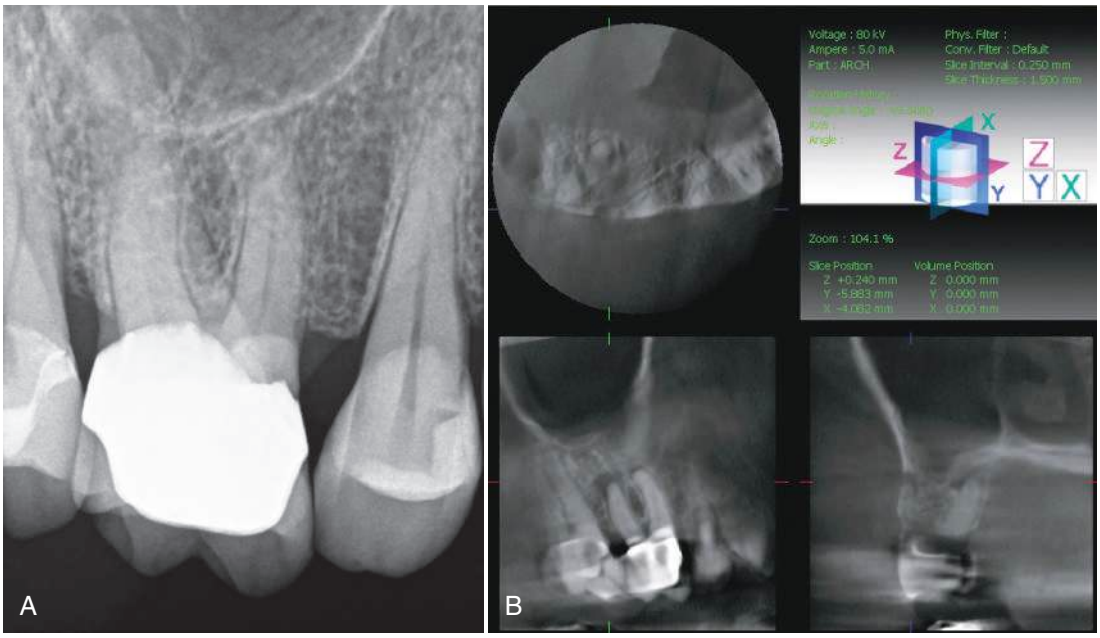


Fig. 2.1 A, A well-angulated periapical radiograph of the maxillary right first molar taken during a diagnostic appointment for endodontic evaluation of the maxillary right quadrant. At first glance, there is little radiographic evidence of significant or periradicular change. **B,** Contemporaneous CBCT image of same tooth gives an entirely different perspective; periapical changes are visible on all three roots in all three anatomic planes of section. (*B* taken with J. Morita Veraviewepocs 3D [J. Morita, Osaka, Japan].)

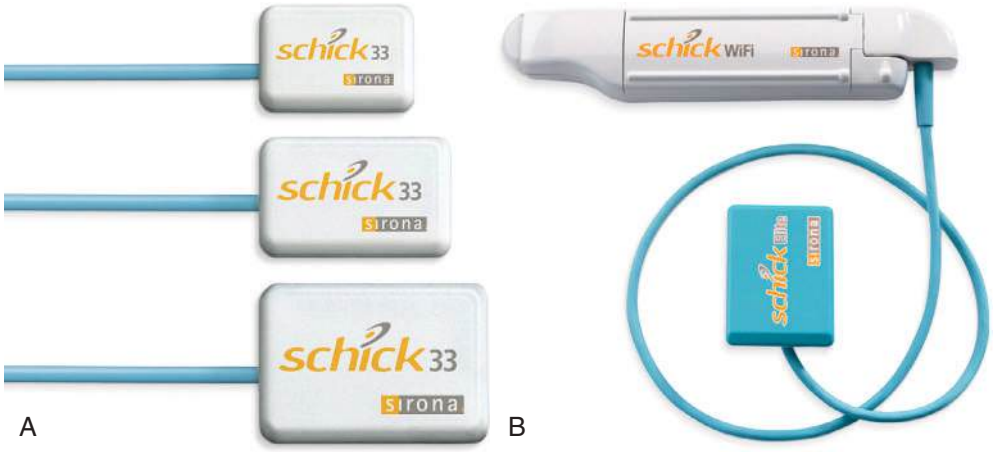


Fig. 2.2 A and **B,** High-resolution complementary metal oxide semiconductor (CMOS) sensors are available from many manufacturers. Note that Figure In **B,** wireless CMOS sensors transmit images to the chairside workstation by 2.4 GHz radiofrequency. (Courtesy SIRONA DENTAL SYSTEMS, Long Island City, NY.)

needed to capture an image, a light-sensitive array was developed that uses a scintillation layer laid on top of the CCD chip or added with a fiberoptic coupling. The generated charge is read out in a "bucket brigade" fashion and transferred to an analog-to-digital converter in the frame grabber assembly of the workstation. The digital information is processed, and an image is formed. CMOS-based sensors, on the other hand, have an active transistor at each element location. The area available for signal generation is relatively less, and there is a fixed pattern noise. These sensors are less expensive to manufacture and have been shown to be equally useful for specific diagnostic tasks.[114] Unlike the CCD, the CMOS chip requires very little electrical energy; therefore, no external power supply is needed to support USB utilization, and wireless applications are feasible. Wireless sensors are available (Fig. 2.2). However, radiofrequency interference may be a problem with these sensors. The current Wi-Fi sensor is less bulky and has a wire attached to it that enables transmission via 802.11 b/g standard. It uses a lithium-ion polymer battery that can last for approximately 100 exposures.

Yet another type of sensor uses PSPs for image capture. PSP technology is also referred to as *computed radiography* (CR).[114,199] Unlike the CCD and CMOS sensors, PSP sensors are wireless. The phosphor is activated by a process called *doping,* which enables charges to be generated and stored when exposed to radiation. A latent image is stored in the sensor, and a PSP reader with a laser beam of specific

wavelength is used to read out the image. Previously captured images can be erased by exposing the PSP sensor to white light. PSP plates can be damaged easily by scratching, but they are not as expensive as CCD or CMOS sensors. Incomplete erasure of the image can lead to ghost images when the plate is reused, and delayed processing can result in a decrease in image clarity.[6] PSP-based sensors are used in high-volume scenarios. Spatial resolution is lower with this type of sensor, but it has a wider dynamic range. These sensors can tolerate a wider range of exposures to produce a diagnostically useful image.

Radiation dose continues to be a concern with all imaging studies. The lowest possible dose must be delivered for each study. Most dental offices would not be in compliance with the latest recommendations of the National Council for Radiation Protection (NCRP) on reducing the radiation dose from intraoral radiographs (Box 2.1). Two terms have been specifically defined in the NCRP's report. The terms *shall* and *shall not* indicate that adherence to the recommendation would be in compliance with the standards of radiation safety. The terms *should* and *should not* indicate prudent practice and acknowledge that exceptions may be made in certain circumstances. In addition, the report establishes nine new recommendations for image processing of conventional film.

A strong argument can be made for clinicians to switch to a direct digital radiography (DDR) system to avoid all the drastic changes necessary to ensure compliance with the new recommendations. Even though restriction of an intraoral dental x-ray beam is mandated by federal law to a circle no greater than 7 cm, rectangular collimation has been proven to significantly reduce the radiation dose to the patient and is strongly recommended in the NCRP Report #145.

The American Dental Association (ADA) Council on Scientific Affairs has made the following statement:

Tissue area exposed to the primary x-ray beam should not exceed the minimum coverage consistent with meeting diagnostic requirements and clinical feasibility. For periapical (PA) and bitewing radiography, rectangular collimation should be used whenever possible because a round field beam used with a rectangular image receptor produces segments of the beam circle that are not used in receptor exposure, which causes unnecessary radiation exposure to the patient.[4]

Box 2.1 Recommendations of the National Council on Radiation Protection

1. Dentists must examine their patients before ordering or prescribing x-ray images (this is not a new guideline).
2. The use of leaded aprons on patients shall not be required if all other recommendations in this report are rigorously followed (read full Report #145).
3. Thyroid shielding shall be used for children and should be provided for adults when it will not interfere with the examination (e.g., panoramic imaging).
4. Rectangular collimation of the beam, which has been recommended for years, shall be routinely used for periapical radiographs. Each dimension of the beam, measured in the plane of the image receptor, should not exceed the dimension of the image receptor by more than 2% of the source-to-image receptor distance. Similar collimation should be used, when feasible, for bitewing radiographs.
5. Image receptors of speeds slower than ANSI speed Group E films shall not be used for intraoral radiography. Faster receptors should be evaluated and adopted if found acceptable. For extraoral radiography, high-speed (400 or greater) rare earth screen-film systems or digital-imaging systems of equivalent or greater speed shall be used.
6. Dental radiographic films shall be developed according to the film manufacturer's instructions using the time-temperature method. In practical application, this means that sight development (reading wet x-ray films at the time of the procedure) shall not be used.
7. Radiographic techniques for digital imaging shall be adjusted for the minimum patient dose required to produce a signal-to-noise ratio sufficient to provide image quality to meet the purpose of the examination.
8. Clinicians designing new offices or remodeling existing locations will need shield protection to be provided by a qualified expert.

Modified from the National Council on Radiation Protection and Measurements: Radiation protection in dentistry, Report #145, Bethesda, MD, 2003. Available at www.ncrppublications.org/Reports/145.

IMAGE CHARACTERISTICS AND PROCESSING

Spatial resolution achieved with current generation digital sensors is equally good or better than that of conventional intraoral radiographic film. Intraoral film has a resolution of 16-line pairs per millimeter (lp/mm) as measured using a resolution tool, and it increases to 20 to 24 lp/mm with magnification. *Spatial resolution* is defined as the ability to display two objects that are close to each other as two separate entities. *Contrast resolution* is defined as the ability to differentiate between areas on the image based on density. Most diagnostic tasks in endodontics require a high-contrast resolution.[142] However, image quality is not only a function of spatial resolution. The choice of appropriate exposure parameters, sensor properties, the image processing used, and viewing conditions and modalities directly affect diagnostic accuracy.

Postprocessing of images may be carried out to alter image characteristics. Radiographs need not be reexposed if image quality is not adequate. Diagnostic information can be teased out of the image if appropriate image processing is used. However, the original image must be acquired with optimal exposure parameters to accomplish meaningful image processing.[212] Suboptimally exposed images cannot be processed to yield diagnostic information, which may lead to a reduction in the diagnostic accuracy of the image. Image enhancement must be task specific. Signal-to-noise ratio (SNR) must be optimized to extract necessary information from the image. The bit-depth of images also has a direct relationship to image quality. It indicates the number of shades of gray that the sensor can capture for display. For example, an 8-bit image can depict 256 shades of gray. Most sensors are 12 or 14 bits in depth, capturing 4096 or 65,536 shades of gray, respectively. If the sensor captures several thousand shades of gray, the image can be manipulated through enhancement techniques to display those shades of gray that best depict the anatomy of interest. The human visual system is limited in the number of shades of gray that can be read at any point in time. Therefore, image

enhancement is a must for all images, so as to delineate signals of interest through manipulation of the grayscale. Most endodontic tasks require a high contrast and thus a shorter grayscale.

Digital radiographs can be saved in different file formats. Several file formats are available: DICOM (*D*igital *I*maging and *C*ommunications in *M*edicine); tiff (*t*agged *i*mage *f*ile *f*ormat); jpeg (*j*oint *p*hotographic *e*xperts' *g*roup); gif (*g*raphics *i*nterchange *f*ormat); BMP (Windows' bitmap image file); PNG (*p*ortable *n*etwork *g*raphics); and so on. There also are several proprietary formats. "Lossy" and "lossless" compression schemes can be used for saving images, although lossless compression is preferred.[72]

DIGITAL IMAGING AND COMMUNICATIONS IN MEDICINE

DICOM is a set of international standards established in 1985 by the American College of Radiology (ACR) and the National Electrical Manufacturers Association (NEMA)[60,205] to address the issue of vendor-independent data formats and data transfers for digital medical images. The ADA has promoted the interoperability of dental images through the efforts of its Working Group 12.1.26. DICOM serves as a standard for the transferal of radiologic images and other medical information between computers, allowing digital communication between systems from various manufacturers and across different platforms (e.g., Apple iOS or Microsoft Windows).[60] The DICOM standard provides for several hundred attribute fields in the record header, which contains information about the image (e.g., pixel density, dimensions, and number of bits per pixel), in addition to relevant patient data and medical information. Although earlier versions did not specify the exact order and definition of the header fields, each vendor is required to publish a DICOM conformance statement, which gives the location of pertinent data. The big hurdle is to support medical and dental consultations between two or more locations with different imaging software.[60] With DICOM in place, dental clinicians can change vendors and maintain database interoperability. Most software vendors are striving to achieve full DICOM compliance, and some have achieved at least partial compliance. However, proprietary DICOM images are still produced in different systems, with the capability to export in universal DICOM format as needed. Diagnostic images are best saved as DICOM files to preserve image fidelity or as tiff files with no compression. Diagnosis suffers when images undergo lossy compression.[64,123,217]

Based on the DICOM model, the ADA Standards Committee on Dental Informatics has identified four basic goals for electronic standards in dentistry: (1) interoperability, (2) electronic health record design, (3) clinical workstation architecture, and (4) electronic dissemination of dental information.[10] The dental profession must continue to promote DICOM compatibility so that proprietary software and file types do not hinder communication and risk making data obsolete.

Diagnostic Tasks in Endodontics

It is important to analyze the type of sensor used, software, processing, video card and monitor, and viewing conditions to determine whether the sensor is good for a specific diagnostic task. Calibration improves diagnostic accuracy.[128] Likewise, the use of optimal processing parameters improves image quality to the extent of making a significant difference in the diagnostic outcome. For instance, density plot analysis was shown to help with endodontic file measurements.[172] The major advantage of direct digital radiography (CCD, CMOS) is that the dose is significantly less compared with that required for film. The use of DDR, therefore, is justified when its performance is comparable to that of film with no statistically significant differences.[120]

The three types of measurement generally available with digital imaging software are (1) linear measurement, the distance between two points in millimeters (Fig. 2.3); (2) angle measurement, the angle between two lines; and (3) area measurement, the area of the image or a segment of the image. Because magnification and distortion errors play a significant role in the accuracy of two-dimensional (2D) radiographic measurement, both film and digital systems are subject to parallax error. However, a study that compared endodontic file length images of human teeth

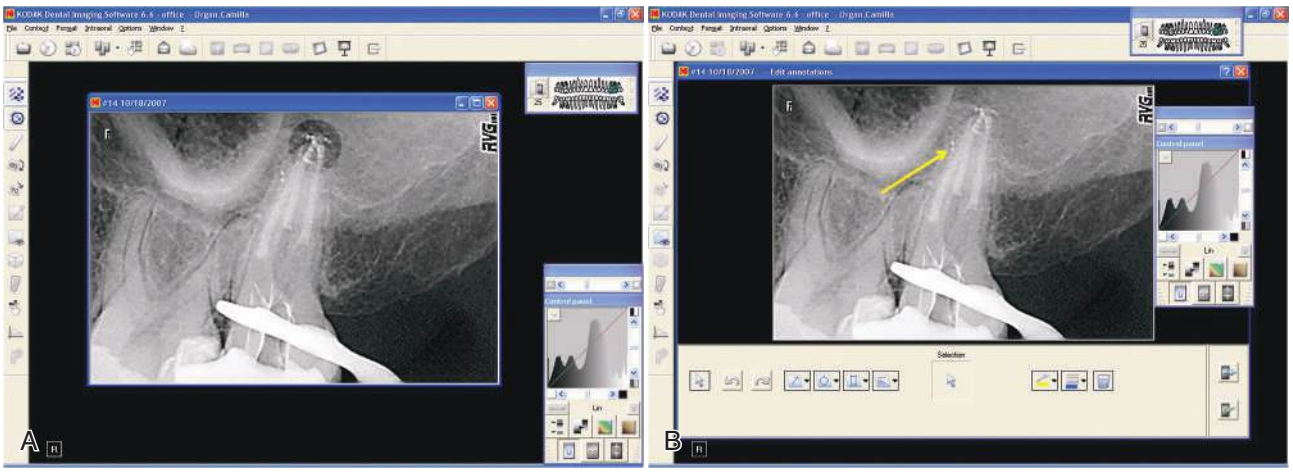


Fig. 2.3 A, Certain regions of interest (ROI) can be highlighted with a preset contrast tool that can be moved around the image. **B,** Preprogramed filters that enhance sharpness and contrast can be selected to optimize the image acquired. (Courtesy Carestream Dental LLC. Atlanta, GA.)

taken with a custom jig suggested that "measurement error was significantly less for the digital images than the film-based images."[63] This was true even though, as the authors pointed out, the measurement differences may not have been clinically significant. Sophisticated calibration algorithms are under development, and accurate measurement of parallel images should be more feasible in the future.[41]

DIAGNOSIS AND HEALING

Image enhancement of direct and indirect digital radiographs based on the diagnostic task at hand has been shown to increase diagnostic accuracy compared to film-based images, which cannot be enhanced.[6,226] Posttreatment endodontic evaluation of healing of apical radiolucent areas is a challenge. Early changes indicating healing and bone fill are difficult to detect on conventional or digital radiographs. However, bone fill can be detected using more sensitive techniques, such as digital subtraction radiography (DSR), in which two images, separated in time but acquired with the exact same projection geometry and technique factors, can be subtracted from one another to tease out subtle changes in the periodontium and surrounding bone. Subtraction techniques are difficult to carry out in routine clinical practice because they are technique sensitive and can yield incorrect information if not performed accurately. Several studies have shown the usefulness of subtraction radiography using digital sensors.[138,150,231]

THREE-DIMENSIONAL IMAGING

Computed tomography (CT) was introduced by Sir Godfrey Hounsfield in the 1970s. *Tomography* refers to "slice imaging," in which thin slices of the anatomy of interest are captured and synthesized manually or using an algorithm. CT makes use of automated reconstruction. Medical-grade CT used a translate-rotate image acquisition scheme as the technology developed, but the modality always resulted in higher radiation dose delivery because of redundancy of data capture, in addition to longer scan times with the potential for motion artifact. Multiple detectors and x-ray sources were used in later generations of CT units to reduce scanning times. The increased radiation dose, artifacts from metallic restorations, cost of scanning, long acquisition times, and lack of adequate dental-specific software have been drawbacks limiting the use of the technology in dentistry until recently. The advent of cone-beam volumetric computed tomography (CBVCT) introduced a faster, low-dose, low-cost, high-contrast imaging modality that could capture information in three dimensions using a limited field of view (FOV).

CBVCT, or cone beam CT (CBCT), is a relatively new diagnostic imaging modality that has been recently added to the endodontic imaging armamentarium. This modality uses a cone beam instead of a fan-shaped beam in multidetector computed tomography (MDCT), acquiring images of the entire volume as it rotates around the anatomy of interest. Compared with MDCT images, CBCT offers relatively high-resolution, isotropic images, allowing effective evaluation of root canal morphology and other subtle changes within the root canal system. Even though the resolution is not as high as that of conventional radiographs (18 microns), the availability of three-dimensional (3D) information, the relatively higher resolution, and a significantly lower dose compared to MDCT make CBCT the imaging modality of choice in challenging situations demanding localization and characterization of root canals.

The adoption of advanced imaging modalities such as CBCT for select diagnostic tasks is becoming popular with clinicians performing endodontic procedures. Two-dimensional grayscale images, whether conventional film-based or digital, cannot accurately depict the full 3D representation of the teeth and supporting structures. In fact, traditional images are poor representations of even the pulpal anatomy. They grossly underestimate canal structure and often cannot accurately visualize PA changes, especially where there is thick cortical bone, as in the presence of anatomic obstructions (Fig. 2.4, *A*). CBCT, however, allows the clinician to view the tooth and pulpal structures in thin slices in all three anatomic planes: axial, sagittal, and coronal. This capability alone allows visualization of PA pathoses and root morphology that was previously impossible to assess (see Fig. 2.4, *B* and *C*). Several tools available in CBCT, such as the ability to change the vertical or horizontal angulation of the image in real time, in addition to thin-slice, grayscale data of varying thicknesses, will never be available for conventional or even digital radiography. Furthermore, the use of CBCT data to view the region of interest in three anatomic planes of section at very low x-ray doses has never been as easy or accessible as it is today.

Microcomputed tomography (micro-CT) has also been evaluated in endodontic imaging.[103,169,170] Comparison of the effects of biomechanical preparation on the canal volume of reconstructed root canals in extracted teeth using micro-CT data was shown to assist with characterization of morphologic changes associated with these techniques.[170] Peters et al.[169] used micro-CT to evaluate the relative performance of nickel-titanium (Ni-Ti) instruments after the shaping of root canals of varying preoperative canal geometry. A study to examine the potential and accuracy of micro-CT for imaging of filled root canals showed it to be a highly accurate and nondestructive method for the evaluation of root canal fillings and its constituents. The qualitative and quantitative correlations between histologic and micro-CT examination of root canal fillings were high.[103] However, it is important to note that micro-CT remains a research tool and cannot be used for human imaging in vivo.

This chapter discusses the principles, applications, imaging attributes, image artifacts, and potential liability of adopting CBCT technology for endodontic procedures. Given this information, the student of endodontics will begin to realize the significant advantages, limitations, and diagnostic and treatment planning capabilities of this radiographic imaging modality.

Principles of Cone Beam Computed Tomography

Two important parameters of cone beam imaging are described in the following sections:

- Voxel size
- Field of view (FOV)

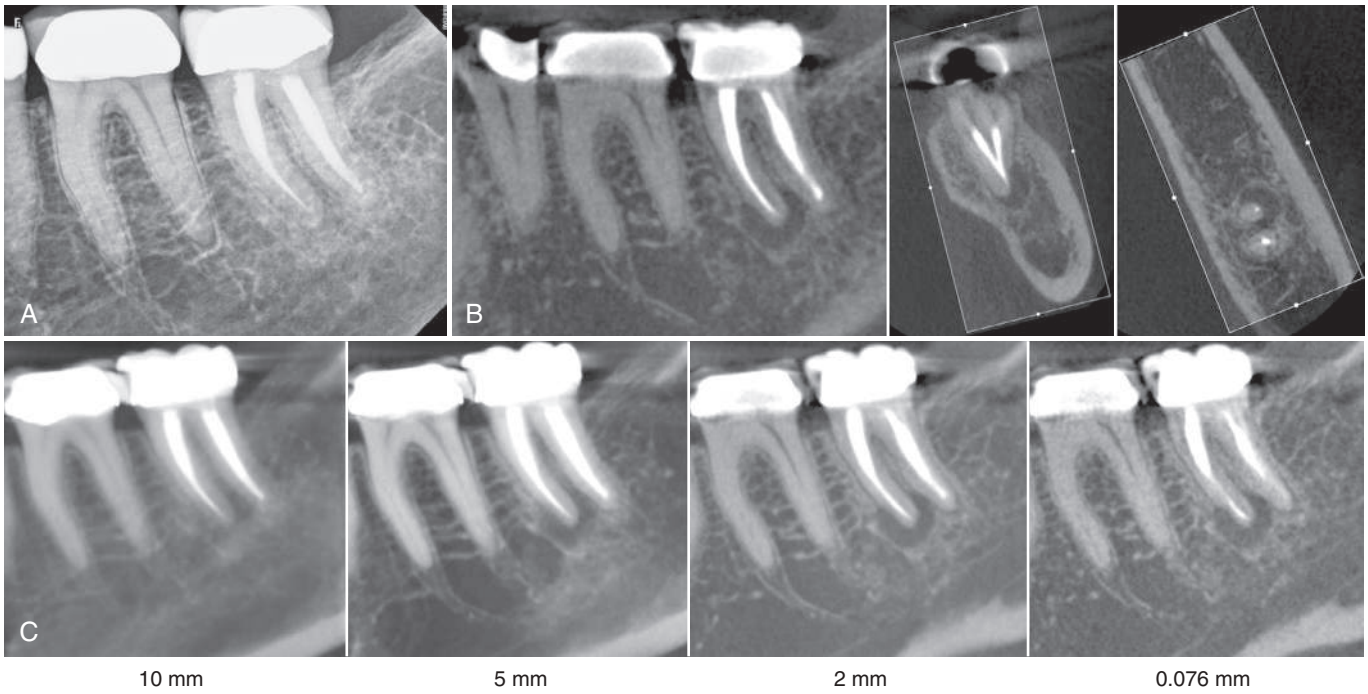


Fig. 2.4 This case demonstrates the difficulty in assessing lesions in the mandibular posterior region when there is a dense cortex. **A,** This well-angulated periapical radiograph shows a normal periapical region associated with the mandibular left second molar, for which the patient has presented for evaluation and possible retreatment. **B,** The CBCT corrected sagittal, coronal, and axial reconstructed multiplanar views, (*left to right,* sagittal, coronal, and axial views) show previous endodontic treatment, with a 6 mm diameter radiolucency with a well-defined, mildly corticated border, centered over a point on the buccal aspect of the root, 2 mm coronal to the apex; these are features consistent with an apical rarefying osteitis. **C,** The ray sum images of the sagittal view, where the image is "thinned" by decreasing the number of adjacent voxels using postprocessing software, simulates a curvilinear projection, showing diminishing superimposition (*left to right,* image layer of 10 mm, 5 mm, 2 mm, and 0.076 mm). (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

VOXELS AND VOXEL SIZES

Voxels are cuboidal elements that constitute a 3D volume, unlike pixels, which are 2D. Data are acquired and represented in three dimensions using voxels. Unlike with MDCT, typically used in medicine, cone beam units acquire x-ray information using low kV and low mA exposure parameters in a single pass from 180 to 360 degrees of rotation around the anatomy of interest. Medical scanners use higher voltages of 120 kV or more and a current of about 400 mA. Several units used in maxillofacial imaging use significantly lower exposure parameters (Figs. 2.5 to 2.7). The x-ray dose for all cone beam units is significantly lower than the dose received from a MDCT unit. Image attributes are also different in that volumes are reconstructed from isotropic voxels; that is, the images are constructed from volumetric detector elements that are cubical in nature and have the same dimensions of length, width, and depth. These voxel sizes can be as small as 0.076 to 0.6 mm.[139] By comparison, MDCT slice data are 0.5 mm to 1 cm thick. Fig. 2.8 illustrates the difference between a pixel and a voxel, the difference between an anisotropic pixel of MDCT and an isotropic pixel (voxel) of CBCT, and how the pixel data are acquired from both modalities.



Fig. 2.5 i-CAT unit. (Courtesy Imaging Sciences International, Hatfield, PA.)

The patient is positioned on a gantry in an MDCT unit, and images are acquired multiple slices at a time, which prolongs the acquisition time. The number of slices acquired is a direct function of the sensor array configuration. Spiral CT uses continuous translator motion of the gantry as images are acquired, thus shortening the acquisition time. This results in significantly higher absorbed x-ray doses for the patient. A typical CBCT examination would expose the patient to only about 20 to 500 μSv in a single study, whereas a typical medical examination of the head

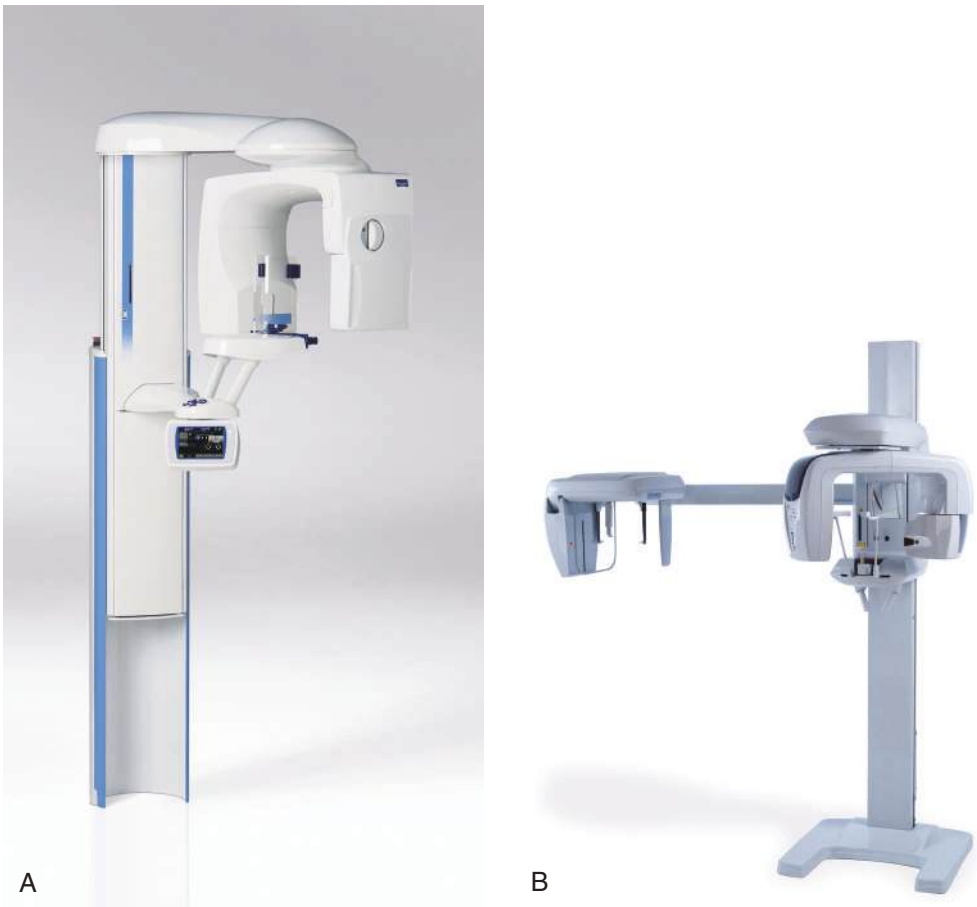


Fig. 2.6 A, Planmeca ProMax 3D. **B,** J. Morita Veraviewepocs 3D. (*A,* Courtesy Planmeca Oy, Helsinki, Finland. *B,* Courtesy, JMorita USA, Irvine, CA).



Fig. 2.7 A, CS 9000 3D and CS 8100 extraoral imaging systems. **B,** Morita Accu-i-Tomo 170. (A, Courtesy, Carestream Dental LLC, Atlanta, GA;. *B,* Courtesy J Morita USA, Irvine, CA).

would approach 2100 μSv because the image data are gathered one section at a time. Therefore, soft-tissue imaging is better with MDCT because the signal intensity is higher. However, this is not a requirement for dental diagnostic tasks because hard tissue visualization is more important. Consequently, CBCT data have a much higher resolution than MDCT data for hard tissue visualization because of the smaller voxel sizes that medical-grade scanners are incapable of achieving at a significantly lower dose. Increased noise is observed as a result of volumetric acquisition, but the SNR is maintained at a desirable level that facilitates adequate diagnosis based on hard tissue signals.

FIELD OF VIEW

The FOV (Figs. 2.9 and 2.10) ranges from as small as a portion of a dental arch to an area as large as the entire head. The selection of the FOV depends on several factors. Among the most important are the following:

- Diagnostic task
- Type of patient
- Spatial resolution requirements

Diagnostic Task

The diagnostic task is the single most important determinant of the FOV in any imaging study. Based on the outcome

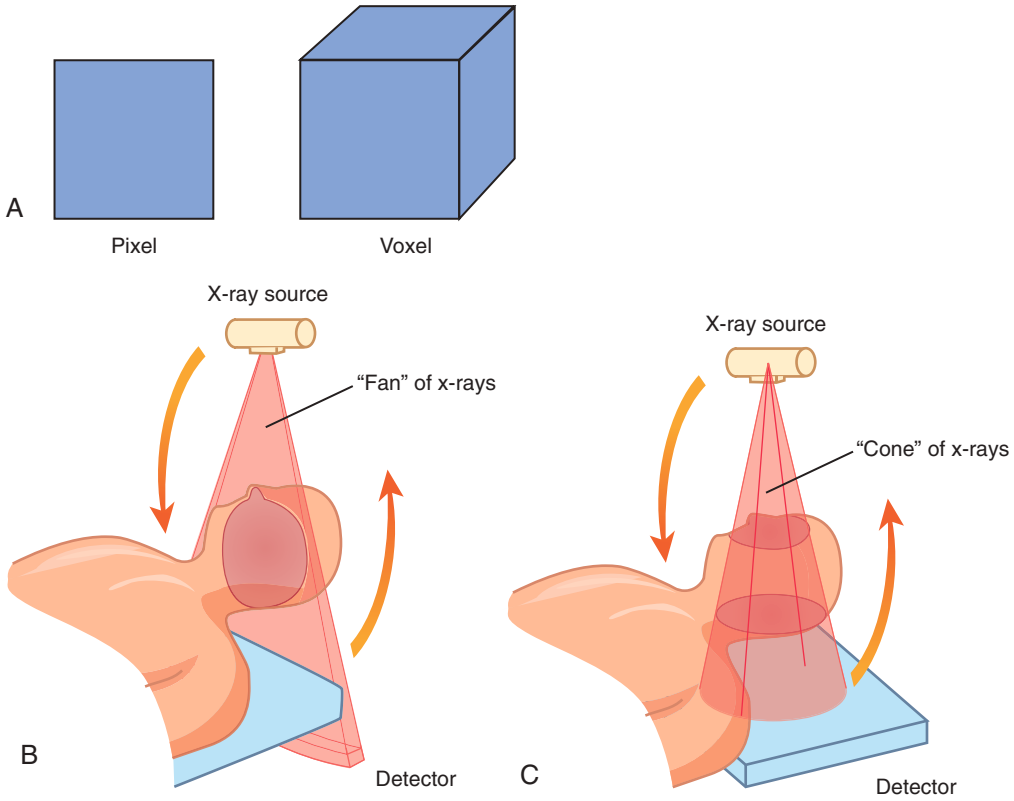


Fig. 2.8 A, *Left,* Drawing represents a *pixel* (picture element), the image capture and display element of any traditional digital image displayed on the computer. Shades of gray or color are displayed in these pixels to represent a 2D image. *Right,* Drawing represents a *voxel* (volume element). Voxels in CBCT are isometric and have the same dimension or length on all sides. They are very small (from 0.076 to 0.6 millimeters) and are the capture elements for cone beam imaging devices. Principles of conventional fan beam and cone beam computed tomography are presented in **B** and **C,** respectively. (**B** and **C,** From Babbush CA: *Dental implants: the art and science,* ed 2, St Louis, 2011, Elsevier/Saunders.)

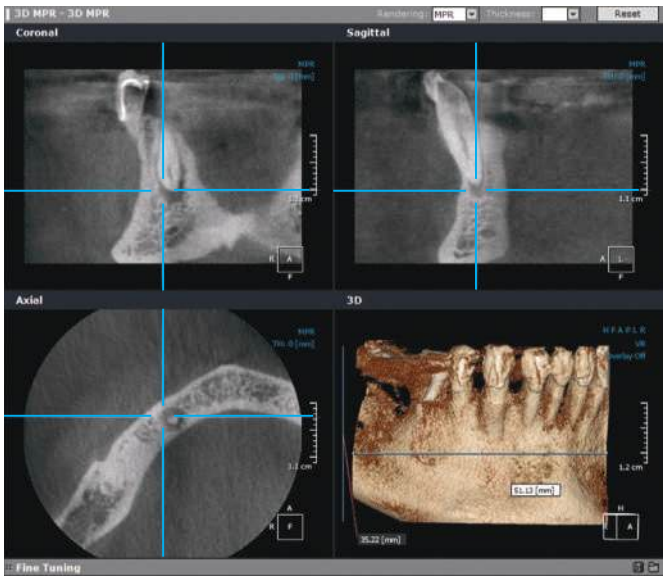


Fig. 2.9 Multiplanar and 3D color reconstructed views of the mandibular quadrant taken on a CBCT machine with a volume size of 37 × 50 mm. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

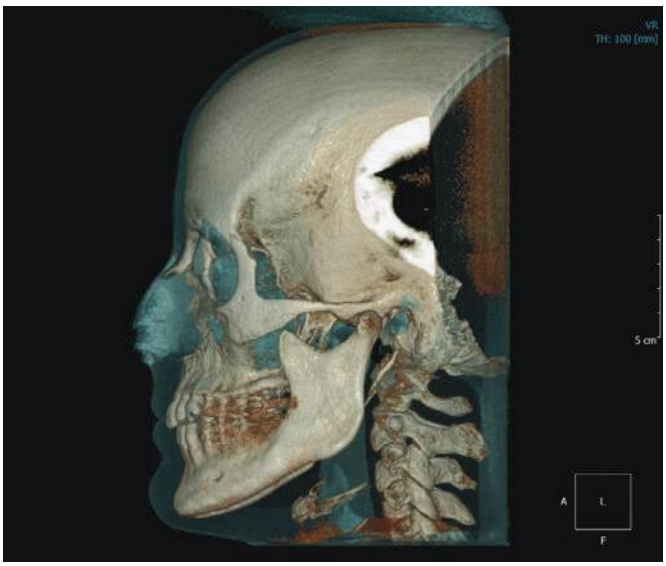


Fig. 2.10 Image of entire head (17 × 23 cm) from a large FOV unit. (Image acquired with iCAT unit [Imaging Sciences International, Hatfield, PA].)

of the clinical assessment, history, and evaluation of previous and other available imaging studies, a segment of the jaw or a larger area may need to be imaged using an appropriate FOV. If systemic conditions or generalized disorders are suspected, a larger FOV is sometimes required. For most endodontic purposes, a limited FOV can be used if no signs or symptoms of systemic conditions are reported or noted. Under no circumstances should a screening study be done using a large FOV in the absence of signs and symptoms justifying the procedure. Several multifunctional cone beam

units are available that allow the clinician to acquire several image types. Image quality has a direct impact on the diagnostic outcome; therefore, the choice of a FOV should be made carefully. Fig. 2.11 illustrates the advantages of using multiple image types for an endodontic case.

Additional benefits of CBCT imaging software include allowing the clinician to format the volume to generate an image that looks like a panoramic radiograph. Conventional panoramic machines, although not commonly used by endodontists, use the focal trough, or zone of sharpness, to position patients so as to minimize distortion along multiple axes. All inherent problems associated with panoramic imaging, including distortion, magnification, blurring, ghost shadows, and other artifacts, can be expected on the resulting image if patient positioning is not accurate. With CBCT, such artifacts are not generated, resulting in a distortion-free panoramic reconstruction (Fig. 2.12). However, it must be noted that CBCTs should not be generated in patients requiring a panoramic radiograph alone because of dose concerns.

Newer hybrid units, such as the CS 9300 3D Extraoral Imaging System (Carestream Dental, Atlanta, GA), have a wide range of FOV choices for a variety of diagnostic tasks, in addition to a conventional panoramic imaging option (Fig. 2.13). The CS 9000 unit offers the lowest voxel size of 76 microns, whereas the CS 9300 can resolve down to 90 microns, with a range extending to 500 microns for larger FOV studies. Likewise, the Morita 3D Accuitomo 80 (J. Morita USA, Irvine, CA) generates isotropic voxels of 80 microns. Although not necessary for use in every case, this technology, when appropriate, improves visualization and ultimately leads to better care in select situations. A record of exposure and doses must be maintained for each patient.

Type of Patient

Patient size and thus the amount of regional anatomy captured in the study also help determine the FOV. The smallest possible FOV must be chosen for the task at hand. Just because a clinician owns a cone beam machine does *not* mean that every patient should be exposed to a cone beam study.[140] If previous studies are available, they need to be evaluated first in a recall patient. Use of imaging in children must be minimized. Cone beam units with smaller FOVs can somewhat limit the radiation dose to critical organs and tissues of the head and neck in these cases.

Several CBCT units allow the use of a wheelchair to position special needs patients or a regular chair for patients to sit down during image acquisition. The potential to reduce motion artifacts may be an advantage (Figs. 2.14 and 2.15).

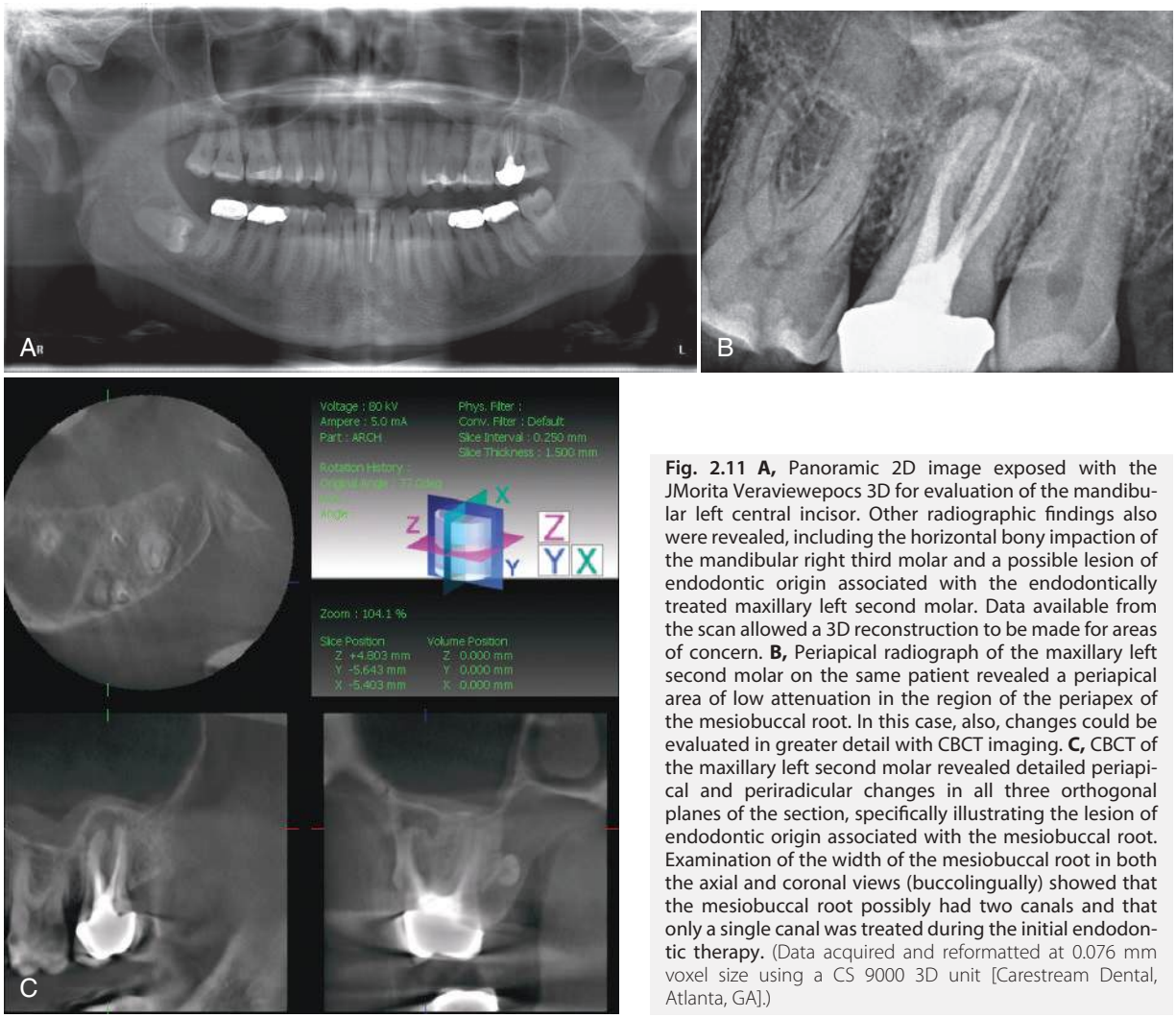


Fig. 2.11 A, Panoramic 2D image exposed with the JMorita Veraviewepocs 3D for evaluation of the mandibular left central incisor. Other radiographic findings also were revealed, including the horizontal bony impaction of the mandibular right third molar and a possible lesion of endodontic origin associated with the endodontically treated maxillary left second molar. Data available from the scan allowed a 3D reconstruction to be made for areas of concern. **B,** Periapical radiograph of the maxillary left second molar on the same patient revealed a periapical area of low attenuation in the region of the periapex of the mesiobuccal root. In this case, also, changes could be evaluated in greater detail with CBCT imaging. **C,** CBCT of the maxillary left second molar revealed detailed periapical and periradicular changes in all three orthogonal planes of the section, specifically illustrating the lesion of endodontic origin associated with the mesiobuccal root. Examination of the width of the mesiobuccal root in both the axial and coronal views (buccolingually) showed that the mesiobuccal root possibly had two canals and that only a single canal was treated during the initial endodontic therapy. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

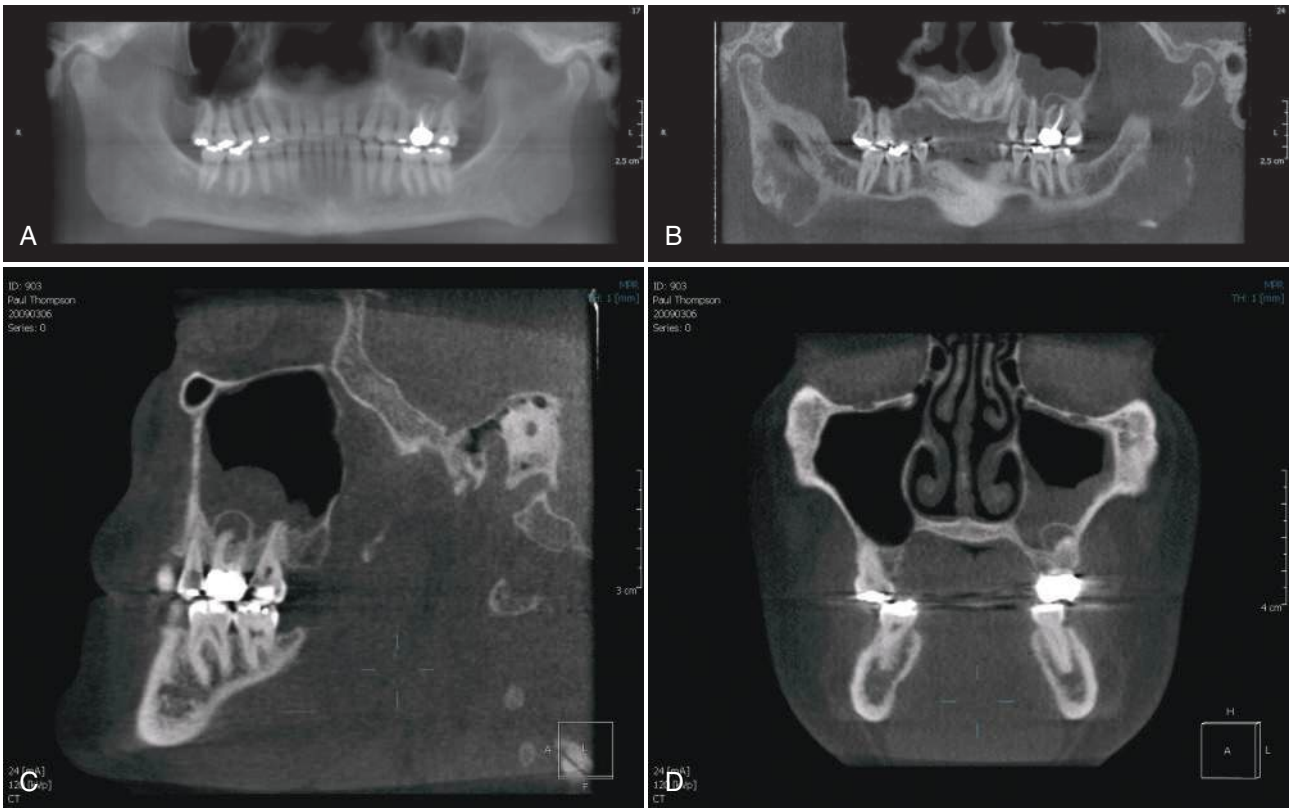


Fig. 2.12 **A,** This reconstructed panoramic image from CBCT data approximates the view one would see with a conventional panoramic radiograph. It is somewhat difficult to see the lesion on the maxillary left first molar. **B,** This thin-slice pseudopanoramic image (0.01 mm) shows the lesion without most of the anatomic superimposition. These slices of the maxillary left first molar show the lesion in sagittal **(C)** and coronal **(D)** views, confirming the features seen in the pseudopanoramic view.



Fig. 2.13 CS 9300 3D Extra-Oral Imaging System. (Courtesy Carestream Dental LLC, Atlanta, GA.)



Fig. 2.14 Veraviewepocs 3D R100. (Courtesy JMorita USA, Irvine, CA.)



Fig. 2.15 CS 8100-3D. (Courtesy Carestream Dental LLC, Atlanta, GA.)

Spatial Resolution Requirements

All endodontic imaging procedures require high spatial resolution. Assessment of canal structure, canal length, and lesions of endodontic origin (LEOs)[189] showing apical change, in addition to an understanding of possible retreatment cases, are important tasks requiring minute detail. If CBCT is used, the data acquisition should be performed at the smallest voxel size: the smaller the voxel size, the higher the spatial resolution. Many of the larger stand-alone cone beam machines, such as the i-CAT (Imaging Sciences International, Irvine, CA), default to a 0.4 mm voxel size. This voxel size is inadequate for high spatial detail. However, these units often have a voxel size selection option that allows smaller voxel sizes to be used during the image acquisition. The absolute maximum voxel size for endodontic imaging should be 0.2 mm.[44] Units typically use voxel sizes of 0.075 to 0.16 mm for their native image capture (Fig. 2.16).

Imaging Tasks Improved or Simplified by Cone Beam Volumetric Computed Tomography

The Executive Opinion of the American Academy of Oral and Maxillofacial Radiology and, later, the position paper on the use of CBCT in endodontics jointly developed by the American Association of Endodontists (AAE) and the American Academy of Oral and Maxillofacial Radiology (AAOMR), list indications for potential use in selected cases, including evaluation of the anatomy and complex morphology, differential diagnosis of complex pathoses with certain qualifiers, intraoperative or postoperative assessment of endodontic treatment, dentoalveolar trauma, resorption, presurgical case planning, outcomes assessment, and dental implant case planning.[1] Use of CBCT must be determined on a case-by-case basis only. These indications do not in any way mandate the use of CBCT for every case that falls into one of the preceding categories. For endodontic treatment and assessments, there are at least five primary imaging tasks in which CBCT scans have a distinct advantage over traditional 2D radiographs. These tasks include the evaluation of the following factors:

1. Differential diagnosis
   a. Lesions of endodontic origin
   b. Lesions of nonendodontic origin
2. Evaluation of anatomy and complex morphology
   a. Dental anomalies
   b. Root canal system morphology

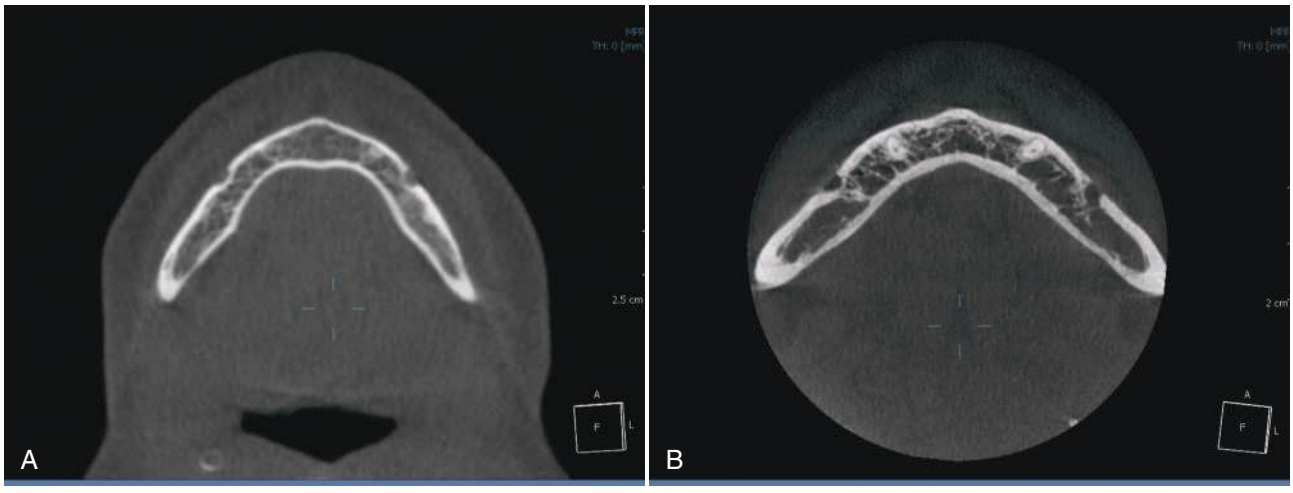


Fig. 2.16 **A,** Axial slice from data obtained with a 0.4 mm voxel size. **B,** Compare the trabecular pattern and outline of the mental foramina to the same location in an axial slice from data obtained with a 0.16 mm voxel size.

3. Diagnosis of endodontic treatment failures
4. Intraoperative or postoperative assessment of endodontic treatment complications
   a. Vertical root fractures
   b. Overextended root canal obturation material
   c. Separated endodontic instruments
   d. Calcified canals
   e. Perforations
5. Presurgical treatment planning
6. Dentoalveolar trauma
7. Internal and external root resorption
8. Endodontic treatment outcomes assessment
9. Dental implant case planning
10. 3D guided endodontics

DIFFERENTIAL DIAGNOSIS

Lesions of Endodontic Origin

Clinical endodontic diagnosis relies on subjective and objective information collected during patient examinations. Diagnosis of the pulpal status of the teeth can sometimes be challenging if adequate radiographic information is unavailable. It is fundamental to understand that LEOs arise secondary to pulpal breakdown products and form adjacent to canal portals of exit.[183,190] These radiolucent lesions, formed as a result of loss of bone mineralization, can and do form three dimensionally anywhere along the root surface anatomy.[184] A 30% to 40% mineral content loss is needed for these lesions to be visualized on conventional radiographs.[137] Furthermore, the thickness of the cortical plate covering the lesion may significantly affect the radiographic appearance of the lesion on a conventional image.[229] DSR has been observed to increase diagnostic capability with observers identifying incipient PA lesions in more than 70% of the cases.[137]

Before the advent of CBCT, clinicians were unable to routinely visualize the presence, specific location, and extensiveness of PA bone loss using conventional radiography.[127] This was especially true in areas with superimposition of anatomic structures. Visual obstruction from anatomic features, such as buccal bone and the malar process over the apices of maxillary roots, simply "disappears" when the examiner can scroll through the slices of the bone from facial to palatal in 0.1 mm sections while also changing axial orientations. CBCT showed significantly higher rates of detection of PA lesions in maxillary molars and premolars compared to PA radiography.[129] CBCT also showed a significantly higher diagnostic accuracy to detect apical periodontitis (AP) when compared to PA radiographs using human histopathology as the gold standard.[108]

In a comparative investigation of the use of CBCT and PA radiography in the evaluation of the periodontal ligament (PDL), Pope et al.[176] showed that necrotic teeth examined with CBCT had widened PDLs but 60% of vital teeth showed a PA widening of 0.5 mm or more. They called for further investigation to determine whether health and disease can be appropriately judged using CBCT in epidemiologic investigations. See Chapter 3 for more information.

Lesions of Nonendodontic Origin

Differential diagnosis of PA pathology is crucial to endodontic treatment planning. Substantial evidence in the literature points to a significant chance that lesions of the tooth-supporting structures are nonendodontic in origin, such as PA cemento-osseous dysplasia; central giant cell granulomas; lateral periodontal cyst, simple bone cysts; odontogenic cysts, tumors, or malignancies; and neuropathic pain (Figs. 2.17 and 2.18).[39,68,100,168,179,180]

Neuropathic orofacial pain or atypical odontalgia (AO), also known as *chronic continuous dentoalveolar pain* (CCDAP)[153] and *persistent dentoalveolar pain* (PDAP),[98] is related to a tooth, teeth, or pain at an extraction site where no clinical or radiographic pathosis is evident. Two systematic reviews of AO showed the incidence of persistent pain of more than 6 months' duration after nonsurgical and surgical endodontic treatment, excluding local inflammatory causes, was 3.4%.[151] The pathophysiology of this pain is uncertain, but it is hypothesized to involve deafferentation of peripheral sensory neurons in predisposed patients. The diagnosis of AO is challenging and depends on the patient history and clinical examination findings in addition to the absence of radiographic findings. In some cases, the symptoms from AP and AO are closely related. Pigg et al.[173] conducted a study of 20 patients with AO. All the patients had at least one tooth in the region of discomfort that had undergone invasive treatment; 21 of 30 teeth had undergone endodontic treatment. These researchers found that 60% had no PA lesions, and among those who did, CBCT showed 17% more PA lesions than conventional radiography. This study demonstrated that CBCT may be a useful supplement to 2D radiography (see Fig. 2.18).

The 3D radiographic appearance of a PA lesion provides additional information about the lesion's relationship to the tooth and other anatomic structures (e.g., the vascular bundle) and about the aggressiveness of the lesion. This information, along with pulp sensitivity testing, is useful for adequate treatment planning and management of these conditions (see also Chapter 3).

EVALUATION OF ANATOMY AND COMPLEX MORPHOLOGY

The precise location and visualization of dental anomalies, root morphology, and canal anatomy are vastly improved with CBCT data. Root curvature, additional roots, and anomalies within the canals themselves (e.g., obstructions, narrowing, bifurcation) are made more apparent when all three anatomic planes of section are available for review, especially with the capability of narrowing the slice thickness to as little as 0.075 mm. Visual obstruction from anatomic features such as buccal bone and the malar process over the apices of maxillary roots simply will "disappear" when you can scroll through the slices of the bone from facial to palatal in 0.076 mm sections while also changing axial orientations (Fig. 2.19).

Dental Anomalies

The use of CBCT technology has been reported in the diagnosis and treatment planning of various dental anomalies (e.g., dens invaginatus) that often have complex morphologic presentations.[152] The prevalence of dens invaginatus was as high as 6.8% in the adolescent Swedish population studied.[21] The complex nature of the anomaly presents a diagnostic challenge when conventional radiographs are

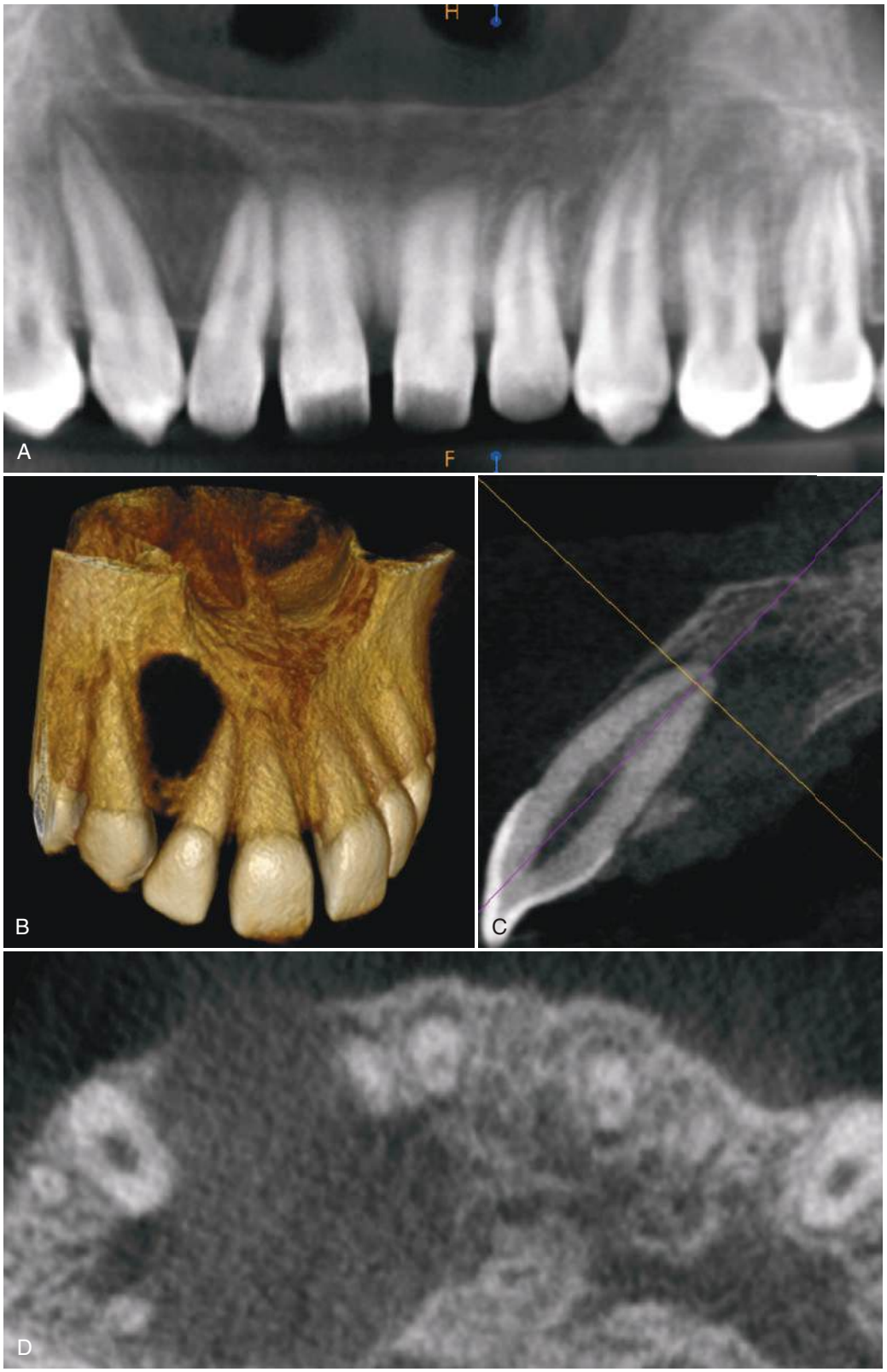


Fig 2.17 Lateral periodontal cyst. A 12-year-old male presented for orthodontic consultation with an elevated lesion in the maxillary right anterior region that was fluctuant under pressure; the patient was referred for endodontic evaluation and radiographic assessment. The pulp vitality was normal for the maxillary anterior dentition. A preoperative CBCT of the anterior maxilla was exposed (**A,** pseudopanoramic reformation; **B,** 3D surface reformation; **C,** cross-sectional reformation of the maxillary right lateral incisor) showing a unilocular/ovoid area of uniform low density that measured approximately 18.0 mm × 15.0 mm. **D,** Axial reformation showing a partially well-defined, noncorticated border mesial to the canine and a poorly defined border between the distal of the right lateral and central incisor. Surrounding tissues showed locally destroyed buccal and palatal cortical plates of the right anterior side of the maxilla with 6 mm of expansion of the buccal and palatal plates in the area between the canine and lateral incisor. Roots of the canine and lateral incisor are displaced mesially and distally, respectively. There was a 3 to 5 mm thickening of the sinus membrane of the overlying floor of the maxillary sinus. Seven months after surgical intervention, the patient was asymptomatic and a follow-up CBCT study was acquired, showing normal healing.



Fig 2.17, cont'd E, 3D surface. **F,** Cross-sectional. **G,** Axial reformation. (Courtesy Dr. Rina Gonzalez-Albazzaz and Dr. Barry Pass.)

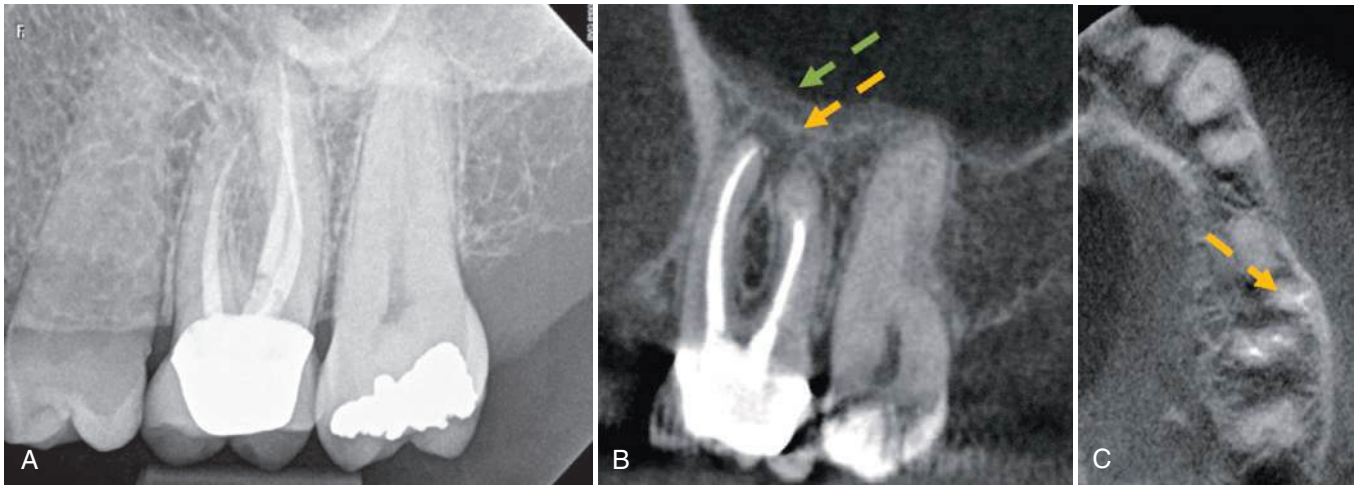


Fig. 2.18 A 62-year-old female was referred for evaluation and possible treatment for persistent dentoalveolar pain (PDAP) in the maxillary left region. The patient could alleviate this condition by placing a moist cotton roll in the adjacent vestibule to prevent the buccal mucosa from contacting the alveolus supporting the maxillary teeth in this area. This condition began after a history of local anesthetic administration, persisted for 1 year, and led to three successive new crowns and endodontic treatment on the "offending tooth," in an effort to remedy the condition without improvement. The response to endodontic tests, TMJ, and myofascial evaluations were normal. Application of topical xylocaine resulted in cessation of pain for 15 minutes. **A,** A PA radiograph showed a root-treated, maxillary left first molar with no apparent radiographic lesion. **B,** A limited FOV CBCT of the maxillary left posterior was acquired. The corrected sagittal view showed an approximately 4 mm, well-defined, ovoid, mildly corticated area of low attenuation (radiolucent) centered over the apex of the mesiobuccal root and extending to the junction of the middle and apical third of the mesiobuccal root *(yellow arrow)*. There was mild mucositis *(green arrow)*. **C,** There was a previously untreated mesioaccessory canal *(yellow arrow)* and mild mucositis. A diagnosis of neuropathic pain and a chronic apical periodontitis was made. Daily application of topical ketamine, gabapentin, and clonidine was prescribed. Endodontic revision of the maxillary left first molar was performed 3 months after the patient was stabilized with the topical medications. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D [Carestream Dental, LLC. Atlanta, GA].)

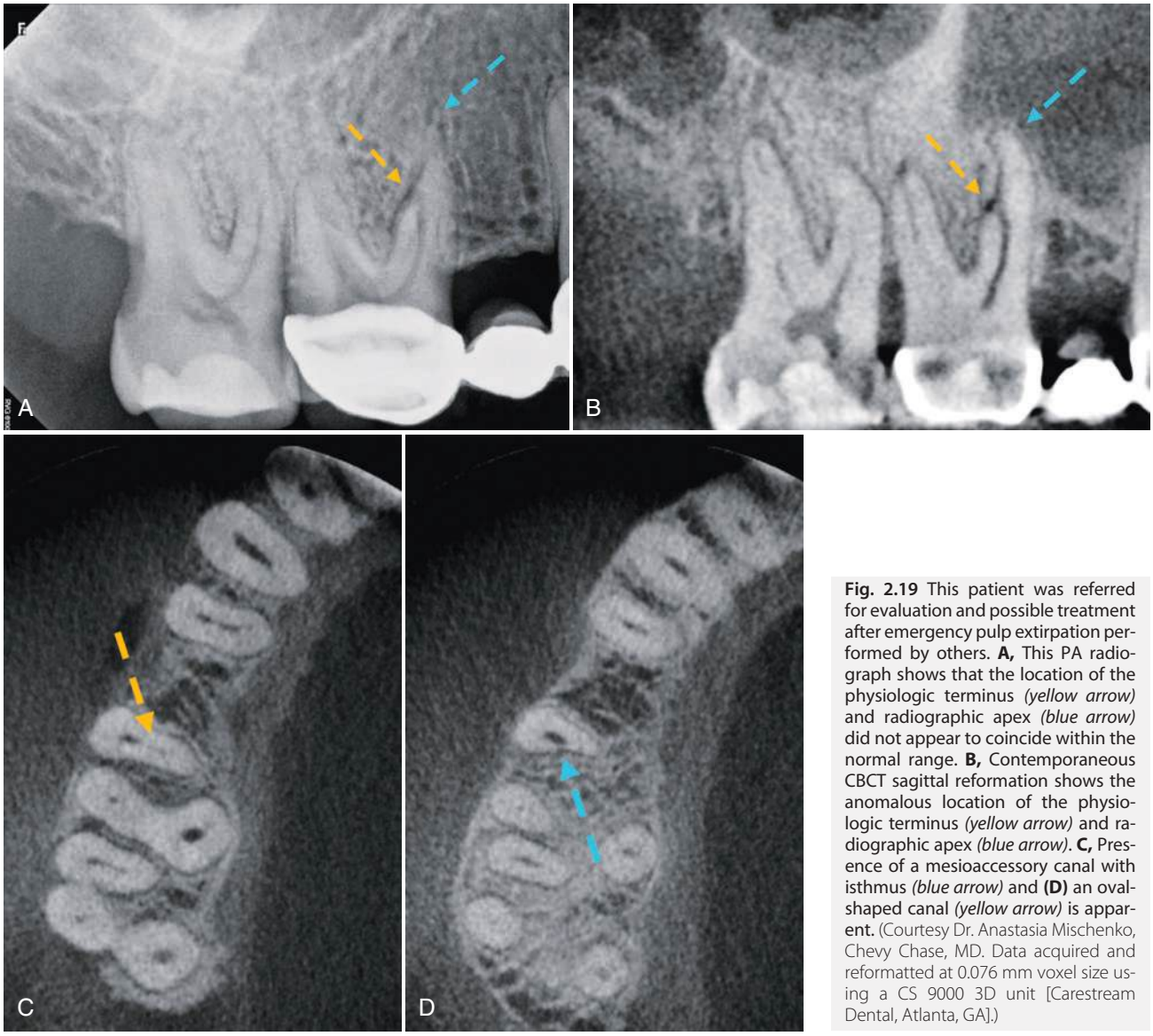


Fig. 2.19 This patient was referred for evaluation and possible treatment after emergency pulp extirpation performed by others. **A,** This PA radiograph shows that the location of the physiologic terminus *(yellow arrow)* and radiographic apex *(blue arrow)* did not appear to coincide within the normal range. **B,** Contemporaneous CBCT sagittal reformation shows the anomalous location of the physiologic terminus *(yellow arrow)* and radiographic apex *(blue arrow)*. **C,** Presence of a mesioaccessory canal with isthmus *(blue arrow)* and **(D)** an oval-shaped canal *(yellow arrow)* is apparent. (Courtesy Dr. Anastasia Mischenko, Chevy Chase, MD. Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

used.[160] In case reports in which CBCT was used for diagnosis and treatment planning, treatment options included conservative endodontic treatment of the invagination, surgical treatment of the PA pathology, and complete revascularization of the dens after removal of the invagination (Fig. 2.20).[144,222]

Root Canal System Morphology

As the adage goes, nature seldom makes a straight line and never makes two of the same. This statement is dramatically illustrated in the evaluation of root canal system morphology. With ever-present unusual and atypical root shapes and numbers, there is sometimes a need to look further than what a clinician can see or imagine with 2D radiography (Fig. 2.21). Variations in root canal morphology have been studied using various in vitro techniques.[70,174,220,221] The results of these studies point to the fact that there is significant variation in the root canal morphology among various ethnic population groups.[8,90,91,149,224] CBCT has been reported to be comparable to canal staining and clearing techniques for identification of the root canal morphology,[145] and CBCT studies report variation in the root canal morphology among various ethnic groups.[146,204,236,238] Accuracy of CBCT imaging for detection of second mesiobuccal canal in maxillary molars has been shown to be 96%, and the prevalence of midmesial canals in mandibular molars was found to be 16.4%.[203]

DIAGNOSIS OF ENDODONTIC TREATMENT FAILURES

The failure of previous endodontic therapy can be attributed to various factors, such as procedural errors, missed canals, or persistent PA pathosis. Knowledge of the cause of failure is pertinent to the treatment of these cases because it allows the cause to be adequately rectified. With the advent of CBCT, in select cases of retreatment in which the cause of failure is otherwise undetectable, adequate information may be collected to apply to the treatment plan. Use of CBCT in these situations will aid in treatment planning

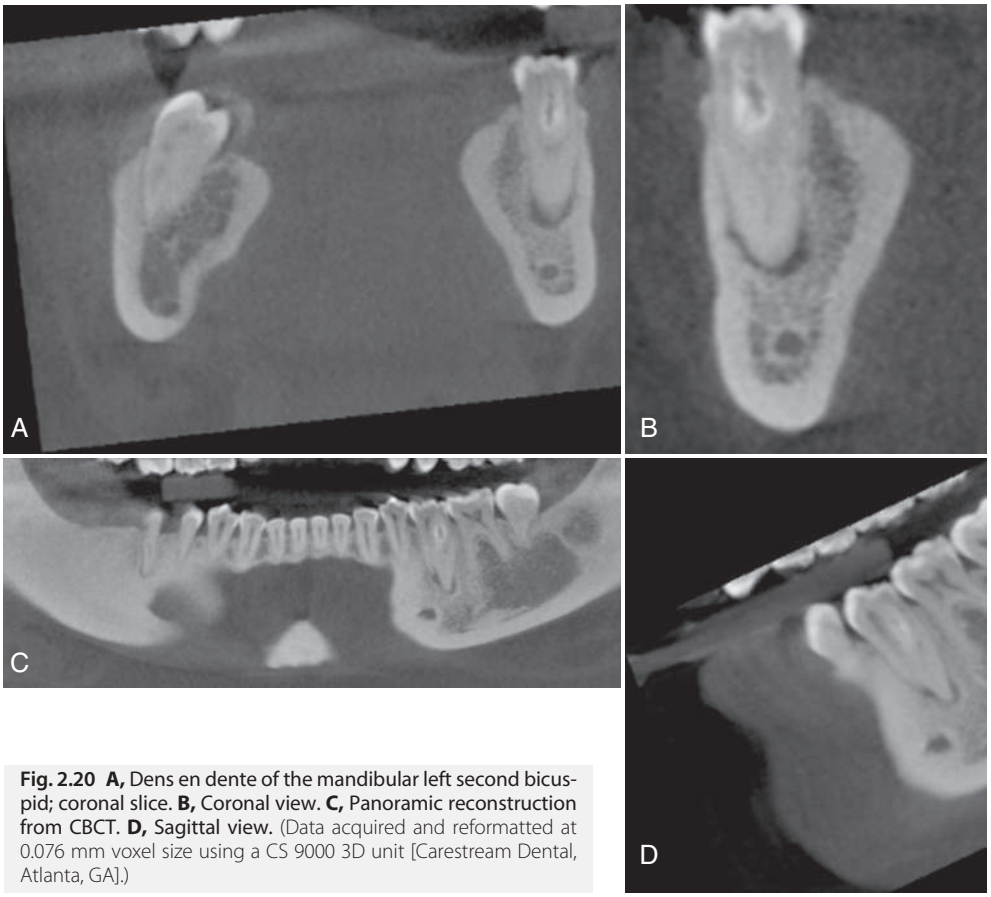


Fig. 2.20 A, Dens en dente of the mandibular left second bicuspid; coronal slice. **B,** Coronal view. **C,** Panoramic reconstruction from CBCT. **D,** Sagittal view. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

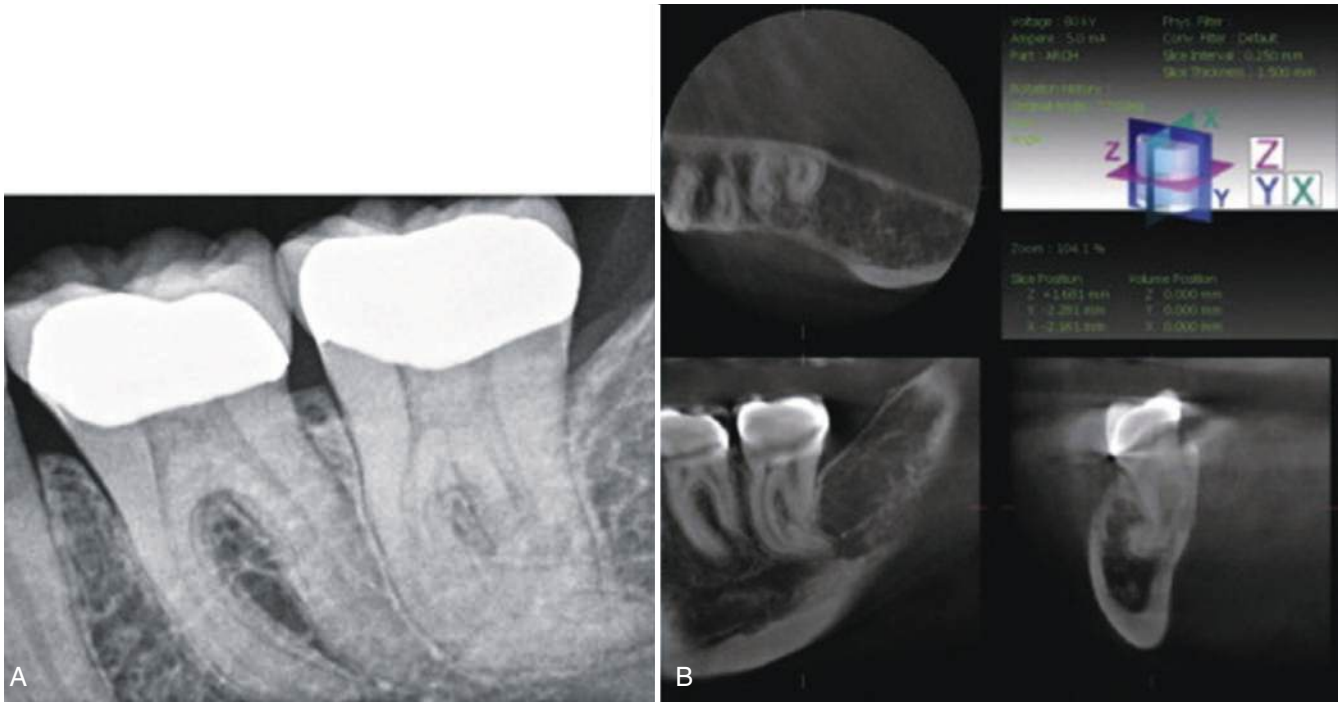


Fig. 2.21 A, Mandibular left second molar referred for endodontic evaluation and possible treatment. This 2D radiograph reveals significant pulp stones and canal calcification developing not only in the coronal aspect of the root canal system, but also extending down the visible distal canal. The apical third of the canal system appears unusual and dilacerated. Cone-beam computed tomography (CBCT) would be beneficial in visualizing the root canal anatomy, to create the ideal endodontic access. **B,** Single slice of the CBCT image for the same tooth. Information about the direction of the root canal anatomy is provided in all three planes of the section: axial, coronal, and sagittal. Interestingly, the axial slice shows that the mesial lingual root actually traverses buccally as it approaches its terminal extent. This is valuable information for the clinician before the entire root canal system is cleaned and shaped; it may also establish a higher degree of treatment predictability.

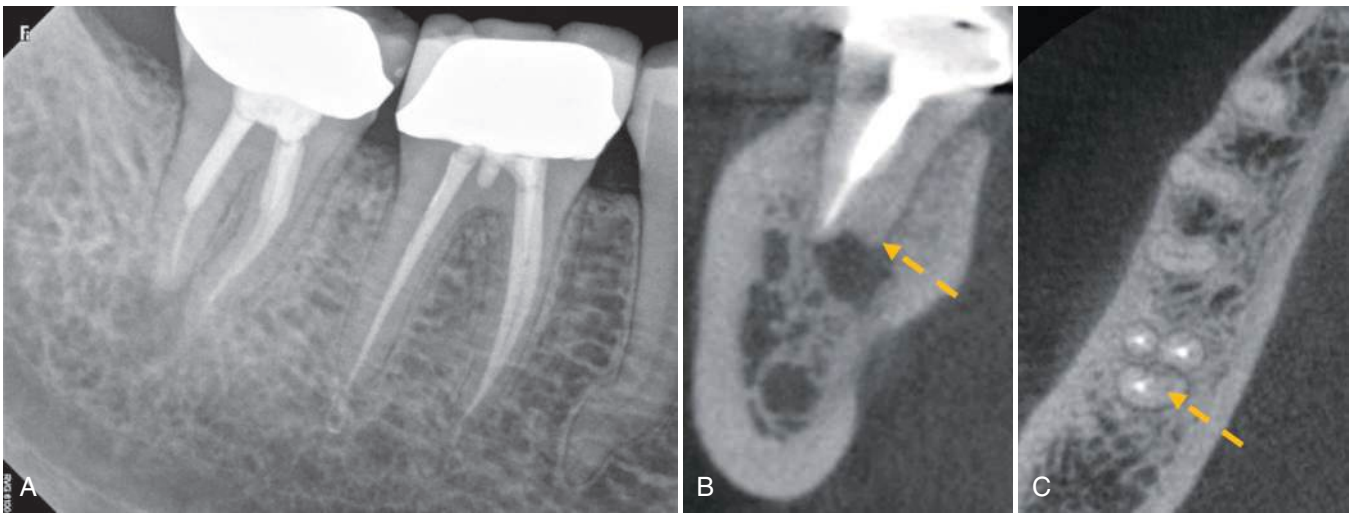


Fig. 2.22 This 38-year-old female patient presented for evaluation and treatment of a symptomatic mandibular right second molar, which had been endodontically treated more than 10 years previously. This tooth was sensitive to percussion and biting forces; periodontal findings were normal. Microscopic examination of the exposed dentin was negative for a vertical fracture. **A,** The PA radiograph showed the previous endodontic treatment; a post present in the distal canal; and an approximately 5 mm diameter, unilocular, well-defined area of uniform low density centered at the periapex of the distal root—features consistent with an apical periodontitis. Contemporaneous **(B)** CBCT sagittal and **(C)** axial reformations revealed a previously untreated distobuccal canal *(yellow arrow)*. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

and improve outcomes (Fig. 2.22).[109] A study by Rodriguez et al. showed that the endodontic retreatment strategies were altered in 49.8% of cases after CBCT images were included for diagnosis by general dentists and endodontists.[181]

Use of CBCT in evaluation of previously endodontically treated teeth aids in visualizing missed canals, canal deviations, resorption, procedural errors such as perforation, and other pathology.

The technology is most useful in detecting uninstrumented and unfilled canals, extension of the root canal filling, and the presence and extent of periradicular bone loss. The incidence of missed canals in failed endodontic treatments were found to be 23.04% with the highest incidence in upper molars. These teeth had 4.38 more likelihood of having a PA lesion.[109] The sensitivity of CBCT and PA radiographs for diagnosing strip perforations in root-filled teeth has been shown to be low, although CBCT showed a significantly higher sensitivity when compared to PA radiographs.[194] Radiopaque filling materials in the root canals of endodontically treated teeth can produce streak artifacts, which can mimic fracture lines or perforations.[194,237] Use of lower exposure parameters and nonmetallic fiber posts can significantly reduce these artifacts, allowing for an improved diagnosis.[59]

Intraoperative or Postoperative Assessment of Endodontic Treatment Complications

VERTICAL ROOT FRACTURES

Vertical root fractures (VRFs) that run along the long axis of a tooth are often difficult to diagnose clinically. The prevalence of VRF in endodontically treated teeth has been reported to range from 8.8% to 13.4%.[77,207,233] These fractures typically run in the buccolingual direction and are confined to the roots, making it difficult to visualize the fracture. Visualizing the fracture on a conventional radiograph is possible when the x-ray beam is parallel to the plane of the fracture.[182] Challenges in diagnosis with regard to the extent and exact location of the fracture often lead to unwarranted extraction of teeth. Since the introduction of CBCT to dentistry, various reports of the application of the technology to detect vertical root fractures have been published. The reported sensitivity for detection of VRFs has ranged from 18.8% to 100%[136]; by comparison, conventional radiographs have a reported sensitivity of approximately 37% (Fig. 2.23).[62,95] CBCT has been used to visualize VRFs in controlled clinical studies in which clinical diagnosis was difficult.[62] VRFs were successfully detected at a spatial resolution ranging from 76 to 140 microns. However, only a limited number of units provide such high resolution. A comparison of various CBCT units for the detection of VRFs demonstrated that the units with flat panel detectors (FPDs) were superior to the image intensifier tube (IIT)/CCD–based detectors; the smaller FOV and the ability to view axial slices also improved detection of VRFs.[96] Continued improvement of sensor technology, including the use of FPDs, has resulted in enhanced resolution. Voxel dimensions are smaller in these units. Detection of vertical root fractures with thickness ranging from 0.2 to 0.4 mm was found to be more accurate with CBCT than with digital radiography.[155,161] The presence of root canal filling in the teeth lowers the specificity of CBCT in detecting vertical root fractures[95,96,111]; this has been attributed to the radiopaque material causing streak artifacts that mimic fracture lines.[237]

Zhang et al. reported a low sensitivity of 33.3% for detection of subtle vertical root fractures in a clinical study. However, the study showed the presence of vertical bone loss in a high number of the cases.[235] Bone loss pattern around teeth with vertical root fractures viewed on CBCT images may assist clinicians in diagnosis of VRF.[211]



Fig. 2.23 Root fracture in an endodontically obturated maxillary right central incisor. **A,** Axial view with artifacts from a highly attenuating (opaque) obturant. **B,** View without artifacts from obturant. **C,** Oblique parasagittal view. **D,** Paracoronal view. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

OVEREXTENDED ROOT CANAL OBTURATION MATERIAL

CBCT scans provide the opportunity to map endodontic treatment complications through the examination of 3D representations of the teeth and supporting structures in different planes. Few high-level studies related to the effects of endodontic treatment complications have been published in the endodontic literature.[206] However, it is generally recognized that the overfilling of the root canal, causing damage to vital structures such as the inferior alveolar neurovascular (IAN) bundle (Fig. 2.24) or the maxillary sinus, can cause significant morbidity.[34,35,76,80]

Endodontic therapy undertaken in close proximity to the IAN bundle should receive special attention because direct trauma, mechanical compression, chemical neurotoxicity, and an increase in temperature greater than 10°C may cause irreversible damage.[65,85,88,214] Scolozzi et al. reported that sensory disturbances can include pain, anesthesia, paresthesia, hypoesthesia, and dysesthesia.[191] The IAN bundle is located in the cribriform bone-lined mandibular canal and courses obliquely through the ramus of the mandible and horizontally through the mandible body to the mental foramen and the incisive foramen.[11] There are many anatomic variations of the IAN bundle, including the anterior loop and bifid mandibular canals.[45] Kovisto et al.[116] used CBCT measurements from 139 patients to show that the apices of the mandibular second molar were closest to the IAN bundle. In females, the mesial root of the second molar was closer than in males, and the distances in all roots measured increased with the age of the patient. There was a high correlation between the measurements from left to right side in the same patient, and an average distance of 1.51 to 3.43 mm in adults.[116] Procedures involving the mandibular second molar were most likely to cause nerve damage.[126] Further research is required to clarify the risks and benefits of CBCT when endodontic treatment is contemplated on teeth with a proximal relationship between the IAN and root apices. Porgrel treated 61 patients with involvement of the IAN bundle after root canal therapy during a 7-year period. Eight patients were asymptomatic; 42 patients were seen for mild symptoms or were examined more than 3 months postoperatively, with only 10% experiencing improvement. Five patients underwent surgical treatment before 48 hours elapsed and recovered completely. Six patients underwent surgery between 10 days and 3 months, with four experiencing partial recovery; the remaining two had no improvement.[175]

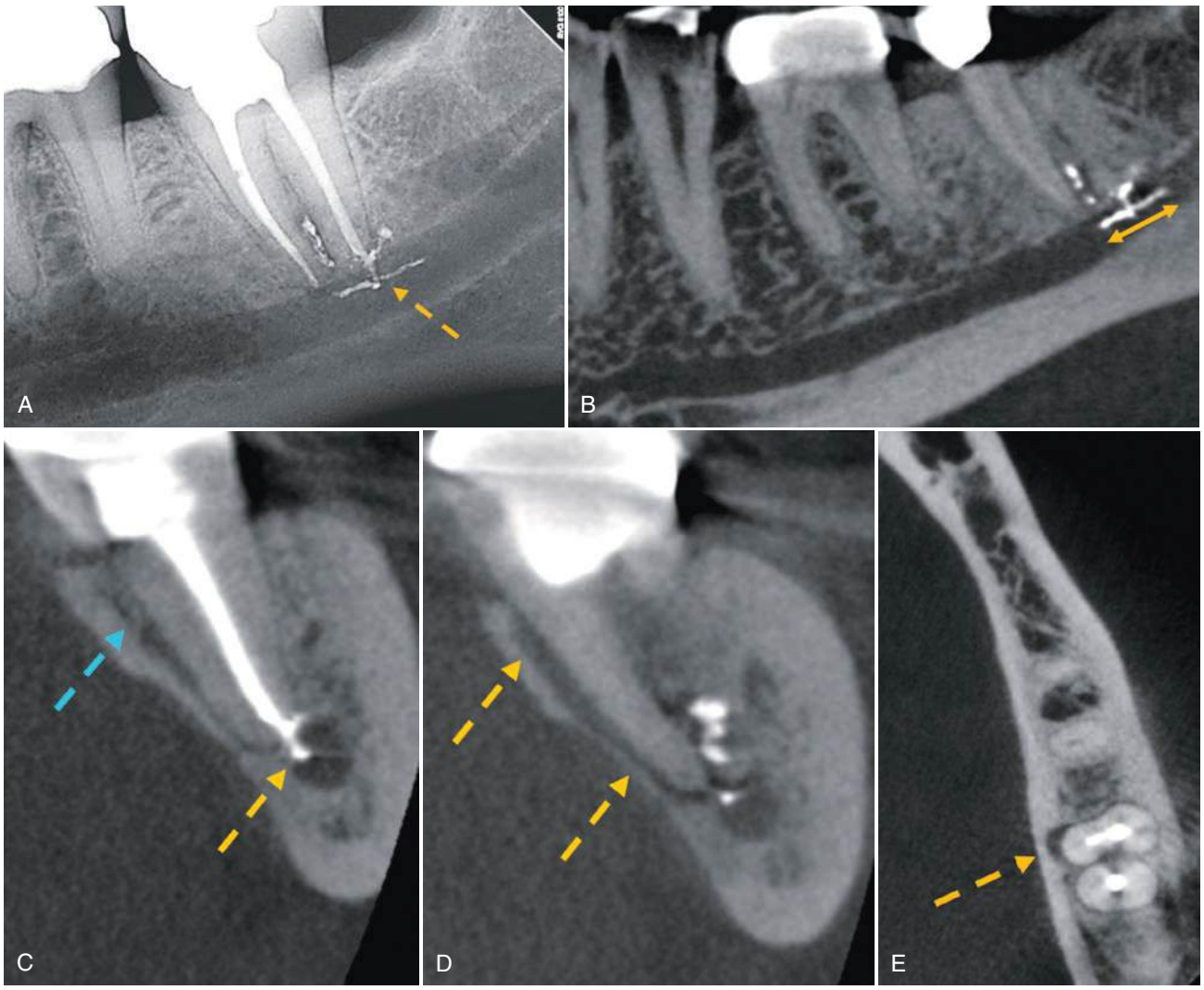


Fig. 2.24 This 64-year-old male patient presented with sensitivity when biting on the mandibular left second molar. The history included endodontic revision more than 6 months earlier and subsequent transient paresthesia and dysesthesia along the distribution of the inferior alveolar nerve, which persisted for 1 week after the retreatment. Epicritic and protopathic sensibility was normal. The results of periodontal probing, staining and microscopic examination of the exposed root surface at the sulcus were normal. **A,** Initial PA radiograph showed the approximate location of excess radiopaque material, a feature consistent with extruded root canal sealer *(yellow arrow)*. **B,** Contemporaneous CBCT corrected sagittal reformation shows approximate length of extruded material, measuring 3.4 mm *(yellow arrow)*. **C and D,** Corrected cross-sectional reformation shows the foreign material interior to the inferior alveolar canal *(yellow arrow)*, an area of low attenuation extending from the apex superiorly to an area near the alveolar crest *(blue arrow)*, and erosion of the lingual endosseous surface. **E,** The same feature was shown in the corrected axial view *(yellow arrow)*, bisected by a dark line extending from the obturant to the surface of the root—features suggestive of a vertical fracture. Examination of the extracted tooth revealed a vertical fracture at the lingual aspect of the distal root. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

New imaging technologies, such as high-resolution magnetic resonance imaging (MRI-HR) and magnetic resonance neurography (MRN), promise to improve isolation of the IAN from the neighboring artery and vein contained within the inferior alveolar bony canal. MRN studies have documented the ability to demonstrate nerve continuity and localize extraneural nerve compression before surgical nerve exploration. Postoperative PA radiographs should be exposed on the day of endodontic treatment completion or a suspected iatrogenic event, and any suspected compromise of the IAN bundle or other vital structures should be evaluated immediately. In all cases in which trauma to the IAN bundle is suspected from PA or panoramic radiography or by the report of symptoms consistent with nerve injury, exposure of a CBCT image volume should be considered. It is generally accepted that immediate surgical debridement should be attempted to maximize recovery.[65,177] With the introduction of MRI algorithms for dental diagnostic purposes, it is expected that this imaging modality will be increasingly used in diagnostic and treatment planning. MRI has the capability to demonstrate vascularity to the tooth of interest, in addition to the presence of inflammatory exudates in the apical regions, without exposing the patient to ionizing radiation. Receiver coils are being developed to enhance the image quality of maxillofacial and dental magnetic resonance studies.

The accidental introduction of root canal instruments, irrigating solutions, obturation material, and root tips into

the maxillary sinus has been reported. Serious consequences associated with the intrusion of foreign bodies into the maxillary sinus include pain, paresthesia, and aspergillosis, a rare but well-documented complication of endodontic treatment.[24] Guivarc'h et al. reported that the overextension of heavy metal–containing root canal sealers, such as zinc oxide eugenol cement, may promote fungal infection in immunocompromised patients, leading to bone destruction and damage to adjacent structures. This case report described the use of CT to assess the patient before surgery and at 6 months.[89] The use of CBCT as an aid in the localization and retrieval of an extreme overextension of thermoplasticized injectable gutta-percha into the sinus and contiguous soft tissues has been described by Brooks and Kleinman.[37]

SEPARATED ENDODONTIC INSTRUMENTS

Instrument fracture can occur at any stage of endodontic treatment, and in any canal location. The incidence of this complication, reported in clinical studies on a per canal or per tooth basis, ranges from 0.39% to 5.0%.[56,157] Molars are predominantly affected by instrument fracture, with the highest incidence found in the apical third of mandibular molars.[7,52,123,147]

A systematic review and meta-analysis showed that when endodontic treatment was performed at a high technical standard, instrument fracture did not significantly reduce the prognosis. More specifically, when no initial radiographic PA lesion was present, 92.4% of cases remained healthy; when a PA lesion was present initially, 80.7% of PA lesions showed radiographic healing. However, the presence or absence of periradicular lesions on preoperative and postoperative examinations was based on planar radiographic assessments, which calls these findings into question.[156] Other studies showed that the chances of endodontic failure increased if the canal system could not be thoroughly disinfected, if a periradicular periodontitis was present, or if technical standards were compromised.[51,110,198,200] Use of CBCT to triangulate the retained instrument and assess the canal shape, especially in cases in which the operating microscope does not allow direct visualization, can be helpful in formulating a removal strategy. If the fractured instrument is lodged in the lingual aspect of a ribbon-shaped canal, for example, an instrument may be inserted toward the buccal to bypass and remove the imbedded instrument without forcing the fragment further apically. Without the use of CBCT, intracanal instruments can be reliably removed or bypassed in 85.3% of cases if straight-line access is possible; however, reliable removal or bypass is possible in only 47.7% of cases if the instrument is not visible (Fig. 2.25).[147] When a separated instrument is lodged in the apical third of a root canal, the chances of retrieval are the lowest, but the apical terminus may be adequately sealed by treatment of an anastomosing canal, if present.[79] The possibility of instrument removal based on CBCT triangulation has not been published to date.

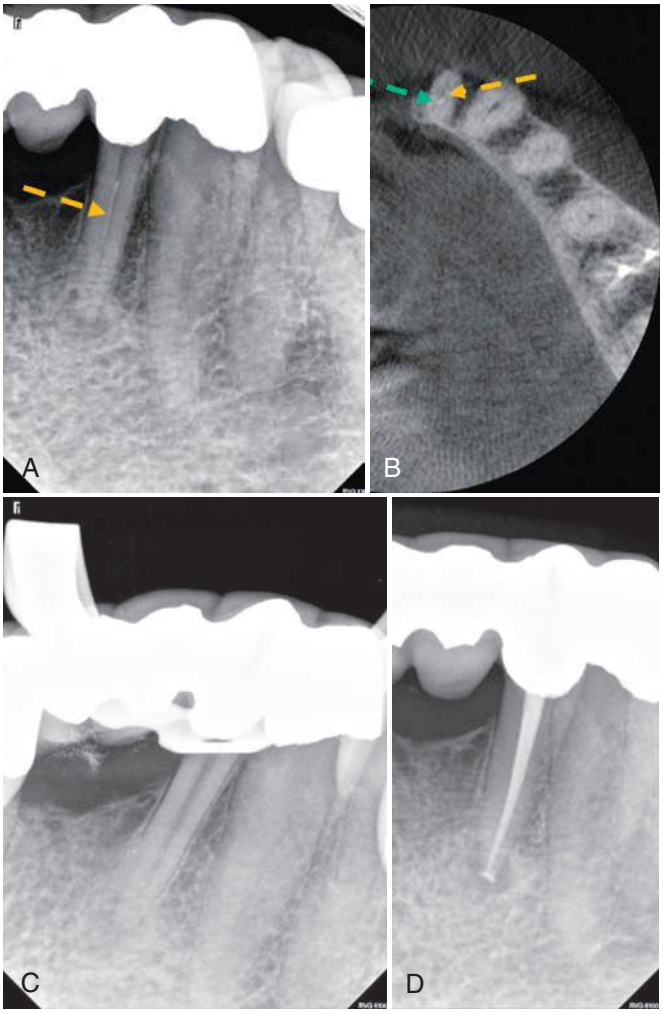


Fig. 2.25 Unexpected torsional and flexural failure of endodontic instruments can occur during instrumentation. **A,** PA radiograph shows a separated file *(yellow arrow)* located at the midroot of the mandibular left lateral incisor in a patient referred for revision treatment. **B,** To aid development of a retreatment strategy, CBCT was used to localize the instrument *(yellow arrow)* in the lingual canal, with identification of the buccal canal *(green arrow)* to facilitate bypassing the instrument and **(C),** subsequent removal. **D,** Obturation was completed without complication. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

CALCIFIED CANALS

According to the Pew Research Center, 10,000 US individuals will reach the age of 65 every day until 2030, and the nation's 65-year-old and over cohort will grow to 81 million in 2050, up from 37 million in 2005.[171] This aging population will present increasing challenges for dental clinicians because calcification of the root canal system increases as part of the natural aging process,[86] possibly leading to more untreated canals that may serve as a niche for microorganisms.[28,104] Pulp chambers in the crown of the tooth decrease in size, forming more rapidly on the roof and floor of posterior teeth.[219] Typically, root canals calcify at the coronal aspect first, with decreasing calcification as the canal travels apically. Magnification and illumination are essential tools for the identification and treatment of calcified canals, but CBCT can assist in the perioperative treatment of such conditions.[23] Preoperative assessment of calcified teeth using CBCT can suggest the best tactic for locating calcified canals in the chamber floor and roots

using software-based measurement tools. The insertion of radiopaque markers, such as instruments or obturation material, can facilitate reliable canal localization using available multiplanar reformations. The increased sensitivity and specificity provided by CBCT can also assist in the determination of the PA status of calcified root canals that may not require measures that can lead to procedural errors, such as off-course access, instrument fracture, or root perforation.[118] The difficulty in locating calcified canals can be further compounded by morphologic anomalies associated with gender and ethnic origin.[192] CBCT can be an important adjunct to magnification and illumination in these cases.

PERFORATIONS

A *perforation* is defined as a "mechanical or pathologic communication between the root canal system and external tooth surface"[9]; it is usually associated with an iatrogenic event, accounting for about 10% of all nonhealed cases.[102] Root perforations can be caused by a post preparation, the search for a calcified canal, a strip perforation, or an attempt to retrieve a fractured instrument. They are often difficult to localize with conventional imaging because no information about the buccolingual dimension can be obtained.[232] Shemesh et al.[194] compared the sensitivity and specificity of CBCT scans with two-angulation PA imaging using phosphor plates to assess the likelihood of detecting root or strip perforations after root canal treatment with laterally compacted gutta-percha and sealer. They found that the two methods showed similar specificity, but the CBCT image volumes showed higher sensitivity. Single-angulated PA radiographs showed 40% of the perforations, and two-angled PAs showed 63%, suggesting that if PA radiographs alone are used, two-angled images were superior. There was no significant difference in the detection of root perforations between PA and CBCT radiography. The researchers noted that the results may have been affected by the small size of the perforations and that the method of obturation did not favor extravasation of obturation material.[194] CBCT images suffer from beam hardening artifact resulting from root canal obturation and restorative materials (e.g., gutta-percha, posts, and perforation repair materials), which creates challenges to the interpretation of root integrity. An approach advocated by Bueno et al.[40] suggested that a map-reading strategy of viewing sequential axial slices reduces the beam hardening effect. Improved CBCT software algorithms are expected to reduce artifact formation in the future.

Presurgical Treatment Planning

Surgical endodontic treatment is often performed in cases of endodontic nonhealing when nonsurgical retreatment is not possible. In the past, conventional and digital 2D PA radiographs were the only means of assessing the apical region. Unfortunately, the information available from these images may not adequately prepare the clinician to treat the pathosis surgically. For example, the clinician may be unable to observe whether the lesion has perforated the buccal or palatal cortical plates, as in the example that follows, or even observe which root or roots are involved. Presurgical confusion is resolved with cone beam imaging. Multiplanar views allow the clinician to see the defect and suspected causes from the axial, sagittal, and coronal aspects; 3D grayscale or color imaging helps the clinician visualize the entire defect before the incision is made. This is an immense improvement over conventional imaging and helps with better assessment of the prognosis prior to the surgical treatment (Fig. 2.26).

The relationship of the teeth and the associated pathology to important anatomic landmarks must be taken into consideration in the treatment planning for endodontic surgical procedures. These anatomic landmarks include, but are not limited to, the maxillary sinus, the mandibular canal, the mental foramen, the incisive canal, and the buccal and lingual/palatine cortical plate. The close proximity of the maxillary posterior teeth to the sinus has been linked to maxillary sinusitis of odontogenic origin; changes in the maxillary sinus have ranged from thickening of the Schneiderian membrane to actual accumulation of fluid in the sinuses.[130,131,134] The relationship of the roots of the posterior teeth to the sinus during presurgical treatment planning and the changes within the sinus can be best appreciated with the use of CBCT images.[32,131,193]

The relationship of the roots of the mandibular posterior teeth and associated PA pathoses to the mandibular canal, the presence of an anterior loop, and the distance of the mandibular canal from the buccal and lingual cortical plates are pertinent pieces of information when surgical procedures in mandibular posterior teeth are planned.[31] The 3D nature of this relationship can be best assessed using CBCT.[113] A potential difference in the location of the mandibular canal with respect to age and gender has been reported.[116,197]

The determination of the thickness of the buccal cortical plates and anatomy of the root apex prior to endodontic surgical procedures enables the clinician to preplan the osteotomy and the root end resection. The thickest buccal cortical plate was determined to be in the second molar area as per a recent study.[234]

DENTOALVEOLAR TRAUMA

Dental trauma is common during the lifespan, with prevalence in the primary dentition of approximately 30% and in the permanent dentition of approximately 20%.[12] Systematic epidemiologic data at a large medical center suggest a prevalence of 48.2% for dental injuries in all facial trauma, resulting in injuries to the dentition in 57.8% of household and play accidents, 50.1% of sports accidents, 38.6% of work-related accidents, 35.8% of acts of violence, and 34.2% of traffic accidents, with 31% unspecified.[78] The prevalence of traumatic dental injuries varies according to the population studied, but these injuries occur most commonly in children 7 to 10 years of age (also see Chapter 21).[18] Dental traumatic injuries affect one fourth of all schoolchildren, with most injuries occurring before the age of 19.[81] Maxillary central incisors sustain approximately 80% of all dental traumatic injuries, followed by maxillary lateral incisors and mandibular incisors.[16] The most common type of traumatic dental injuries in the primary dentition are luxation injuries, whereas

crown fractures are the predominant dental injury to the permanent dentition.[117] Determination of the extent of injury to the dentinopulpal complex requires a methodical approach that evaluates the teeth, periodontium, and associated structures (Fig. 2.27) and may result in significant long-term complications.[55]

Trauma to the orofacial complex can result in the following injuries to the permanent dentition, based on the International Association of Dental Traumatology (IADT) guidelines: (1) infraction; (2) enamel fracture; (3) enamel-dentin fracture; (4) enamel-dentin-pulp fracture; (5) crown-root fracture without pulp exposure, (6) crown-root fracture with pulp exposure; (7) root fracture; (8) alveolar fracture; (9) concussion; (10) subluxation; (11) extrusive luxation; (12) lateral luxation; (13) intrusive luxation; and (14) avulsion.[13,58] (These guidelines can be reviewed at www.dentaltraumaguide.org.) The IADT guidelines for assessment of permanent teeth at initial visit recommend exposing several projections and angulations, with additional radiographs indicated when foreign objects become embedded in soft tissue. However, the recommendations suggest that CBCT may be beneficial when used to assess patients after dental traumatic injuries, especially in cases of lateral luxation, root fracture, complications, and monitor healing. The heterogeneity of dental traumatic injuries and difficulty in assessing affected teeth using intraoral radiography, especially when in the plane of PA radiographs, is evident from the guideline recommendations.[58]

Intra-alveolar root fractures generally affect the permanent dentition of males and are relatively uncommon, accounting for 0.5% to 7% of dental impact injuries.[19,54,84,154] Root-fractured teeth present diagnostic challenges, and the limitations of planar radiography have been well documented in the dental literature.[50,58,115,159] A systematic retrospective study showed that maxillary central (68%) and lateral (27%) incisors were primarily affected, with only limited occurrence in mandibular incisors (5%). This retrospective study concluded that CBCT allowed for improved treatment planning and showed the true nature of the injury compared with PA imaging alone.[228] At least seven systematic laboratory studies and one systematic in vivo animal study reported significantly improved accuracy for the detection of root fractures when CBCT was compared with PA radiography.[95,96,101,135,155,225] In a systematic clinical study, Bornstein et al.[33] examined 38 patients with 44 permanent teeth that sustained intra-alveolar fractures. In the study sample reported, 68.2% of teeth had oblique fractures that extended to the cervical third of the root, contradicting the findings of previous studies conducted with PA imaging alone. CBCT imaging offered improved visualization of the location and angulation of root fractures compared to PA and occlusal intraoral radiographs.[33]

Injuries that include fractures and comminution injures to the supporting structures or are difficult to visualize with intraoral imaging because of compression of overlying tissues, may be evident on CBCT scans. Several of these

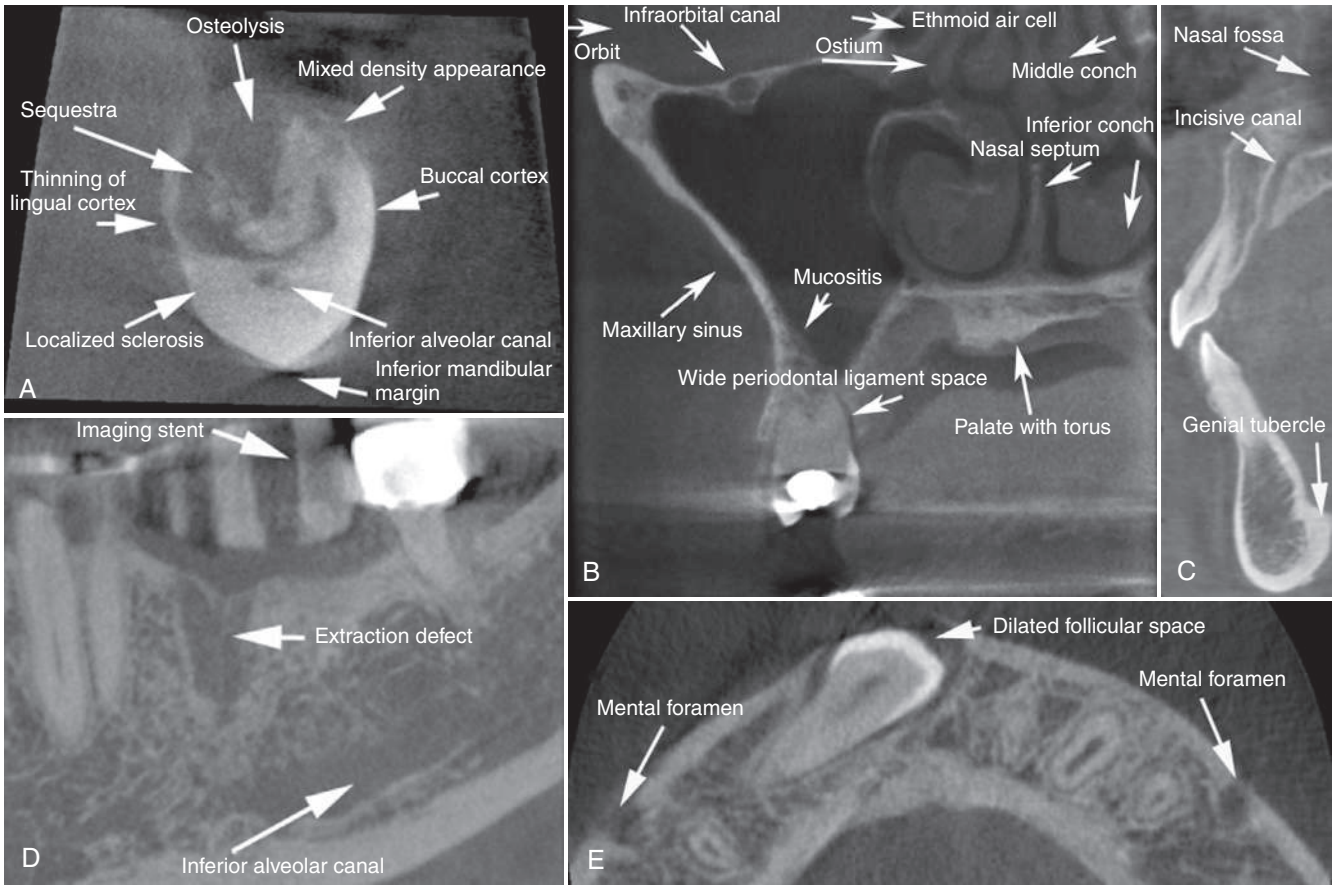


Fig. 2.26 A to W, Anatomic landmarks on CBCT images acquired using CS 9000, CS 9300, and iCAT units. (Carestream Dental, Atlanta, GA [CS 9000, CS 9300]; and Imaging Sciences International, Hatfield, PA [iCAT].)

Continued

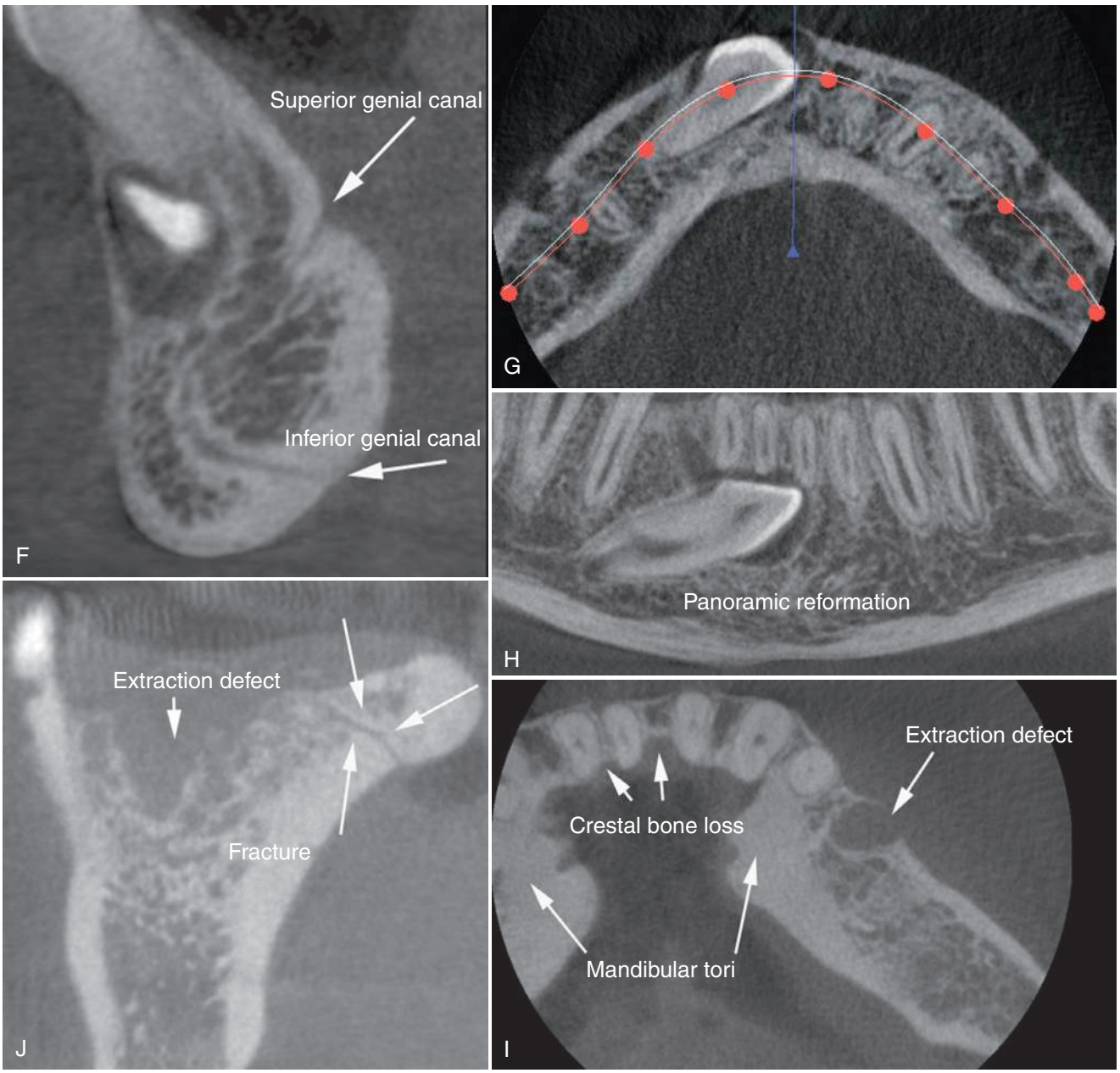


Fig. 2.26, cont'd

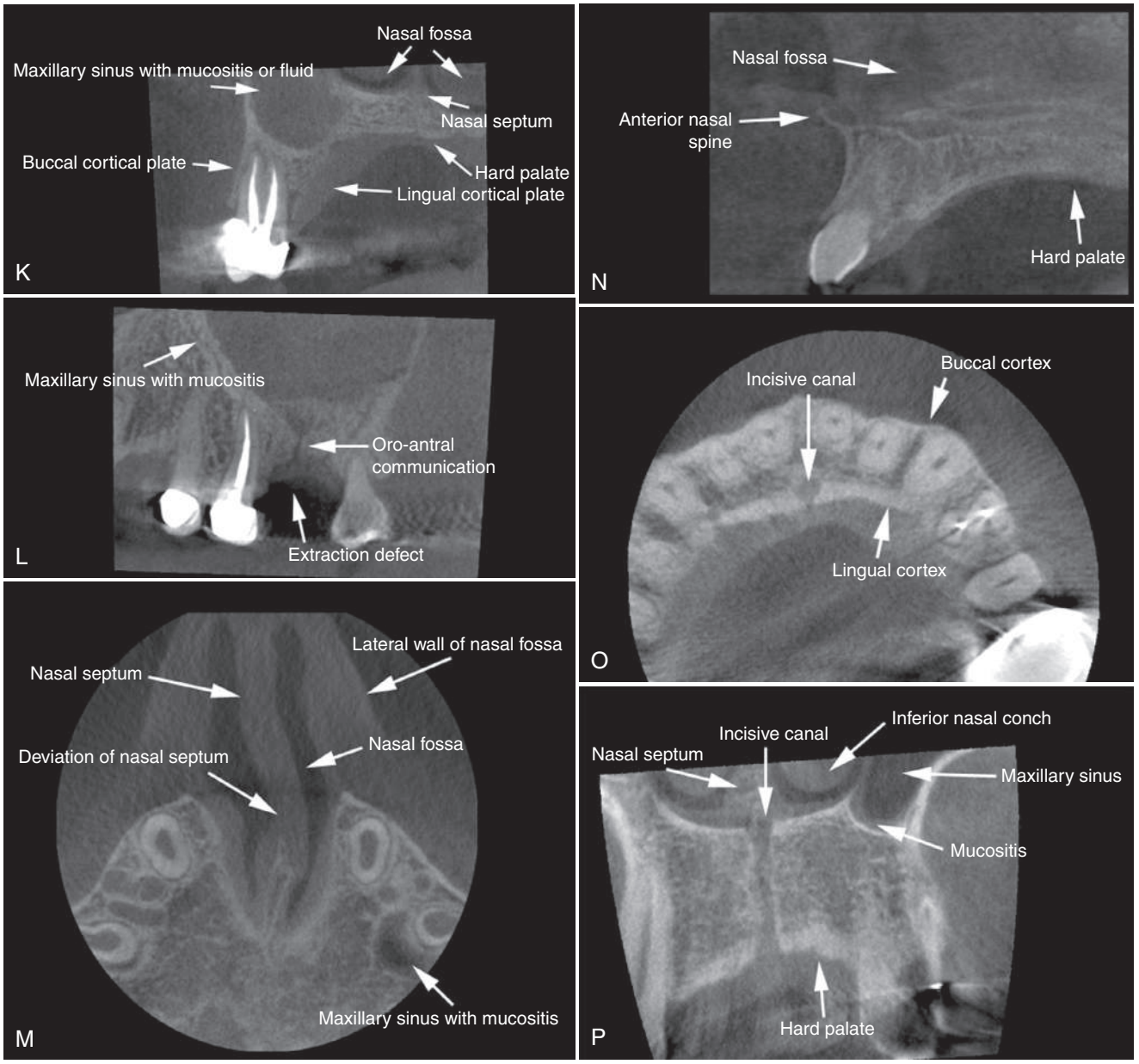


Fig. 2.26, cont'd

Continued

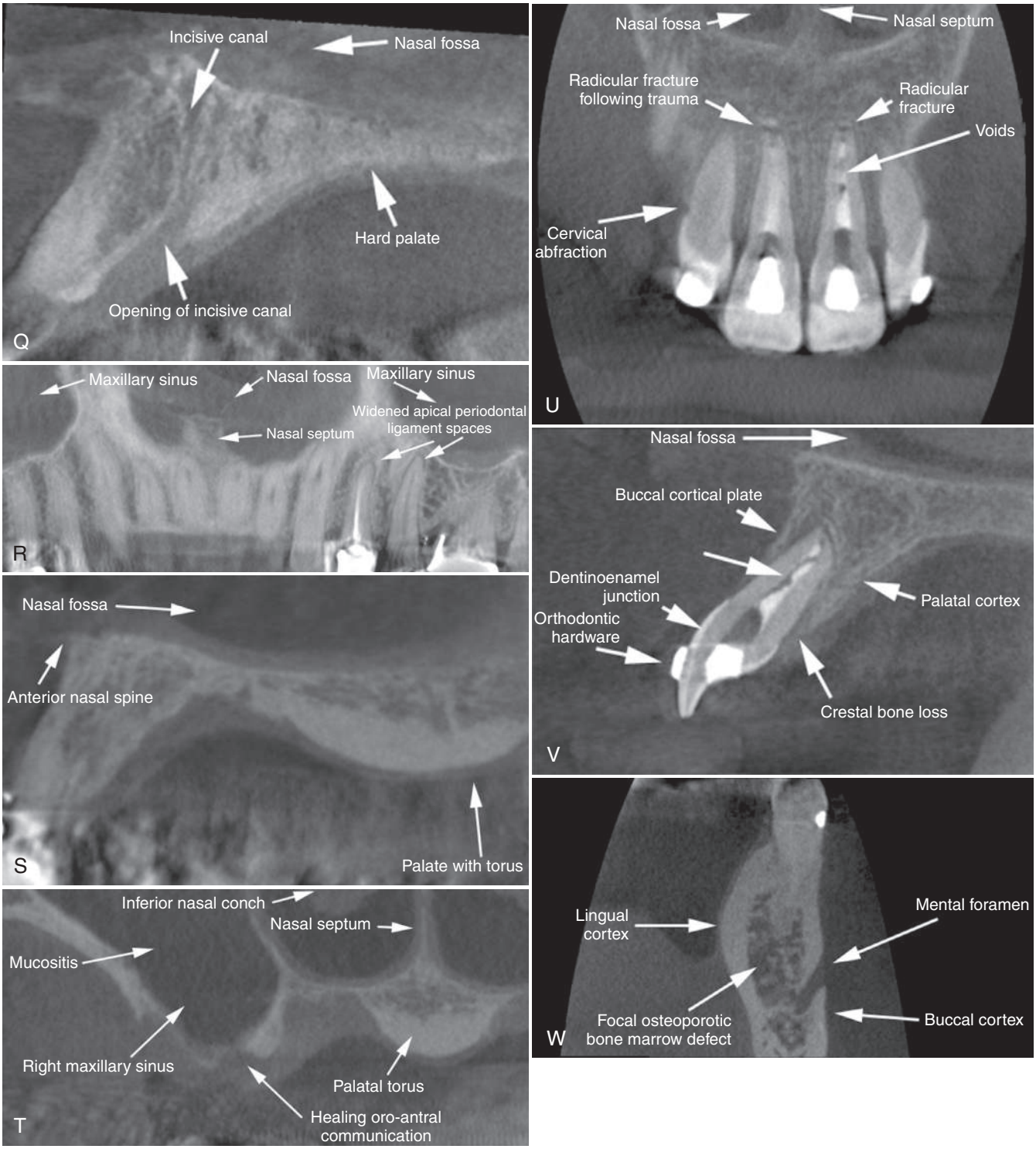


Fig. 2.26, cont'd

studies suggest that lower resolution scans using voxel sizes in excess of 0.3 mm may not improve radiographic assessments.[96,225] A study by Wang et al.[223] showed that the sensitivity and specificity of PA radiography for root fractures were 26.3% and 100%, respectively; the findings for CBCT were 89.5% and 97.5%, respectively. CBCT images of root-filled teeth with fractures showed lower sensitivity and unchanged specificity, whereas 2D images showed the same sensitivity and specificity.[223] CBCT allows for the management of traumatic injuries in which a root fracture or alveolar fracture is suspected by providing undistorted multiplanar views of the dentition and supporting bone without the superimposition of anatomic structures.[47,127,187] CBCT image volumes provide superior sensitivity in detecting intra-alveolar root fractures compared with multiple PA radiographs; this allows for the detection of dental and alveolar displacements, including damage to other perioral structures, such as the maxillary sinus and nasal floor.[105] The presence of root canal

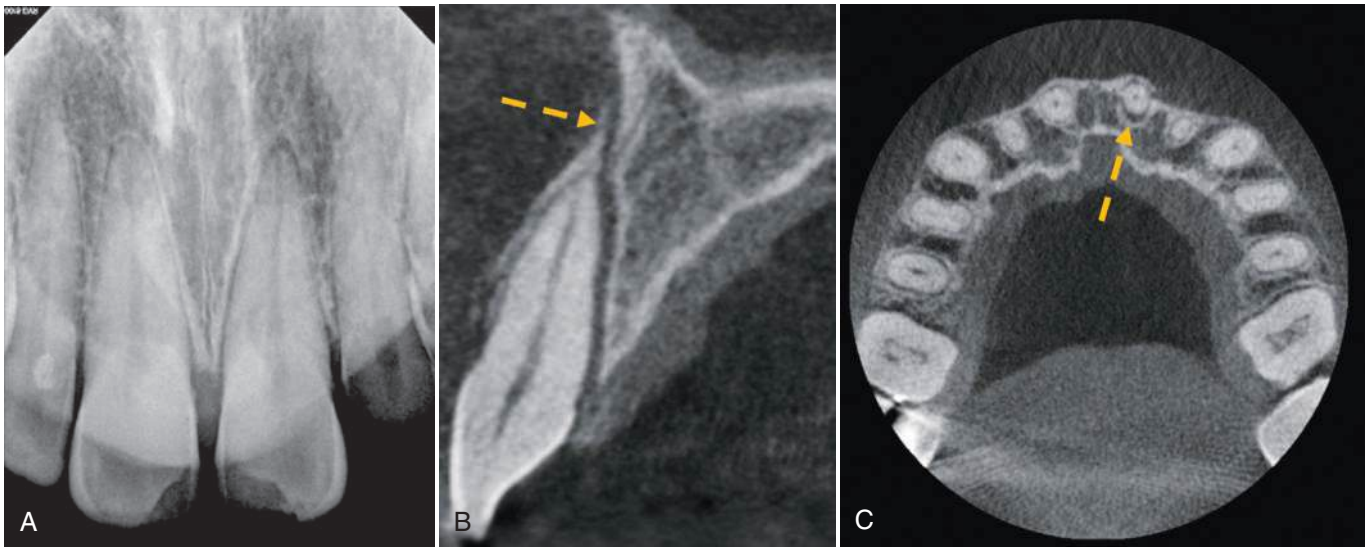


Fig. 2.27 After a traumatic injury to the maxillary right and left central incisors, crown fractures were noted. The PA radiograph showed an extrusive luxation injury in the maxillary left central incisor **(A).** The alveolar fracture (**B,** *yellow arrow*) and true extent of the displacement (**C,** *yellow arrow*) became evident with CBCT. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

fillings and posts affected the specificity of the findings as a result of artifact generation.[96,135] Outcome measurements of a region of interest can be compared over time with greater geometric accuracy using CBCT.[94]

Injury to the soft tissues, such as the lips, cheek and tongue, by embedded debris and tooth fragments, usually require additional radiographic images when relying on intraoral radiography. CBCT scans will aid in wound management by locating the precise size and shape of the foreign body, possibly reducing the scope of surgical interventions.[195]

The healing of root fractures is influenced by many factors, most prominently the stage of root development, with immature roots showing better healing than mature roots.[73] Other factors that influence healing are the extent of dislocation and repositioning, the type of splinting, the use of antibiotics, and the location of the fracture on the root. The long-term survival of teeth with intra-alveolar root fractures was evaluated in a systematic review by Andreasen et al.[17] This study showed that the type of healing (e.g., hard tissue fusion, PDL interposition with and without bone) and the location of the fracture on the root had the most influence on tooth loss. Yearly clinical and intraoral radiographic follow-up examinations are recommended by the IADT for some injuries for up to 5 years posttrauma.[58] Further evidence-based comparisons of outcomes for traumatic dental injuries after treatment interventions were addressed by the IADT with the development of a Core Outcome Set for children and adults.

Limited FOV CBCT should be considered when placement of individual PA radiographs will adversely affect patient management or result in retakes; PA radiographs will produce a higher radiation dose for assessment of the region of interest[105]; intra-alveolar fracture of the root or supporting structures is suspected and sufficient information cannot be obtained with conventional radiography; or foreign bodies are present in the lip, cheek, or tongue.[164] The decision to use CBCT imaging for assessment of traumatic injuries should be based on the diagnostic yield expected and in accordance with the "as low as reasonably achievable" (ALARA) principle. CBCT scan volumes that use the most appropriate detector size and shape, beam projection geometry, and beam collimation should be selected to produce high-resolution images and reduce x-radiation exposure whenever possible.[188] In all cases, it should be recognized that children and young adults are more susceptible to the effects of radiation than adults, and CBCT studies should answer specific clinical questions that cannot be answered by lower dose PA and panoramic imaging techniques.[97] In compliance with the Image Gently Alliance recommendations for pediatric imaging, always select x-rays when essential for diagnosis and treatment; use the fastest image receptor available, use CBCT only when necessary, collimate beam to area of interest, always use thyroid shield, and reduce exposure time and technique factors, such as mA.[227] New technologies that allow for comparison of matched CBCT images in a serial fashion at a reduced dose have been tested. This technology promises a dose reduction of 10 to 40 times by using the initial scan as prior knowledge and adaptive prior image constrained compressed sensing (APICCS) algorithms to greatly reduce the number of projections and x-ray tube current levels required (Fig. 2.28).

INTERNAL AND EXTERNAL ROOT RESORPTION

As described in Chapter 18, root resorption results in the loss of dentin, cementum, or bone by the action of clastic cells.[22] In the primary dentition, root resorption is caused by normal physiologic processes, except when resorption is premature; in the permanent dentition, root resorption is caused by inflammatory processes.[46,66,165] Root resorption can be classified according to location as either internal, appearing on the wall of the root canal, and external, affecting the outer surface of the root. Both internal and external root resorption have subtypes that show specific radiographic characteristics.[208]

The successful management of root resorption in the adult dentition depends on clinical and radiographic examination



Fig. 2.28 A, Horizontal root fractures resulting from trauma were evident in this 22-year-old male patient, who was referred with a contemporaneous film–based PA radiograph **(B)** for evaluation and possible treatment 9 months after trauma to his maxillary lateral and central incisors. Since the trauma, the teeth had been stabilized with a ribbon-type splint on the palatal surface; they were of normal color and responded normally to all sensibility tests. There was slight mobility of the traumatized teeth. **C,** The true nature of the root fractures are shown in the 3D segmented reformation and the corrected cross-sectional views of the **(D)** maxillary right lateral and **(E)** central incisors, the **(F)** maxillary left central and **(G)** lateral incisors, and the 3D surface-rendered reformation **(H).** Temporal reevaluation at 7 years showed minimal changes when compared to the initial presentation *(left to right in each group)*. Task-specific exposure parameters allowed each successive CBCT image volume to be exposed with lower mA, resulting in a 20% radiation dose reduction. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

leading to early detection and accurate diagnosis.[166] Unfortunately, teeth affected by root resorption have a poor prognosis if the causative lesion is not treated.[163]

2D radiographic projections suffer from inherent limitations that mask lesions: superimposition of 3D structures,[27] geometric distortion,[215] and misrepresentation.[127] Although parallax intraoral imaging techniques can be helpful in localizing root resorption[15] by observing lesion shift on changing angulation, only CBCT assessments can provide the true size and position of all resorptive defects in the region of interest.[46,163] Intraoral radiography is not reliable when lesions are small or located on the labial or lingual/palatal surfaces.[15] In ex vivo studies, CBCT showed improved accuracy in the detection of simulated internal and external resorptive lesions of all sizes.[107] Intraoral imaging produced false-negative results in 51.9% of cases studied and false-positive results in 15.3%.[143]

The use of CBCT in the evaluation of root resorption eliminates superimposition and compression of 3D features, is accurate in all multiplanar views, and can reliably represent the anatomy of three-dimensional structures. Patel et al.[163] compared the sensitivity and specificity of PA imaging with CBCT scans using the receiver operating characteristic (ROC) analysis, a standard measure of diagnostic performance. PA imaging showed satisfactory accuracy (Az 0.78), whereas CBCT showed perfect results (Az 1.00).[163]

Internal root resorption (IRR) is usually idiopathic and can be subdivided into either internal inflammatory or internal replacement resorption. IRR is a relatively rare occurrence that is usually detected on routine diagnostic radiographs.[122,162] It is characterized by internal structural changes in the tooth that can appear in any location and can exhibit varied features, such as smooth or irregular borders and radiodensity of uniform or mixed attenuation.[99] IRR is usually asymptomatic, associated with pulpal necrosis coronal to the resorptive lesion and vital or partially vital pulps where active.[163] When viewed with intraoral radiographs, IRR can easily be confused with external cervical resorption (ECR) owing to the similarity of the two lesions.[92] Parallax intraoral radiographs may aid identification because the lesion will maintain its position relative to the root canal. However, assessment may be affected by radiographic superimposition and by teeth with complex anatomy. CBCT is helpful for diagnosing the location and exact size of IRR. In a study by Estrela et al.,[66] 48 PA radiographs and CBCT scans were exposed on 40 individuals.[66] IRR was detected in 68.8% of PA radiographs, whereas CBCT scans showed 100% of the lesions. Conventional radiographs were able to detect only lesions between 1 and 4 mm in 52.1% of the images, whereas CBCT was able to show 95.8% of the lesions. This finding agreed with other studies that demonstrated the value of tomographic analysis.[46,125] In a study by Kim et al.,[112] the extent and location of the IRR was accurately reproduced with the fabrication of a rapid prototyping tooth model. Although relatively few systematic studies on artificially induced IRR have been reported because of the difficulty in creating such defects, Kamburoglu and Kursun[106] concluded that high-resolution CBCT imaging performed better than low-resolution CBCT imaging in detecting simulated small internal resorptive lesions.

External root resorption (ERR) can be divided into four subtypes: external surface resorption, external inflammatory resorption, external replacement resorption, and external cervical resorption (ECR). ERR may progress rapidly, and early treatment is recommended.[61]

The recommended timing of radiographic examinations using CBCT has not been established with a strong evidence grade. Many ERR cases involve young patients in whom the radiation dose is a critical factor and multiple scans would be difficult to justify. To minimize radiation dose to children the following guidelines have been suggested: Select radiographs based on a patient's individual needs, not as a routine; use the fastest image receptor possible; collimate the x-ray beam to expose only the area of interest; use thyroid collars; reduce the exposure as appropriate for the size of the child; use only when necessary, never as a screening tool.[227]

External surface resorption depends on the nature of the stimulus and can vary from small, concave lesions to extensive destruction of the root.[14] Commonly caused by orthodontic treatment, proximity to other teeth, or cysts and tumors, radiographic imaging with CBCT allows for accurate multiplanar mapping of the lesion. These excavations normally show small irregularities on the root surface with intact PDL and lamina dura. The appearance of the ERR may help indicate the nature of the lesion, where benign lesions tend to displace teeth and where more aggressive lesions, such as malignant tumors, tend to cause extensive root resorption, showing either blunting of the root apex or spiked/knife-edge destruction (Fig. 2.29).[165]

External inflammatory resorption resulting from pulpal necrosis or traumatic injuries is radiographically characterized by concave and occasionally patchy excavations of low density that affect the root surface and the supporting bone. Lesion borders are irregular, show loss of lamina dura, and are often associated with the apical region due to the proximity of the apical terminus.

External replacement resorption is caused by damage to the periodontium, which may result from dental trauma or orthodontic treatment.[74] The teeth are immobile, have a distinctive tone on percussion, and show disappearance of the normal periodontal membrane—features consistent with a direct union with alveolar bone.

External cervical resorption is a result of damage to the cementum layer of the root by clastic activity. Further destruction of cementum and then dentin[93] is dependent on stimulating factors that include microorganisms from the periodontium.[82] ECR is a complex and aggressive process and usually begins at the cervical region and may result in a pink discoloration of the overlying enamel. Patients affected by ECR are usually asymptomatic because the pericanal resorption-resistant/retarding sheet protects the pulp until the later stages.[209] Orthodontic treatment, trauma, parafunctional habits, and malocclusion account for most of these lesions, with a combination of factors resulting in increased occurance.[167] Intraoral radiography shows significant variability, with areas of low density affecting the dentin extending coronally and/or apically with sometimes greater destruction distant from the origin on the buccal or lingual/palatal aspect of the root, confounding 2D radiographic assessment. The radiographic appearance of these lesions is often mixed owing to the level

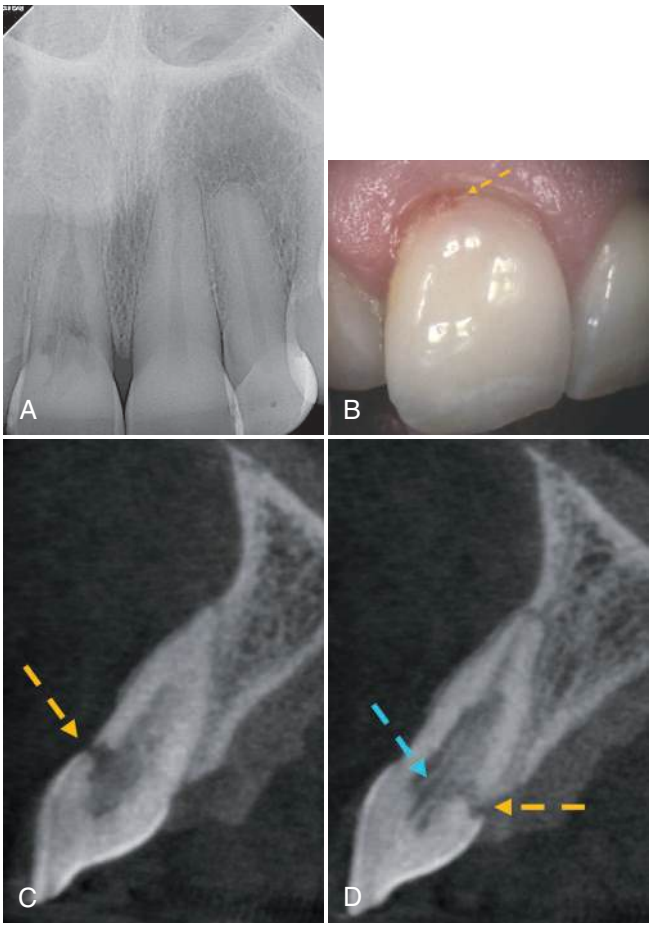


Fig. 2.29 A, External cervical resorption was evaluated in this patient after PA radiographic image, and **(B)** visual examination showed pathognomonic signs of this lesion *(yellow arrow)*. A CBCT image was exposed to determine the true extent of resorption and restorability. **C,** The lesion showed a perforative defect at the CEJ on the facial *(yellow arrow)* and **(D)** palatal aspects *(yellow arrow)*. **D,** The presence of an intact layer of mineralized dentin, the "pericanalar resorption-resistant sheet" *(blue arrow)*, is a hallmark of this condition. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA].)

of fibro-osseous and granulomatous tissue present. External cervical resorption is located on the external surface of the root, showing an intact root canal wall that can be traced through the lesion, whereas IRR is continuous with the canal wall.

External cervical resorption is always associated with bony resorption, making comparability of laboratory studies problematic because the lesions lack the changes in the periodontal membrane and associated bony changes that would improve visualization. The early stages of ERR are difficult to view with conventional radiography, and lesions less than 0.6 mm in diameter and 0.3 mm in depth could not be detected. Medium-sized lesions were visible in 6 of 13 cases, with improved visualization for proximal lesions without regard to the root third being examined.[15] Patel et al. showed that the accuracy of parallax and single parallel PA radiographs were similar, and that more teeth were considered unrestorable when assessed with CBCT (78.7%) than with PA imaging (49.3%).[158] This study further highlights the benefits of using CBCT to assess ECR.[158] Vaz de Souza et al. showed that CBCT was significantly better than PA radiography at diagnosing ECR according to the Heithersay classification.[216]

ERR is difficult to detect if the lesion is confined to the buccal, palatal, or lingual surfaces of the root.[30,83] Liedke et al.[125] conducted systematic diagnostic performance tests and showed similar sensitivity and specificity among the different voxel sizes studied (0.4, 0.3, and 0.2 mm); however, the likelihood ratio showed better probability of correct identification of ERR with either 0.3 or 0.2 mm scans. These researchers suggested the use of a 0.3 mm voxel size protocol rather than a 0.2 mm one, to reduce scanning time and the resulting dose.[125] Although voxel size is an important consideration, the SNR of different detectors, the radiation dose, viewing conditions, and the processing algorithms also affect detection probability.

Even though many in vitro studies have been performed on the ability of CBCT to detect root resorption, additional evaluations that use in vivo methodology will add to the cumulative knowledge.

Endodontic Treatment Outcomes Assessment

The reduction of inflammation and healing through regeneration or repair are the ideal outcomes for endodontic treatment.[148] However, endodontic treatment outcome predictors using PA radiographs and CBCT imaging have been shown to vary and are influenced by factors such as patient inclusion and exclusion criteria.[124] Historically, PA radiographs and physical examinations were used to determine the success of endodontic treatment, and the absence of posttreatment periradicular radiolucencies and symptoms was considered by many to be the criterion for success. However, these planar imaging-based studies have resulted in an overestimation of successful outcomes, compared to CBCT assessments,[230] because AP confined within the cancellous bone or lesions covered by a thick cortex may be undetectable with conventional radiographic assessments.[26] Additional discrepancies between PA radiography and CBCT have resulted from geometric distortion, limiting comparisons of temporal evaluations, even with careful attention to paralleling technique factors.[133]

A clinical study comparing the sensitivity, specificity, predictive values, and accuracy of PA and panoramic radiography and CBCT imaging in 888 consecutive patients showed that the prevalence of AP in root-treated teeth was 17.6%, 35.3%, and 63.3%, respectively. Conventional radiography showed increased accuracy when the larger lesions were assessed.[67]

Using histologic examination as the gold standard, a systematic review and meta-analysis of the literature by Dutra et al.[121] compared different radiographic modalities in the binary assessment of AP versus no lesion in humans—a very important measure in outcome studies. Significantly, only nine studies were ultimately selected for qualitative synthesis and only six for meta-analysis. All included studies were limited by using only artificially induced apical lesions, possible bias, variation in bone lesion sizes, poor or no inter- and intraobserver agreement, and except for one study, did not evaluate the maxilla, where more tissue compression occurred. This study showed the accuracy values of 0.72 for digital PA radiographs, 0.73 for film-based PA radiographs, and 0.96 for CBCT.[121]

Outcome predictors identified with PA radiographs and CBCT scans may be different depending on the research performed. Liang et al.[124] retrospectively evaluated 115 endodontically treated teeth with vital pulps 2 years after treatment. The authors noted that the recall rate of 36% was comparable to that in other studies, but the result may have been affected because patients with symptomatic or already extracted teeth may not have responded. This lost-to-follow-up cohort significantly reduced the level of evidence.[75] PA radiography identified PA lesions in only 12.6% of teeth compared to CBCT images, which identified 25.9% of teeth with PA lesions. In multivariate logistic regression analysis, the extent of root fillings and density were outcome predictors when using PA imaging, whereas density of root fillings and quality of the coronal restoration were outcome predictors when using CBCT scans.

The predictive value and diagnostic accuracy of radiologic assessments are critical to the practice of dentistry, and the diagnostic value of radiographs depends on the radiograph's ability to show the histology of AP. De Paula-Silva et al.[53] evaluated the periapex of 83 root-treated and untreated dogs' teeth using PA radiography, CBCT scans, and histopathologic analysis. PA radiography detected AP in 71% of roots; CBCT scans detected AP in 84%; and histologic analysis detected AP in 93%. These findings, corroborated by other studies,[38,87,178] emphasized the low negative predictive value (NPV) of PA radiography at 0.25, showing that when the PA tissues had a normal appearance, 75% actually had AP. CBCT scans resulted in an NPV almost two times higher than that for PA radiography; however, CBCT scans were not able to detect some AP that was confined to the apical foramen or had little volumetric bone loss. The positive predictive value was the same for PA radiography and CBCT scans compared with histologic examination, but true positive and true negative diagnosis of AP using CBCT scans occurred in 92% of cases.

Continuing development of sophisticated postprocessing software now allows for modeling of AP and anatomic structures in 3D. Using either manual or semiautomatic segmentation, the resulting virtual models allow for alignment and analysis of temporal changes, with some limitations. Bony changes can be depicted in quantitative and qualitative datasets, showing bone density changes such as repair or osteolysis. One of these programs, ITK-SNAP, an open-source software originally developed for MRI assessments, allows for construction of virtual 3D models from DICOM datasets, showing volumetric measurements in voxels and cubic millimeters. Additional tools allow for calculation of the mean intensity of the PA rarifying lesion(s) of interest. The resulting scan data can then be exported in STerioLithography format (STL) and shown in 3DMeshMetric software (Figs. 2.30 and 2.31).

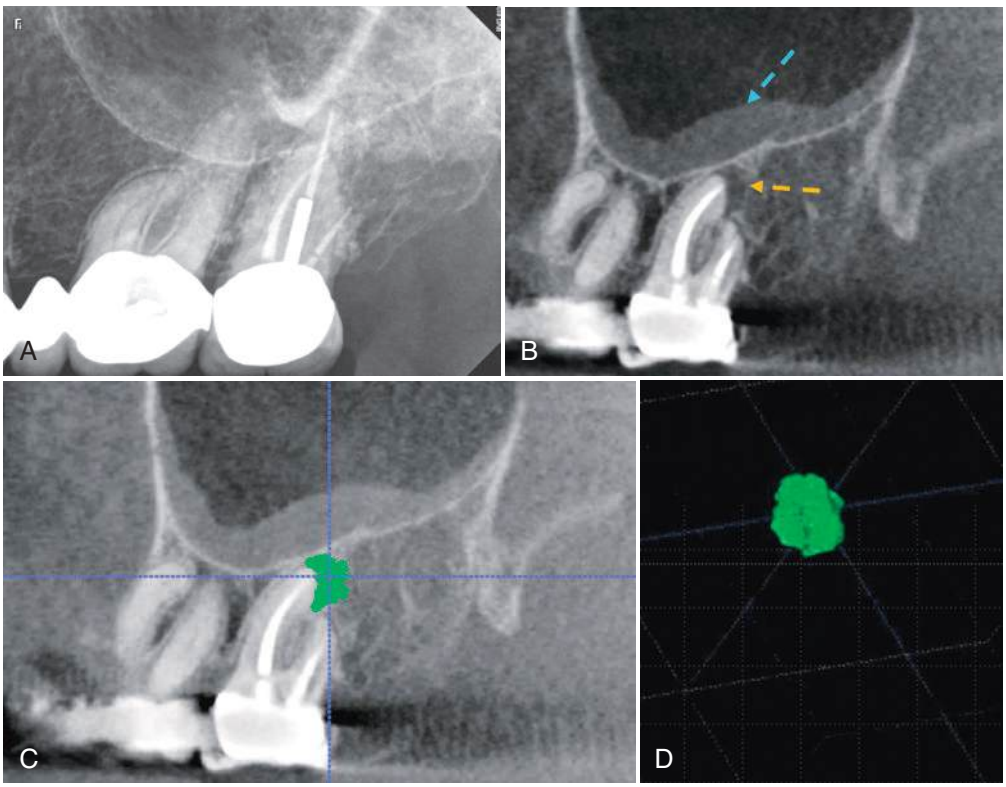
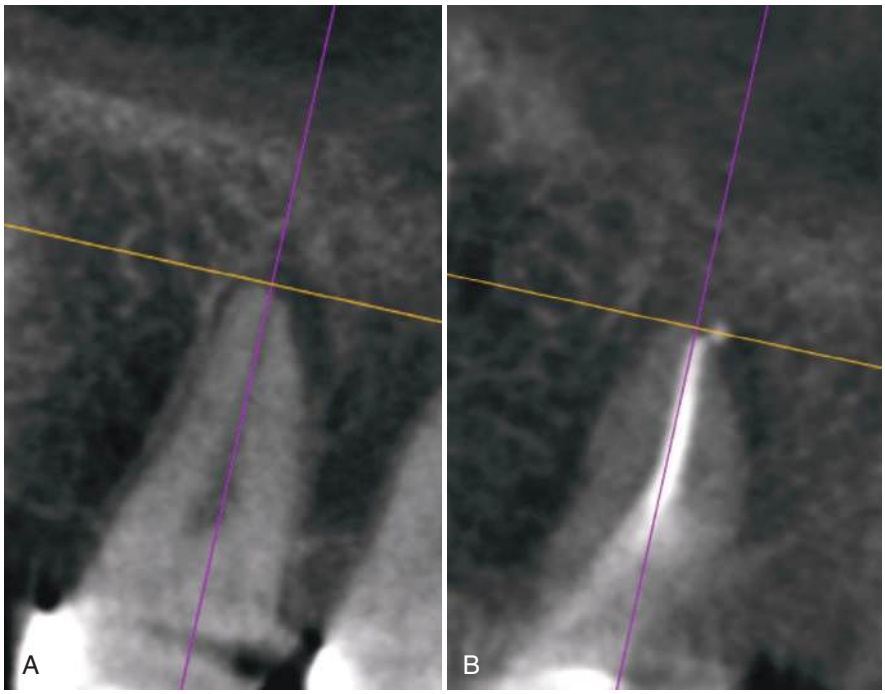
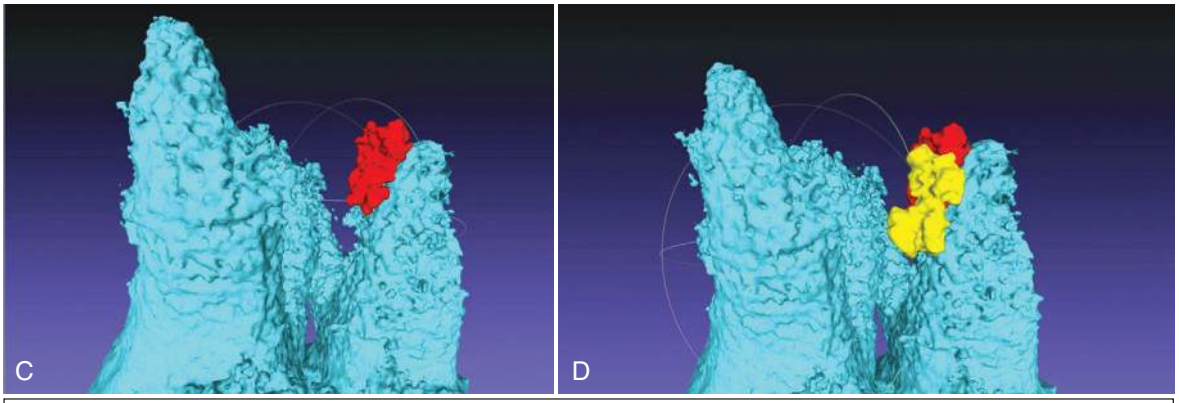


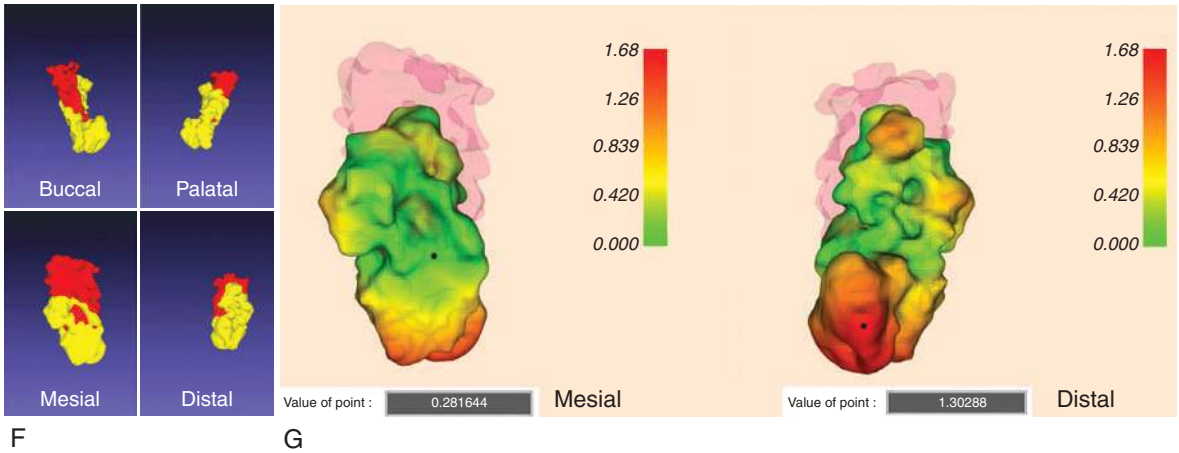
Fig. 2.30 The problem of superimposition of unrelated structures onto the features of interest is reduced when tomographic slices are used instead of images in which an entire volume of data is compressed into a planar image. **A,** This PA radiographic image of the maxillary left second molar shows no radiographic indicators of pathosis. A contemporaneously exposed CBCT image (**B,** corrected sagittal view) shows a 4.3 × 1.9 mm, well-defined unilocular, noncorticated area of low attenuation centered over the apex of the mesiobuccal root, consistent with a periradicular periodontitis *(yellow arrow)*. There is a moderate mucositis in the region of the maxillary sinus adjacent to this tooth *(blue arrow)*. Experimental semiautomated segmentation of this image using active contour methods (ITK-SNAP) allowed for measurement of the true volume of the lesion **(C)** and for future temporal comparisons based on volumetric measurements **(D).** This lesion measured 85,112 voxels and 38.1044 cubic mm. (Data acquired and reformatted at 0.076 mm voxel size using a CS 9000 3D unit [Carestream Dental, Atlanta, GA]; volumetric segmentation and measurement using ITK-SNAP [Radiology Department, School of Medicine, University of Pennsylvania].)





E

Label Id	Label name	Number of Voxels	Volume (mm^3)	Image mean	Image Stdev
0	Clear Label	13716676	6146.24	565.658	1074.02
1	Vol1	15704	7.03673	–45.4508	58.9991
2	Vol2	19059	8.54006	–70.2623	77.7926



F G

Fig. 2.31 Radiographic interval change is shown by comparing two volumetric datasets exposed at different times. **A,** Initial CBCT corrected sagittal reformation of maxillary right second molar, distobuccal root and **(B)** corrected sagittal reformation of same root at 13-month checkup. **C** and **D,** ITK-SNAP segmented sagittal reformation of same root. Alignment and overlap of the 3D models: tooth-blue, initial lesion is red, lesion at 13-month check-up is yellow, respectively. **E,** Voxel counting, volumetric measurement (mm) and mean intensity of the periapical lesion. **F,** Models were exported to the .stl format and then viewed with MeshLab software. **G,** 3D Mesh Metric Qualitative (color-coded map) and quantitative (point value) assessments of the lesions before (translucent), and after (colored). The color map shows changes with green color in the mesial region (0.28 mm) and changes with dark red color in the distal region (1.30 mm). (Courtesy Dr. Ane Poly.)

Risks and benefits for patients who are medically complex and subject to possible increased morbidity from AP, such as patients with an altered immune system (i.e., chemotherapy or anti-HIV protocols or prosthetic joints and/or infective endocarditis), should be considered in the decision about whether to expose a CBCT scan. The American Academy of Periodontology has published a position paper stating that periodontal disease may contribute to adverse systemic health conditions.[186] The scientific basis for the relationship of AP and adverse systemic health conditions has not been established[43]; however, new associations between AP and systemic health should be predicated on research that uses CBCT in the detection of endodontic disease.[49]

The determination of successful outcomes is nuanced and complex. New AAE-sponsored initiatives to study outcomes is underway and will ultimately result in an agreed standardized set of criteria, called a "core outcomes set," facilitating data synthesis and possibly resulting in more transparent and complete reporting.

Dental Implant Case Planning

Successful endosseous implant site assessment requires the development of a prosthetically driven approach,[2] with special emphasis on the evaluation of bone volume, the osseous topography, and the location of anatomic structures in relation to the positioning of the implant. The AAOMR has published advisory recommendations suggesting all radiographic studies should interface with the dental and medical histories, clinical examinations, and treatment planning. Panoramic radiography, which may be supplemented by PA radiography, should be used for initial imaging assessments. Cross-sectional imaging, including CBCT, should not be used as an initial imaging examination. The AAOMR affirms the need for cross-sectional imaging in the preoperative diagnostic phase, recommending CBCT because it provides the highest diagnostic yield at an acceptable radiation dose risk. CBCT should be used at the smallest FOV necessary, with optimized technique factors, to minimize radiation dose in accordance with the ALARA principle.[213]

CBCT allows for precise planning and delivery of implants that can reproduce the anatomy with submillimetric accuracy, leading to improved outcomes.[201] The use of CBCT to assess linear measurements, proximity to vital anatomic structures, mapping of the alveolar ridge topography, and fabrication of surgical guides is supported by the dental literature. The use of CBCT to gauge bone density, provide intraoperative surgical navigation, and assess implant integration is generally considered an area that requires further research.[25]

Virtual implant planning using CBCT data allows clinicians to visualize the result before the commencement of treatment, facilitating the virtual investigation of multiple treatment scenarios until the best plan is attained. The evaluation of bone dimensions, bone quality, the long axis of the alveolar bone, internal anatomy, and jaw boundaries; the detection of pathologic features; and the transfer of radiographic information are the main imaging goals. Use of static, CT-generated guide stents and dynamic navigation systems provide improved accuracy when compared with freehand methods.[29]

Pathoses of the jaws, such as retained root tips, inflammatory lesions, cysts, and tumors, in addition to extraoral structures, such as the sinuses and temporomandibular joints (TMJs), must also be assessed.[97] CBCT imaging should be considered to evaluate implant sites in the region of teeth with a high likelihood of periradicular pathosis.[239]

3D-GUIDED ENDODONTICS

3D-guided surgical and nonsurgical endodontic treatment is an emerging application using guides manufactured with the aid of CBCT images and intraoral scanning of the area of interest. Although studies with high level of evidence is lacking in this area, several in vitro studies and case reports have been published.[5,48,118,132]

The technology is currently being utilized in endodontic treatment of teeth with calcified canals and teeth that have anatomical variations of the root canal system. A template is created by superimposition of CBCT images on intraoral scanned images with sleeves placed in the images for creating a 3D printed guide with a directed sleeve. Most often implant treatment planning software is used for the creation of the guide (Fig. 2.32).[48,118,132] Recent study evaluating access on calcified teeth showed a higher detection of canal (91.7% vs. 41.7%) using guided approach. There was significantly lower amount of tooth substance loss using the guided access (49.9 mm vs. 9.8 mm).[48]

Endodontic microsurgery aims to create conservative osteotomy utilizing magnification and microsurgical instruments. Various studies have shown improved prognosis with the microsurgical procedures when compared to traditional endodontic surgery.[210] Surgical treatment planning with CBCT images have aided clinicians in identifying anatomical structures in the surgical area and increasing precision of the procedure.[119] Surgical endodontics with 3D-guided templates has further helped in making microsurgical procedures more precise.[3,5,71] A recent study comparing "freehand" CBCT-aided surgery to surgery using template guides showed a deviation of 1.743 mm when using guides from the target location identified on preoperative CBCT. Freehand surgery showed a deviation of 2.638 mm.[3] Use of prefabricated grids placed while acquiring CBCT images were used as guides for creating osteotomy during endodontic surgery in a recent study. Deviation from the target point was significantly lower when compared to nonguided drilling (Fig. 2.33).[71]

Image Perception and Viewing Environment

Medical image perception is an important area of knowledge, and ongoing research relies on an understanding of perceptual issues, such as psychological factors, dwell time, visual search physiology, search tactics, appreciation for the reading environment, and fatigue factors, to increase search satisfaction. Understanding these issues may improve the ability to interpret and report on dental radiographic findings.[185]

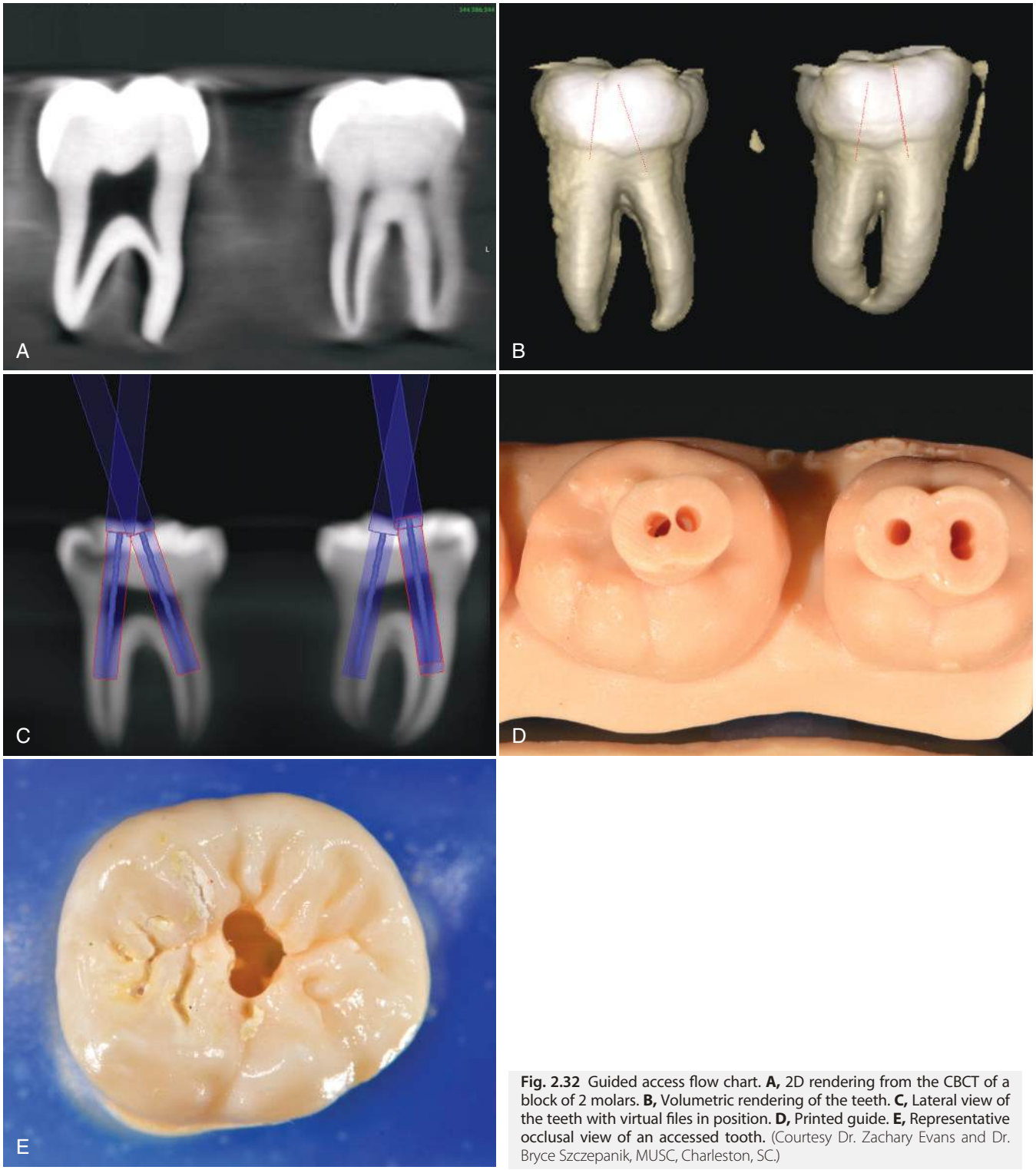


Fig. 2.32 Guided access flow chart. **A,** 2D rendering from the CBCT of a block of 2 molars. **B,** Volumetric rendering of the teeth. **C,** Lateral view of the teeth with virtual files in position. **D,** Printed guide. **E,** Representative occlusal view of an accessed tooth. (Courtesy Dr. Zachary Evans and Dr. Bryce Szczepanik, MUSC, Charleston, SC.)

The increasing use of digital radiography in the dental environment has led to a sea change in workflow and the necessity for new ways to view and document radiographic images. Simple to accomplish, but important to improve, are the viewing conditions for softcopy interpretation, including moderately reduced ambient lighting, ranging from 25 to 40 lux.[36]

The Future of Cone-Beam Computed Tomography

The first decade of the 21st century saw the development of a wide range of CBCT applications, especially in dentistry. Lower radiation dose, higher spatial resolution, smaller FOV, and relatively lower cost may contribute to CBCT

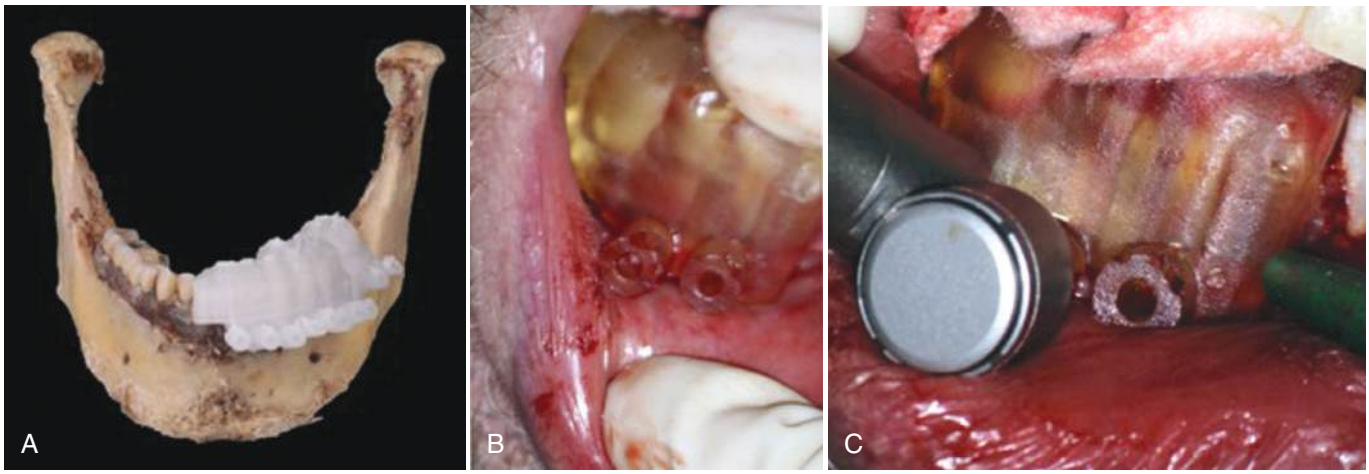


Fig. 2.33 Guided endodontic surgery. **A,** 3D printed guide. **B,** Guide placed in position prior to endodontic surgery. **C,** Surgical access with printed guide in position. (Courtesy Dr. Ackerman and Dr. Jalali, Texas A&M College of Dentistry, Dallas, TX.)

becoming the standard of care in 3D dentomaxillofacial imaging in selected cases. CBCT systems are increasingly being used in medical applications, such as operating rooms, emergency departments, intensive care units, and private otolaryngology offices. Operating room–based C-arm systems have been in use for a number of years, with applications in interventional angiography, cancer surgery, vascular surgery, orthopedic surgery, neurosurgery, and radiotherapy.[69] Applications in otolaryngology and mammography are common, and imaging of extremities in weight-bearing scenarios is under development. Many of these applications rely on systems that use task-specific protocols that benefit from CBCT's 2D FPDs, which allow for a single rotation of the source to generate a study of the region of interest, as opposed to complex MDCTs that use redundant imaging via multiple slice acquisitions to generate a 3D volume.[196]

The introduction of new, high-performance, flat-panel detectors and software algorithms centering on improving the noise-power spectrum and noise-equivalent quanta will continue to increase the utility of CBCT systems in the future. Areas of research include (1) image perception and image quality assessment, to better understand how physicians and dentists analyze radiographic images and thereby improve diagnostic decision making[202]; (2) iterative reconstruction that uses sophisticated algorithms to reduce artifacts; (3) known-component reconstructions that use a model-based 3D image reconstruction and iterative software to reduce image artifacts in the presence of metallic devices such as screws and implants; (4) image registration to align tissues for image-guided surgery and outcomes assessment;[141] (5) image-guided procedures that provide instantaneous surgical navigation; (6) segmentation to allow discrimination between normal and diseased tissues and permit volumetric measurements (see Figs. 2.30 and 2.31); and (7) use of metallic artifact reduction algorithms that help partly remove distracting signals and artifacts from a CBCT study. Note, however, that there is potential for loss of diagnostic information as well, especially in situations where high spatial resolution is required, such as detection of vertical root fractures in roots with posts.

Magnetic Resonance Imaging

MRI has been explored as a potentially useful imaging modality in dentistry, and in particular endodontics. Advantages include nonexposure to ionizing radiation; capability to acquire images in any desired plane without having to reformat data; capability to image edema, effusion, and inflammation in the adjacent bone; spread of infection; better soft-tissue delineation; and assessment of root structure and variations in canal morphology and/or number. Disadvantages include high cost of a scanner, longer acquisition times, interpretation challenges due to differences in appearance of hard and soft tissue as compared to CBCT or conventional radiographs, potential for motion artifacts, limited resolution, lack of easy access, choice of appropriate protocol based on scanner parameters, and generation of artifacts unlike those seen on x-ray–based images. All of these challenges need to be addressed in order for the imaging modality to be widely accepted and used in endodontics. Various pulse sequences have been developed for optimizing image quality for endodontic diagnostic purposes, but a magnetic field strength that is sufficiently high to improve resolution and further fine-tuning of task-specific regional receiver coils are required.[20,57]

Conclusions

Digital radiography has several advantages and has become an indispensable diagnostic tool for many dentists in daily practice. Once the digital image appears on the monitor, the dental x-ray software allows image enhancement, which should be used with caution and be based on the diagnostic task. Inappropriate use of enhancement has been shown to adversely affect diagnosis.[142] If digital radiographs are exported using various software packages created for graphic design and image manipulation, digital information can be altered, added, or removed. The DICOM standard has been accepted as the universal standard for image transmission and archiving, so that each image can be transmitted and stored without the use of proprietary software that would

seriously limit its distribution. DICOM ensures that all images are readable in any viewing software without loss of fidelity or diagnostic information. Image enhancement features of digital radiography allow mishandling of images, leading to potential abuse. Published studies illustrate the potential for fraudulent use of digital radiography.[42]

There is a dearth of studies related to the diagnostic performance of the different sensor types currently available on the market. Slight to moderate differences in spatial resolution capabilities exist. With rapid advancement in sensor technology and frequent software upgrades, selection of one system over another for a specific diagnostic task may appear challenging. A review of the most commonly used solid-state sensors notes that most systems perform comparably with intraoral film and also allow for postprocessing of images, which is not possible with film-based images. Other factors that assume significance in this context are the availability of technical and customer support, frequency of both hardware and software upgrades, dimensions of the sensor and its active area, number of sensors needed in a practice (and thus cost issues), the detector quantum efficiency (DQE), and conformance to the DICOM standard for seamless integration with other systems. CCD/CMOS sensors appear to offer the best contrast and spatial resolution, in addition to facilitating instantaneous image capture, and therefore are recommended for endodontic applications. Careful and appropriate image processing further helps tease out the signal of interest. In an enterprise-wide setting or in larger private practices that have multiple specialty areas, PSP-based sensors may be more cost effective for large volume imaging (e.g., full mouth series of radiographs). However, at least one or two CCD/CMOS-based systems should be available for faster image acquisition, such as for endodontic purposes and intraoperative procedures.

It is recommended that the literature be constantly reviewed for updates on digital radiography and advanced imaging modalities for specific endodontic applications because hardware and software upgrades continue to make rapid progress. Previous studies have shown that most digital images performed comparably to conventional intraoral film for a variety of diagnostic tasks. Most of these studies were done with earlier generation sensors. The advances in sensor technology have resulted in greatly enhanced image quality, and this trend is expected to continue. Also of interest in the future will be the use of task-based, appropriate image processing parameters that result in a reduced radiation dose and significant enhancement of the diagnostic information. Automation of this process will result in faster and more consistent image processing based on the diagnostic task. Such procedures are routinely carried out in medical radiology.

Three-dimensional imaging will continue to be used extensively as sensor characteristics improve and more user-friendly software is introduced. As bit-depth and spatial resolution of images increase, CBCT will continue to be explored for more applications in endodontics. Image interpretation also is important. Occult pathology and incidental findings in adjacent regions can be easily missed or go unrecognized by those who have not received specific training in the interpretation of regional anatomy. Image processing can greatly alter signal characteristics, thus rendering the task fairly challenging. Besides, if other pathoses are discovered, additional imaging may be necessary, including MRI, nuclear medicine studies, or even MDCT, for the evaluation of soft tissues, with and without the use of contrast.

The advent of 3D imaging has provided the endodontist with tools that were not available until now, facilitating interactive image manipulation and enhancement and thus significantly increasing the amount of information that can be gleaned from a volume. Lack of distortion, magnification, artifacts associated with conventional radiography, and the relative low radiation dose compared to medical-grade CT will result in more clinicians adopting such technology to enable accurate diagnoses and treatment planning, in addition to long-term follow-up and the evaluation of healing. Judicious use of CBCT and all other imaging modalities using ionizing radiation is advocated. The AAE/AAOMR position paper provides recommendations for the use of CBCT in endodontics[1] and is presented at the end of this chapter.

When the clinician works with different vendor products, it is important to have a quality assurance program in place. This is not being done currently. Additionally, accreditation of stand-alone imaging laboratories is now a requirement for reimbursement of medical and dental diagnostic procedures from government agencies and some major third-party insurance providers. Several states are considering enforcing this requirement to prevent abuse of CBCT.

Likewise, definitive referral criteria are lacking. Indications, contraindications, and choice of alternate imaging modalities need to be considered before CBCT is used. There is a learning curve to this technology, and appropriate positioning, choice of exposure parameters (and thus the effective dose), reconstruction schemes, choice of post-processing algorithms based on diagnostic task, voxel sizes, and cost must be considered. The literature contains few studies to help us formulate definitive guidelines for the use of CBCT in dentistry.

It is equally important to record the doses associated with each study. Accreditation criteria have been developed by the Intersocietal Accreditation Commission for CT in dentistry that are useful for ensuring the safe use of these units. The lowest possible dose must be imparted to the patient as part of a radiologic examination to minimize stochastic effects that have no known threshold for expression. No dose can be considered a "safe dose." The benefits of any radiographic study must outweigh the risks. All studies must be interpreted fully because signals from adjacent areas may appear in the volume of interest, including small FOV studies. Retakes can be avoided by adhering to protocol selection based on the task at hand. The ALARA principle must be followed, regardless of the dose values reported by the vendor, to optimize the dose for the specific examination. Use of thyroid collars and lead aprons is recommended in the NCRP guidelines, as long as they do not interfere with image acquisition.

JOINT POSITION STATEMENT OF THE AMERICAN ASSOCIATION OF ENDODONTISTS AND THE AMERICAN ACADEMY OF ORAL AND MAXILLOFACIAL RADIOLOGY ON THE USE OF CONE BEAM COMPUTED TOMOGRAPHY IN ENDODONTICS: 2015/2016 UPDATE

This statement was prepared by the Special Committee to Revise the Joint AAE/AAOMR Position Statement on Use of CBCT in Endodontics and approved by the AAE Board of Directors and AAOMR Executive Council in 2016.

Introduction

This updated joint position statement of the American Association of Endodontists (AAE) and the American Academy of Oral and Maxillofacial Radiology (AAOMR) is intended to provide scientifically based guidance to clinicians regarding the use of cone-beam computed tomography (CBCT) in endodontic treatment and reflects new developments since the 2010 statement.[1] The guidance in this statement is not intended to substitute for a clinician's independent judgment in light of the conditions and needs of a specific patient.

Endodontic disease adversely affects quality of life and can produce significant morbidity in afflicted patients. Radiography is essential for the successful diagnosis of odontogenic and nonodontogenic pathoses, treatment of the root canal systems of a compromised tooth, biomechanical instrumentation, evaluation of final canal obturation, and assessment of healing.

Until recently, radiographic assessments in endodontic treatment were limited to intraoral and panoramic radiography. These radiographic technologies provide two-dimensional representations of three-dimensional anatomic structures. If any element of the geometric configuration is compromised, the image may demonstrate errors.[2] In more complex cases, radiographic projections with different beam angulations can allow parallax localization. However, complex anatomy and surrounding structures can render interpretation of planar images difficult.

The advent of CBCT has made it possible to visualize the dentition, the maxillofacial skeleton, and the relationship of anatomic structures in three dimensions.[3] CBCT, as with any technology, has known limitations, including a possible higher radiation dose to the patient. Other limitations include the potential for artifact generation, high levels of scatter and noise, and variations in dose distribution within a volume of interest.[4]

CBCT should be used only when the patient's history and a clinical examination demonstrate that the benefits to the patient outweigh the potential risks. CBCT should not be used routinely for endodontic diagnosis nor for screening purposes in the absence of clinical signs and symptoms. Clinicians should use CBCT only when the need for imaging cannot be met by lower dose two-dimensional (2D) radiography.

Volume Size(s)/Field of View

There are numerous CBCT equipment manufacturers, and several models are available. In general, CBCT is categorized into large, medium, and limited-volume units based on the size of their "field of view" (FOV). The size of the FOV describes the scan volume of CBCT machines. That volume determines the extent of anatomy included. It is dependent on the detector size and shape, beam projection geometry, and the ability to collimate the beam. To the extent practical, FOV should only slightly exceed the dimensions of the anatomy of interest.

Generally, the smaller the FOV, the lower the dose associated with the study. Beam collimation limits the radiation exposure to the region of interest and helps ensure that an optimal FOV can be selected based on disease presentation. Smaller scan volumes generally produce higher resolution images. Because endodontics relies on detecting small alterations such as disruptions in the periodontal ligament space, optimal resolution should be sought.[5]

The principal limitations of large FOV CBCT imaging are the size of the field irradiated and the reduced resolution compared to intraoral radiographs and limited volume CBCT units with inherent small voxel sizes.[4] The smaller the voxel size, the higher is the spatial resolution. Moreover, the overall scatter generated is reduced due to the limited size of the FOV. Optimization of the exposure protocols keeps doses to a minimum without compromising image quality. If a low-dose protocol can be used for a diagnostic task that requires lower resolution, it should be employed, so long as strong indications to the contrary are absent.

In endodontics, the area of interest is limited and determined prior to imaging. For most endodontic applications, limited FOV CBCT is preferred to medium or large FOV CBCT because there is less radiation dose to the patient, higher spatial resolution, and shorter volumes to be interpreted.

Dose Considerations

Selection of the most appropriate imaging protocol for the diagnostic task must be consistent with the "as low as reasonably achievable" (ALARA) principles that every effort should be made to reduce the effective radiation dose to the patient "as low as reasonably achievable." Because radiation dose for a CBCT study is higher than that for an intraoral radiograph, clinicians must consider overall radiation dose over time. For example, will acquiring a CBCT study now eliminate the need for additional imaging procedures

in the future? It is recommended to use the smallest possible FOV, the smallest voxel size, the lowest mA setting (depending on the patient's size), and the shortest exposure time in conjunction with a pulsed exposure-mode of acquisition.

If extension of pathoses beyond the area surrounding the tooth apices or a multifocal lesion with possible systemic etiology is suspected, and/or a nonendodontic cause for devitalization of the tooth is established clinically, appropriate larger field of view protocols may be employed on a case-by-case basis.

There is a special concern with overexposure of children (up to and including 18-year-olds) to radiation, especially with the increased use of computed tomography scans in medicine. The AAE and the AAOMR support the Image Gently Campaign led by the Alliance for Radiation Safety in Pediatric Imaging. The goal of the campaign is "to change practice; to raise awareness of the opportunities to lower radiation dose in the imaging of children." Information on use of CT is available at http://www.imagegently.org/Procedures/ComputedTomography.aspx.

Interpretation

If a clinician has a question regarding image interpretation, it should be referred to an oral and maxillofacial radiologist.[6]

Recommendations

The following recommendations are for limited FOV CBCT scans.

DIAGNOSIS

Endodontic diagnosis is dependent upon thorough evaluation of the patient's chief complaint, history, and clinical and radiographic examination. Preoperative radiographs are an essential part of the diagnostic phase of endodontic therapy. Accurate diagnostic imaging supports the clinical diagnosis.

Recommendation 1: Intraoral radiographs should be considered the imaging modality of choice in the evaluation of the endodontic patient.

Recommendation 2: Limited FOV CBCT should be considered the imaging modality of choice for diagnosis in patients who present with contradictory or nonspecific clinical signs and symptoms associated with untreated or previously endodontically treated teeth.

Rationale

- In some cases, the clinical and planar radiographic examinations are inconclusive. Inability to confidently determine the etiology of endodontic pathosis may be attributed to limitations in both clinical vitality testing and intraoral radiographs to detect odontogenic pathoses. CBCT imaging has the ability to detect periapical pathosis before it is apparent on 2D radiographs.[7]
- Preoperative factors such as the presence and true size of a periapical lesion play an important role in endodontic treatment outcome. Success, when measured by radiographic criteria, is higher when teeth are endodontically treated before radiographic signs of periapical disease are detected.[8]
- Previous findings have been validated in clinical studies in which primary endodontic disease detected with intraoral radiographs and CBCT was 20% and 48%, respectively. Several clinical studies had similar findings, although with slightly different percentages.[9,10] Ex vivo experiments in which simulated periapical lesions were created yielded similar results.[11,12] Results of in vivo animal studies, using histologic assessments as the gold standard, also showed similar results as those observed in human clinical and ex vivo studies.[13]
- Persistent intraoral pain following root canal therapy often presents a diagnostic challenge. An example is persistent dentoalveolar pain also known as atypical odontalgia.[14] The diagnostic yield of conventional intraoral radiographs and CBCT scans was evaluated in the differentiation between patients presenting with suspected atypical odontalgia versus symptomatic apical periodontitis, without radiographic evidence of periapical bone destruction.[15] CBCT imaging detected 17% more teeth with periapical bone loss than conventional radiography.

INITIAL TREATMENT

Preoperative

Recommendation 3: Limited FOV CBCT should be considered the imaging modality of choice for the initial treatment of teeth with the potential for extra canals and suspected complex morphology, such as mandibular anterior teeth, and maxillary and mandibular premolars and molars, and dental anomalies.

Intraoperative

Recommendation 4: If a preoperative CBCT has not been taken, limited FOV CBCT should be considered as the imaging modality of choice for intra-appointment identification and localization of calcified canals.

Postoperative

Recommendation 5: Intraoral radiographs should be considered the imaging modality of choice for immediate postoperative imaging.

Rationale

- Anatomical variations exist among different types of teeth. The success of nonsurgical root canal therapy depends on the identification of canals, the cleaning, shaping and obturation of root canal systems as well as the quality of the final restoration.
- 2D imaging does not consistently reveal the actual number of roots and canals. In studies, data acquired by CBCT showed a very strong correlation between sectioning and histologic examination.[16,17]
- In a 2013 study, CBCT showed higher mean values of specificity and sensitivity when compared to intraoral radiographic assessments in the detection of the MB2 canal.[18]

NONSURGICAL RETREATMENT

Recommendation 6: Limited FOV CBCT should be considered the imaging modality of choice if clinical examination and 2D intraoral radiography are inconclusive in the detection of vertical root fracture (VRF).

Rationale

- In nonsurgical retreatment, the presence of a vertical root fracture significantly decreases prognosis. In the majority of cases, the indication of a vertical root fracture is more often due to the specific pattern of bone loss and periodontal ligament space enlargement than to direct visualization of the fracture. CBCT may be recommended for the diagnosis of vertical root fracture in unrestored teeth when clinical signs and symptoms exist.
- Higher sensitivity and specificity were observed in a clinical study where the definitive diagnosis of vertical root fracture was confirmed at the time of surgery to validate CBCT findings, with sensitivity being 88% and specificity 75%.[19] Several case series studies have concluded that CBCT is a useful tool for the diagnosis of vertical root fractures. In vivo and laboratory studies evaluating CBCT in the detection of vertical root fractures agreed that the sensitivity, specificity, and accuracy of CBCT were generally higher and reproducible.[20,21] The detection of fractures was significantly higher for all CBCT systems when compared to intraoral radiographs. However, these results should be interpreted with caution because detection of vertical root fracture is dependent on the size of the fracture, presence of artifacts caused by obturation materials and posts, and the spatial resolution of the CBCT.

Recommendation 7: Limited FOV CBCT should be the imaging modality of choice when evaluating the nonhealing of previous endodontic treatment to help determine the need for further treatment, such as nonsurgical, surgical or extraction.

Recommendation 8: Limited FOV CBCT should be the imaging modality of choice for nonsurgical retreatment to assess endodontic treatment complications, such as overextended root canal obturation material, separated endodontic instruments, and localization of perforations.

Rationale

- It is important to evaluate the factors that impact the outcome of root canal treatment. The outcome predictors identified with periapical radiographs and CBCT were evaluated by Liang et al.[22] The results showed that periapical radiographs detected periapical lesions in 18 roots (12%) as compared to 37 on CBCT scans (25%); 80% of apparently short root fillings based on intraoral radiographs images appeared flush on CBCT. Treatment outcome, length and density of root fillings, and outcome predictors determined by CBCT showed different values when compared with intraoral radiographs.
- Accurate treatment planning is an essential part of endodontic retreatment. Incorrect, delayed, or inadequate endodontic diagnosis and treatment planning places the patient at risk and may result in unnecessary treatment. Treatment planning decisions using CBCT versus intraoral radiographs were compared to the gold standard diagnosis.[23] An accurate diagnosis was reached in 36% to 40% of the cases with intraoral radiographs compared to 76% to 83% with CBCT. A high level of misdiagnosis was noted in invasive cervical resorption and vertical root fracture. In this study, the examiners altered their treatment plan after reviewing the CBCT in 56% to 62.2% of the cases, thus indicating the significant influence of CBCT.

SURGICAL RETREATMENT

Recommendation 9: Limited FOV CBCT should be considered as the imaging modality of choice for presurgical treatment planning to localize root apex/apices and to evaluate the proximity to adjacent anatomical structures.

Rationale

The use of CBCT has been recommended for treatment planning of endodontic surgery.[24,25] CBCT visualization of the true extent of periapical lesions and their proximity to important vital structures and anatomical landmarks is superior to that of periapical radiographs.

SPECIAL CONDITIONS

a. Implant Placement

Recommendation 10: Limited FOV CBCT should be considered as the imaging modality of choice for surgical placement of implants.[26]

b. Traumatic Injuries

Recommendation 11: Limited FOV CBCT should be considered the imaging modality of choice for diagnosis and management of limited dento-alveolar trauma, root fractures, luxation, and /or displacement of teeth and localized alveolar fractures, in the absence of other maxillofacial or soft-tissue injury that may require other advanced imaging modalities.[27]

c. Resorptive Defects

Recommendation 12: Limited FOV CBCT is the imaging modality of choice in the localization and differentiation of external and internal resorptive defects and the determination of appropriate treatment and prognosis.[28,29]

OUTCOME ASSESSMENT

Recommendation 13: In the absence of clinical signs or symptoms, intraoral radiographs should be considered the imaging modality of choice for the evaluation of healing following nonsurgical and surgical endodontic treatment.

Recommendation 14: In the absence of signs and symptoms, if limited FOV CBCT was the imaging modality of choice at the time of evaluation and

treatment, it may be the modality of choice for follow-up evaluation. In the presence of signs and symptoms, refer to Recommendation #7.

Rationale

- Use of limited FOV CBCT for evaluation of healing following nonsurgical and surgical treatment must be considered on a case-by-case basis, with due consideration given the risks and benefits of exposing the patient to ionizing radiation, the patient's history, clinical findings, mode of endodontic intervention, preexisting radiographic appearance, temporal changes, and any other systemic or extraneous factors that could confound the healing process.
- Accurate diagnosis of complete and uneventful healing as determined radiographically, with unequivocal clinical correlation supporting the radiographic diagnosis, is challenging.[30–33] The healing process is dependent on various factors, both extraneous and host related. In the future, once the parameters associated with healing are fully confirmed by evidence-based studies, the use of limited FOV CBCT for outcome assessment can be better interpreted.

Joint Position Paper References

1. American Association of Endodontists, American Academy of Oral and Maxillofacial Radiology: Use of cone-beam computed tomography in endodontics Joint Position Statement of the American Association of Endodontists and the American Academy of Oral and Maxillofacial Radiology, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 111(2):234–237, 2011.
2. Grondahl HG, Huumonen S: Radiographic manifestations of periapical inflammatory lesions, *Endod Topics* 8:55–67, 2004.
3. Patel S, Durack C, Abella F, et al: Cone beam computed tomography in Endodontics—a review, *Int Endod J* 48:3–15, 2015.
4. Suomalainen A, Pakbaznejad Esmaeili E, Robinson S: Dentomaxillofacial imaging with panoramic views and cone beam CT, *Insights imaging* 6:1–16, 2015.
5. Venskutonis T, Plotino G, Juodzbalys G, et al: The importance of cone-beam computed tomography in the management of endodontic problems: a review of the literature, *J Endod* 40(12):1895–1901, 2014.
6. Carter L, Farman AG, Geist J, et al: American Academy of Oral and Maxillofacial Radiology executive opinion statement on performing and interpreting diagnostic cone beam computed tomography, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 106(4):561–562, 2008.
7. De Paula-Silva FW, Wu MK, Leonardo MR, et al: Accuracy of periapical radiography and cone-beam computed tomography scans in diagnosing apical periodontitis using histopathological findings as a gold standard, *J Endod* 35(7):1009–1012, 2009.
8. Friedman S: Prognosis of initial endodontic therapy, *Endod Topics* 2:59–98, 2002.
9. Patel S, Wilson R, Dawood A, et al: The detection of periapical pathosis using periapical radiography and cone beam computed tomography—part 1: preoperative status, *Int Endod J* 8:702–710, 2012.
10. Abella F, Patel S, Duran-Sindreu F, et al: Evaluating the periapical status of teeth with irreversible pulpitis by using cone-beam computed tomography scanning and periapical radiographs, *J Endod* 38(12):1588–1591, 2012.
11. Cheung G, Wei L, McGrath C: Agreement between periapical radiographs and cone-beam computed tomography for assessment of periapical status of root filled molar teeth, *Int Endod J* 46(10): 889–895, 2013.
12. Sogur E, Grondahl H, Bakst G, et al: Does a combination of two radiographs increase accuracy in detecting acid-induced periapical lesions and does it approach the accuracy of cone-beam computed tomography scanning? *J Endod* 38(2):131–136, 2012.
13. Patel S, Dawood A, Mannocci F, et al: Detection of periapical bone defects in human jaws using cone beam computed tomography and intraoral radiography, *Int Endod J* 42(6): 507–515, 2009.
14. Nixdorf D, Moana-Filho E: Persistent dento-alveolar pain disorder (PDAP): Working towards a better understanding, *Rev Pain* 5(4): 18–27, 2011.
15. Pigg M, List T, Petersson K, et al: Diagnostic yield of conventional radiographic and cone-beam computed tomographic images in patients with atypical odontalgia, *Int Endod J* 44(12):1365–2591, 2011.
16. Blattner TC, Goerge N, Lee CC, et al: Efficacy of CBCT as a modality to accurately identify the presence of second mesiobuccal canals in maxillary first and second molars: a pilot study, *J Endod* 36(5): 867–870, 2010.
17. Michetti J, Maret D, Mallet JP, et al: Validation of cone beam computed tomography as a tool to explore root canal anatomy, *J Endod* 36(7):1187–1190, 2010.
18. Vizzotto MB, Silveira PF, Arús NA, et al: CBCT for the assessment of second mesiobuccal (MB2) canals in maxillary molar teeth: effect of voxel size and presence of root filling, *Int Endod J* 46(9):870–876, 2013.
19. Edlund M, Nair MK, Nair UP: Detection of vertical root fractures by using cone-beam computed tomography: a clinical study, *J Endod* 37(6):768–772, 2011.
20. Metska ME, Aartman IH, Wesselink PR, et al: Detection of vertical root fracture in vivo in endodontically treated teeth by cone-beam computed tomography scans, *J Endod* 38(10):1344–1347, 2012.
21. Brady E, Mannocci F, Wilson R, et al: A comparison of CBCT and periapical radiography for the detection of vertical root fractures in non-endodontically treated teeth, *Int Endod J* 47(8):735–746, 2014.
22. Liang H, Li Gang, Wesselink P, et al: Endodontic outcome predictors identified with periapical radiographs and cone-beam computed tomography scans, *J Endod* 37(3):326–331, 2011.
23. Ee J, Fayad IM, Johnson B: Comparison of endodontic diagnosis and treatment planning decisions using cone-beam volumetric tomography versus periapical radiography, *J Endod* 40(7):910–916, 2014.
24. Venskutonis T, Plotino G, Tocci L, et al: Periapical and endodontic status scale based on periapical bone lesions and endodontic treatment quality evaluation using cone-beam computed tomography, *J Endod* 41(2):190–196, 2015.
25. Low KM, Dula K, Bürgin W, et al: Comparison of periapical radiography and limited cone-beam tomography in posterior maxillary teeth referred for apical surgery, *J Endod* 34(5):557–562, 2008.
26. Tyndall D, Price J, Tetradis S, et al: Position statement of the American Academy of Oral and Maxillofacial Radiology on selection criteria for the use of radiology in dental implantology with emphasis on cone beam computed tomography, *Oral Surg Oral Med Oral Pathol Oral Radiol* 113(6):817–826, 2012.
27. May JJ, Cohenca N, Peters OA: Contemporary management of horizontal root fractures to the permanent dentition: diagnosis, radiologic assessment to include cone-beam computed tomography, *Pediatr Dent* 35:120–124, 2013.
28. Estrela C, Bueno MR, De Alencar AH, et al: Method to evaluate Inflammatory Root Resorption by using Cone Beam computed tomography, *J Endod* 35(11):1491–1497, 2009.
29. Durack C, Patel S, Davies J, et al: Diagnostic accuracy of small volume cone beam computed tomography and intraoral periapical radiography for the detection of simulated external inflammatory root resorption, *Int Endod J* 44(2):136–147, 2011.
30. Cheung G, Wei W, McGrath C: Agreement between periapical radiographs and cone-beam computed tomography for assessment of periapical status of root filled molar teeth, *Int Endod J* 46(10): 889–895, 2013.
31. Estrela C, Bueno MR, Azevedo BC, et al: A new periapical index based on cone beam computed tomography, *J Endod* 34(11): 1325–1331, 2008.
32. Tanomaru-Filho M, Jorge EG, Guerreiro-Tanomaru JM, et al: Two- and tridimensional analysis of periapical repair after endodontic surgery, *Clin Oral Investig* 19(1):17–25, 2015.
33. von Arx T, Janner SF, Hanni S, et al: Agreement between 2D and 3D radiographic outcome assessment one year after periapical surgery, *Int Endod J* 49(1):915–925, 2015.

Special Committee to Revise the Joint AAE/AAOMR Position Statement on Use of Limited FOV CBCT in Endodontics

Mohamed I. Fayad, Co-Chair, AAE
Martin D. Levin, AAE
Richard A. Rubinstein, AAE
Craig S. Hirschberg, AAE Board Liaison
Madhu Nair, Co-Chair, AAOMR
Erika Benavides, AAOMR
Axel Ruprecht, AAOMR
Sevin Barghan, AAOMR

References

1. AAE and AAOMR joint position statement: use of cone beam computed tomography in endodontics 2015 update, *J Endod* 41(9): 1393–1396, 2015.
2. Academy of Osseointegration: 2010 guidelines of the academy of osseointegration for the provision of dental implants and associated patient care, *Int J Oral Maxillofac Implants* 25(3):620–627, 2010.
3. Ackerman S, Aguilera FC, Buie JM, et al: Accuracy of 3-dimensional-printed endodontic surgical guide: a human cadaver study, *J Endod* 45(5):615–618, 2019.
4. ADA Council on Scientific Affairs: An update on radiographic practices: information and recommendations, *J Am Dent Assoc* 132(2): 234–238, 2001.
5. Ahn SY, Kim NH, Kim S, et al: computer-aided design/computer-aided manufacturing-guided endodontic surgery: guided osteotomy and apex localization in a mandibular molar with a thick buccal bone plate, *J Endod* 44(4):665–670, 2018.
6. Akdeniz BG, Sogur E: An ex vivo comparison of conventional and digital radiography for perceived image quality of root fillings, *Int Endod J* 38(6):397–401, 2005.
7. Al-Fouzan KS: Incidence of rotary ProFile instrument fracture and the potential for bypassing in vivo, *Int Endod J* 36(12):864–867, 2003.
8. Alavi AM, Opasanon A, Ng YL, et al: Root and canal morphology of Thai maxillary molars, *Int Endod J* 35(5):478–485, 2002.
9. American Association of Endodontists: *Glossary of endodontic terms*. https://www.aae.org/specialty/clinical-resources/glossary-endodontic-terms/. Chicago, 2019, update.
10. American Dental Association moves forward on electronic standards: *ADA News* 30(15):A, 1999.
11. Anderson LC, Kosinski TF, Mentag PJ: A review of the intraosseous course of the nerves of the mandible, *J Oral Implantol* 17(4): 394–403, 1991.
12. Andersson L: Epidemiology of traumatic dental injuries, *J Endod* 39(3 Suppl):S2–S5, 2013.
13. Andersson L, Andreasen JO, Day P, et al: International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 2. Avulsion of permanent teeth, *Dent Traumatol* 28(2):88–96, 2012.
14. Andreasen FM, Pedersen BV: Prognosis of luxated permanent teeth—the development of pulp necrosis, *Endod Dent Traumatol* 1(6):207–220, 1985.
15. Andreasen FM, Sewerin I, Mandel U, et al: Radiographic assessment of simulated root resorption cavities, *Endod Dent Traumatol* 3(1): 21–27, 1987.
16. Andreasen FM, Andreasen JO: Crown fractures. *Textbook and color atlas of traumatic injuries to the teeth*, ed 3, Copenhagen, 1994, Munksgaard.
17. Andreasen JO, Ahrensburg SS, Tsilingaridis G: Root fractures: the influence of type of healing and location of fracture on tooth survival rates - an analysis of 492 cases, *Dent Traumatol* 28(5): 404–409, 2012.
18. Andreasen JO, Ravn JJ: Epidemiology of traumatic dental injuries to primary and permanent teeth in a Danish population sample, *Int J Oral Surg* 1(5):235–239, 1972.
19. Andreasen JO, Andreasen FM, Andersson L: *Textbook and color atlas of traumatic injuries to the teeth*, Copenhagen, Munksgaard, 2007, John Wiley & Sons.
20. Ariji Y, Ariji E, Nakashima M, et al: Magnetic resonance imaging in endodontics: a literature review, *Oral Radiol* 34(1):10–16, 2018.
21. Backman B, Wahlin YB: Variations in number and morphology of permanent teeth in 7-year-old Swedish children, *Int J Paediatr Dent* 11(1):11–17, 2001.
22. Bakland LK: Root resorption, *Dent Clin North Am* 36(2):491–507, 1992.
23. Ball RL, Barbizam JV, Cohenca N: Intraoperative endodontic applications of cone-beam computed tomography, *J Endod* 39(4):548–557, 2013.
24. Beck-Mannagetta J, Necek D, Grasserbauer M: Solitary aspergillosis of maxillary sinus, a complication of dental treatment, *Lancet* 2(8361):1260, 1983.
25. Benavides E, Rios HF, Ganz SD, et al: Use of cone beam computed tomography in implant dentistry: the international congress of oral implantologists consensus report, *Implant Dent* 21(2):78–86, 2012.
26. Bender IB: Factors influencing the radiographic appearance of bony lesions, *J Endod* 8(4):161–170, 1982.
27. Bender IB, Seltzer S: Roentgenographic and direct observation of experimental lesions in bone: I, *J Am Dent Assoc* 62:152–160, 1961.
28. Bergenholtz G: Micro-organisms from necrotic pulp of traumatized teeth, *Odontol Revy* 25(4):347–358, 1974.
29. Block MS, Emery RW: Static or dynamic navigation for implant placement-choosing the method of guidance, *J Oral Maxillofac Surg* 74(2):269–277, 2016.
30. Borg E, Kallqvist A, Grondahl K, et al: Film and digital radiography for detection of simulated root resorption cavities, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 86(1):110–104, 1998.
31. Bornstein MM, Lauber R, Sendi P, et al: Comparison of periapical radiography and limited cone-beam computed tomography in mandibular molars for analysis of anatomical landmarks before apical surgery, *J Endod* 37(2):151–157, 2011.
32. Bornstein MM, Wasmer J, Sendi P, et al: Characteristics and dimensions of the Schneiderian membrane and apical bone in maxillary molars referred for apical surgery: a comparative radiographic analysis using limited cone beam computed tomography, *J Endod* 38(1):51–57, 2012.
33. Bornstein MM, Wolner-Hanssen AB, Sendi P, et al: Comparison of intraoral radiography and limited cone beam computed tomography for the assessment of root-fractured permanent teeth, *Dent Traumatol* 25(6):571–577, 2009.
34. Bouillaguet S, Wataha JC, Tay FR, et al: Initial in vitro biological response to contemporary endodontic sealers, *J Endod* 32(10): 989–992, 2006.
35. Bratel J, Jontell M, Dahlgren U, et al: Effects of root canal sealers on immunocompetent cells in vitro and in vivo, *Int Endod J* 31(3): 178–188, 1998.
36. Brennan PC, McEntee M, Evanoff M, et al: Ambient lighting: effect of illumination on soft-copy viewing of radiographs of the wrist, *AJR Am J Roentgenol* 188(2):W177–W180, 2007.
37. Brooks JK, Kleinman JW: Retrieval of extensive gutta-percha extruded into the maxillary sinus: use of 3-dimensional cone-beam computed tomography, *J Endod* 39(9):1189–1193, 2013.
38. Brynolf I: A histological and roentgenological study of periapical region of human upper incisors, *Odontol Revy* 18(11):1, 1967.
39. Bueno MR, De Carvalhosa AA, De Souza Castro PH, et al: Mesenchymal chondrosarcoma mimicking apical periodontitis, *J Endod* 34(11):1415–1419, 2008.
40. Bueno MR, Estrela C, De Figueiredo JA, et al: Map-reading strategy to diagnose root perforations near metallic intracanal posts by using cone beam computed tomography, *J Endod* 37(1):85–90, 2011.
41. Burger CL, Mork TO, Hutter JW, et al: Direct digital radiography versus conventional radiography for estimation of canal length in curved canals, *J Endod* 25(4):260–263, 1999.
42. Calberson FL, Hommez GM, De Moor RJ: Fraudulent use of digital radiography: methods to detect and protect digital radiographs, *J Endod* 34(5):530–536, 2008.
43. Caplan D: Epidemiologic issues in studies of association between apical periodontitis and systemic health, *Endod Topics* 8(15), 2008.
44. Carter L, Farman AG, Geist J, et al: American academy of oral and maxillofacial radiology executive opinion statement on performing and interpreting diagnostic cone beam computed tomography, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 106(4):561–562, 2008.
45. Claeys V, Wackens G: Bifid mandibular canal: literature review and case report, *Dentomaxillofac Radiol* 34(1):55–58, 2005.

46. Cohenca N, Simon JH, Mathur A, et al: Clinical indications for digital imaging in dento-alveolar trauma. Part 2: Root resorption, *Dent Traumatol* 23(2):105–113, 2007.
47. Cohenca N, Simon JH, Roges R, et al: Clinical indications for digital imaging in dento-alveolar trauma. Part 1: Traumatic injuries, *Dent Traumatol* 23(2):95–104, 2007.
48. Connert T, Krug R, Eggmann F, et al: Guided endodontics versus conventional access cavity preparation: a comparative study on substance loss using 3-dimensional-printed teeth, *J Endod* 45(3): 327–331, 2019.
49. Cotti E, Dessi C, Piras A, et al: Can a chronic dental infection be considered a cause of cardiovascular disease?, A review of the literature, *Int J Cardiol* 148(1):4–10, 2011.
50. Cotton TP, Geisler TM, Holden DT, et al: Endodontic applications of cone-beam volumetric tomography, *J Endod* 33(9):1121–1132, 2007.
51. Crump MC, Natkin E: Relationship of broken root canal instruments to endodontic case prognosis: a clinical investigation, *J Am Dent Assoc* 80(6):1341–1347, 1970.
52. Cuje J, Bargholz C, Hulsmann M: The outcome of retained instrument removal in a specialist practice, *Int Endod J* 43(7):545–554, 2010.
53. de Paula-Silva FW, Wu MK, Leonardo MR, et al: Accuracy of periapical radiography and cone-beam computed tomography scans in diagnosing apical periodontitis using histopathological findings as a gold standard, *J Endod* 35(7):1009–1012, 2009.
54. Davidovich E, Heling I, Fuks AB: The fate of a mid-root fracture: a case report, *Dent Traumatol* 21(3):170–173, 2005.
55. Day PF, Duggal MS: A multicentre investigation into the role of structured histories for patients with tooth avulsion at their initial visit to a dental hospital, *Dent Traumatol* 19(5):243–347, 2003.
56. Di Fiore PM, Genov KA, Komaroff E, et al: Nickel-titanium rotary instrument fracture: a clinical practice assessment, *Int Endod J* 39(9):700–708, 2006.
57. Di Nardo D, Gambarini G, Capuani S, et al: Nuclear magnetic resonance imaging in endodontics: a review, *J Endod* 44(4):536–542, 2018.
58. Diangelis AJ, Andreasen JO, Ebeleseder KA, et al: International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 1. Fractures and luxations of permanent teeth, *Dent Traumatol* 28(1):2–12, 2012.
59. Diniz de Lima E, Lira de Farias Freitas AP, Mariz Suassuna FC, et al: Assessment of cone-beam computed tomographic artifacts from different intracanal materials on birooted teeth, *J Endod* 45(2): 209–213, 2019.
60. DICOM: *Digital imaging and communications in medicine (DICOM) Part 1: Introduction and overview.* ftp://medical.nema.org/medical/dicom/2008/08_01pu.pdf. Accessed July 26, 2009.
61. Durack C, Patel S, Davies J, et al: Diagnostic accuracy of small volume cone beam computed tomography and intraoral periapical radiography for the detection of simulated external inflammatory root resorption, *Int Endod J* 44(2):136–147, 2011.
62. Edlund M, Nair MK, Nair UP: Detection of vertical root fractures by using cone-beam computed tomography: a clinical study, *J Endod* 37(6):768–772, 2011.
63. Eikenberg S, Vandre R: Comparison of digital dental X-ray systems with self-developing film and manual processing for endodontic file length determination, *J Endod* 26(2):65–67, 2000.
64. Eraso FE, Analoui M, Watson AB, et al: Impact of lossy compression on diagnostic accuracy of radiographs for periapical lesions, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93(5):621–625, 2002.
65. Escoda-Francoli J, Canalda-Sahli C, Soler A, et al: Inferior alveolar nerve damage because of overextended endodontic material: a problem of sealer cement biocompatibility?, *J Endod* 33(12): 1484–1489, 2007.
66. Estrela C, Bueno MR, De Alencar AH, et al: Method to evaluate inflammatory root resorption by using cone beam computed tomography, *J Endod* 35(11):1491–1497, 2009.
67. Estrela C, Bueno MR, Leles CR, et al: Accuracy of cone beam computed tomography and panoramic and periapical radiography for detection of apical periodontitis, *J Endod* 34(3):273–279, 2008.
68. Eversole LR, Leider AS, Hansen LS: Ameloblastomas with pronounced desmoplasia, *J Oral Maxillofac Surg* 42(11):735–740, 1984.
69. Fahrig R, Fox AJ, Lownie S, et al: Use of a C-arm system to generate true three-dimensional computed rotational angiograms: preliminary in vitro and in vivo results, *AJNR Am J Neuroradiol* 18(8): 1507–1514, 1997.
70. Fan W, Fan B, Gutmann JL, et al: Identification of a C-shaped canal system in mandibular second molars. Part III. Anatomic features revealed by digital subtraction radiography, *J Endod* 34(10): 1187–1190, 2008.
71. Fan Y, Glickman GN, Umorin M, et al: A novel prefabricated grid for guided endodontic microsurgery, *J Endod* 45(5):606–610, 2019.
72. Farman AG, Avant SL, Scarfe WC, et al: In vivo comparison of Visualix-2 and Ektaspeed Plus in the assessment of periradicular lesion dimensions, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 85(2):203–209, 1998.
73. Feely L, Mackie IC, Macfarlane T: An investigation of root-fractured permanent incisor teeth in children, *Dent Traumatol* 19(1):52–54, 2003.
74. Finucane D, Kinirons MJ: External inflammatory and replacement resorption of luxated, and avulsed replanted permanent incisors: a review and case presentation, *Dent Traumatol* 19(3):170–174, 2003.
75. Fletcher R, Fletcher S: *Clinical epidemiology: the essentials*, Baltimore, 2005, Lippincott Williams & Wilkins.
76. Forman GH, Rood JP: Successful retrieval of endodontic material from the inferior alveolar nerve, *J Dent* 5(1):47–50, 1977.
77. Fuss Z, Lustig J, Tamse A: Prevalence of vertical root fractures in extracted endodontically treated teeth, *Int Endod J* 32(4):283–286, 1999.
78. Gassner R, Bosch R, Tuli T, et al: Prevalence of dental trauma in 6000 patients with facial injuries: implications for prevention, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 87(1):27–33, 1999.
79. Gencoglu N, Helvacioglu D: Comparison of the different techniques to remove fractured endodontic instruments from root canal systems, *Eur J Dent* 3(2):90–95, 2009.
80. Geurtsen W, Leyhausen G: Biological aspects of root canal filling materials—histocompatibility,cytotoxicity, and mutagenicity, *Clin Oral Investig* 1(1):5–11, 1997.
81. Glendor U: Epidemiology of traumatic dental injuries—a 12 year review of the literature, *Dent Traumatol* 24(6):603–611, 2008.
82. Gold SI, Hasselgren G: Peripheral inflammatory root resorption. A review of the literature with case reports, *J Clin Periodontol* 19(8):523–534, 1992.
83. Goldberg F, De Silvio A, Dreyer C: Radiographic assessment of simulated external root resorption cavities in maxillary incisors, *Endod Dent Traumatol* 14(3):133–136, 1998.
84. Gomes AP, de Araujo EA, Goncalves SE, et al: Treatment of traumatized permanent incisors with crown and root fractures: a case report, *Dent Traumatol* 17(5):236–239, 2001.
85. Gonzalez-Martin M, Torres-Lagares D, Gutierrez-Perez JL, et al: Inferior alveolar nerve paresthesia after overfilling of endodontic sealer into the mandibular canal, *J Endod* 36(8):1419–1421, 2010.
86. Goodis HE, Rossall JC, Kahn AJ: Endodontic status in older U.S. adults. Report of a survey, *J Am Dent Assoc* 132(11):1525–1530, quiz 95–96, 2001.
87. Green TL, Walton RE, Taylor JK, et al: Radiographic and histologic periapical findings of root canal treated teeth in cadaver, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 83(6):707–711, 1997.
88. Grotz KA, Al-Nawas B, de Aguiar EG, et al: Treatment of injuries to the inferior alveolar nerve after endodontic procedures, *Clin Oral Investig* 2(2):73–76, 1998.
89. Guivarc'h M, Ordioni U, Catherine JH, et al: Implications of endodontic-related sinus aspergillosis in a patient treated by infliximab: a case report, *J Endod* 41(1):125–129, 2015.
90. Gulabivala K, Aung TH, Alavi A, et al: Root and canal morphology of Burmese mandibular molars, *Int Endod J* 34(5):359–370, 2001.
91. Gulabivala K, Opasanon A, Ng YL, et al: Root and canal morphology of Thai mandibular molars, *Int Endod J* 35(1):56–62, 2002.
92. Gulabivala K, Searson LJ: Clinical diagnosis of internal resorption: an exception to the rule, *Int Endod J* 28(5):255–260, 1995.
93. Gunst V, Mavridou A, Huybrechts B, et al: External cervical resorption: an analysis using cone beam and microfocus computed tomography and scanning electron microscopy, *Int Endod J* 46(9):877–887, 2013.
94. Gutteridge DL: The use of radiographic techniques in the diagnosis and management of periodontal diseases, *Dentomaxillofac Radiol* 24(2):107–113, 1995.
95. Hassan B, Metska ME, Ozok AR, et al: Detection of vertical root fractures in endodontically treated teeth by a cone beam computed tomography scan, *J Endod* 35(5):719–722, 2009.
96. Hassan B, Metska ME, Ozok AR, et al: Comparison of five cone beam computed tomography systems for the detection of vertical root fractures, *J Endod* 36(1):126–129, 2010.

97. Hatcher DC: Operational principles for cone-beam computed tomography, *J Am Dent Assoc* 141(Suppl 3):3S–6S, 2010.
98. Headache Classification Subcommittee, International Headache Society: The international classification of headache disorders. Second edition. *Cephalgia* 24(1): 2004.
99. Heithersay GS: Management of tooth resorption, *Aust Dent J* 52 (1 Suppl):S105–S121, 2007.
100. Hopp RN, Marchi MT, Kellermann MG, et al: Lymphoma mimicking a dental periapical lesion, *Leuk Lymphoma* 53(5):1008–1010, 2012.
101. Iikubo M, Kobayashi K, Mishima A, et al: Accuracy of intraoral radiography, multidetector helical CT, and limited cone-beam CT for the detection of horizontal tooth root fracture, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108(5):e70–e74, 2009.
102. Ingle JI: A standardized endodontic technique utilizing newly designed instruments and filling materials, *Oral Surg Oral Med Oral Pathol* 14:83–91, 1961.
103. Jung M, Lommel D, Klimek J: The imaging of root canal obturation using micro-CT, *Int Endod J* 38(9):617–626, 2005.
104. Kakehashi S, Stanley HR, Fitzgerald RJ: The effects of surgical exposures of dental pulps in germ-free and conventional laboratory rats, *Oral Surg Oral Med Oral Pathol* 20:340–349, 1965.
105. Kamburoglu K, Ilker Cebeci AR, Grondahl HG: Effectiveness of limited cone-beam computed tomography in the detection of horizontal root fracture, *Dent Traumatol* 25(3):256–261, 2009.
106. Kamburoglu K, Kursun S: A comparison of the diagnostic accuracy of CBCT images of different voxel resolutions used to detect simulated small internal resorption cavities, *Int Endod J* 43(9):798–807, 2010.
107. Kamburoglu K, Kursun S, Yuksel S, et al: Observer ability to detect ex vivo simulated internal or external cervical root resorption, *J Endod* 37(2):168–175, 2011.
108. Kanagasingam S, Lim CX, Yong CP, et al: Diagnostic accuracy of periapical radiography and cone beam computed tomography in detecting apical periodontitis using histopathological findings as a reference standard, *Int Endod J* 50(5):417–426, 2017.
109. Karabucak B, Bunes A, Chehoud C, et al: Prevalence of apical periodontitis in endodontically treated premolars and molars with untreated canal: a cone-beam computed tomography study, *J Endod* 42(4):538–541, 2016.
110. Kerekes K, Tronstad L: Long-term results of endodontic treatment performed with a standardized technique, *J Endod* 5(3):83–90, 1979.
111. Khedmat S, Rouhi N, Drage N, et al: Evaluation of three imaging techniques for the detection of vertical root fractures in the absence and presence of gutta-percha root fillings, *Int Endod J* 45(11): 1004–1009, 2012.
112. Kim E, Kim KD, Roh BD, et al: Computed tomography as a diagnostic aid for extracanal invasive resorption, *J Endod* 29(7):463–465, 2003.
113. Kim TS, Caruso JM, Christensen H, et al: A comparison of cone-beam computed tomography and direct measurement in the examination of the mandibular canal and adjacent structures, *J Endod* 36(7):1191–1194, 2010.
114. Kitagawa H, Scheetz JP, Farman AG: Comparison of complementary metal oxide semiconductor and charge-coupled device intraoral X-ray detectors using subjective image quality, *Dentomaxillofac Radiol* 32(6):408–411, 2003.
115. Kositbowornchai S, Nuansakul R, Sikram S, et al: Root fracture detection: a comparison of direct digital radiography with conventional radiography, *Dentomaxillofac Radiol* 30(2):106–109, 2001.
116. Kovisto T, Ahmad M, Bowles WR: Proximity of the mandibular canal to the tooth apex, *J Endod* 37(3):311–315, 2011.
117. Lamus F, Katz JO, Glaros AG: Evaluation of a digital measurement tool to estimate working length in endodontics, *J Contemp Dent Pract* 2(1):24–30, 2001.
118. Lara-Mendes STO, Barbosa CFM, Machado VC, et al: A new approach for minimally invasive access to severely calcified anterior teeth using the guided endodontics technique, *J Endod* 44(10):1578–1582, 2018.
119. Lavasani SA, Tyler C, Roach SH, et al: Cone-beam computed tomography: anatomic analysis of maxillary posterior teeth-impact on endodontic microsurgery, *J Endod* 42(6):890–895, 2016.
120. Leddy BJ, Miles DA, Newton CW, et al: Interpretation of endodontic file lengths using RadioVisiography, *J Endod* 20(11):542–545, 1994.
121. Leonardi Dutra K, Haas L, Porporatti AL, et al: Diagnostic accuracy of cone-beam computed tomography and conventional radiography on apical periodontitis: a systematic review and meta-analysis, *J Endod* 42(3):356–364, 2016.
122. Levin L, M T: Root resorption. In Hargreaves KM, Goodis HE, editors, *Seltzer and Bender's dental pulp*, Chicago, 2002, Quintessence Publishing, pp 425–448.
123. Li G, Sanderink GC, Welander U, et al: Evaluation of endodontic files in digital radiographs before and after employing three image processing algorithms, *Dentomaxillofac Radiol* 33(1): 6–11, 2004.
124. Liang YH, Li G, Wesselink PR, et al: Endodontic outcome predictors identified with periapical radiographs and cone-beam computed tomography scans, *J Endod* 37(3):326–331, 2011.
125. Liedke GS, da Silveira HE, da Silveira HL, et al: Influence of voxel size in the diagnostic ability of cone beam tomography to evaluate simulated external root resorption, *J Endod* 35(2):233–235, 2009.
126. Littner MM, Kaffe I, Tamse A, et al: Relationship between the apices of the lower molars and mandibular canal—a radiographic study, *Oral Surg Oral Med Oral Pathol* 62(5):595–602, 1986.
127. Lofthag-Hansen S, Huumonen S, Grondahl K, et al: Limited cone-beam CT and intraoral radiography for the diagnosis of periapical pathology, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 103(1):114–119, 2007.
128. Loushine RJ, Weller RN, Kimbrough WF, et al: Measurement of endodontic file lengths: calibrated versus uncalibrated digital images, *J Endod* 27(12):779–781, 2001.
129. Low KM, Dula K, Burgin W, et al: Comparison of periapical radiography and limited cone-beam tomography in posterior maxillary teeth referred for apical surgery, *J Endod* 34(5):557–562, 2008.
130. Lu Y, Liu Z, Zhang L, et al: Associations between maxillary sinus mucosal thickening and apical periodontitis using cone-beam computed tomography scanning: a retrospective study, *J Endod* 38(8):1069–1074, 2012.
131. Maillet M, Bowles WR, McClanahan SL, et al: Cone-beam computed tomography evaluation of maxillary sinusitis, *J Endod* 37(6):753–757, 2011.
132. Marchesan MA, Lloyd A, Clement DJ, et al: Impacts of contracted endodontic cavities on primary root canal curvature parameters in mandibular molars, *J Endod* 44(10):1558–1562, 2018.
133. Marmulla R, Wortche R, Muhling J, et al: Geometric accuracy of the NewTom 9000 Cone Beam CT, *Dentomaxillofac Radiol* 34(1):28–31, 2005.
134. Mehra P, Murad H: Maxillary sinus disease of odontogenic origin, *Otolaryngol Clin North Am* 37(2):347–364, 2004.
135. Melo SL, Bortoluzzi EA, Abreu M Jr, et al: Diagnostic ability of a cone-beam computed tomography scan to assess longitudinal root fractures in prosthetically treated teeth, *J Endod* 36(11):1879–1882, 2010.
136. Metska ME, Aartman IH, Wesselink PR, et al: Detection of vertical root fractures in vivo in endodontically treated teeth by cone-beam computed tomography scans, *J Endod* 38(10):1344–1347, 2012.
137. Miguens SA Jr, Veeck EB, Fontanella VR, et al: A comparison between panoramic digital and digitized images to detect simulated periapical lesions using radiographic subtraction, *J Endod* 34(12):1500–1503, 2008.
138. Mikrogeorgis G, Lyroudia K, Molyvdas I, et al: Digital radiograph registration and subtraction: a useful tool for the evaluation of the progress of chronic apical periodontitis, *J Endod* 30(7):513–517, 2004.
139. Miles D: *Color atlas of cone beam volumetric imaging for dental applications*, Hanover Park, IL, 2008, Quintessence Publishing.
140. Miles DA, Danfort RA: A clinician's guide to understanding cone beam volumetric imaging (CBVI), *Acad Dent Ther Stomatol* (Special Issue):1–13, 2007, www.ineedce.com.
141. Mirota DJ, Uneri A, Schafer S, et al: Evaluation of a system for high-accuracy 3D image-based registration of endoscopic video to C-arm cone-beam CT for image-guided skull base surgery, *IEEE Trans Med Imaging* 32(7):1215–1226, 2013.
142. Nair MK, Nair UP: Digital and advanced imaging in endodontics: a review, *J Endod* 33(1):1–6, 2007.
143. Nance RS, Tyndall D, Levin LG, et al: Diagnosis of external root resorption using TACT (tuned-aperture computed tomography), *Endod Dent Traumatol* 16(1):24–28, 2000.

144. Narayana P, Hartwell GR, Wallace R, et al: Endodontic clinical management of a dens invaginatus case by using a unique treatment approach: a case report, *J Endod* 38(8):1145–1148, 2012.
145. Neelakantan P, Subbarao C, Subbarao CV: Comparative evaluation of modified canal staining and clearing technique, cone-beam computed tomography, peripheral quantitative computed tomography, spiral computed tomography, and plain and contrast medium-enhanced digital radiography in studying root canal morphology, *J Endod* 36(9):1547–1551, 2010.
146. Neelakantan P, Subbarao C, Subbarao CV, et al: Root and canal morphology of mandibular second molars in an Indian population, *J Endod* 36(8):1319–1322, 2010.
147. Nevares G, Cunha RS, Zuolo ML, et al: Success rates for removing or bypassing fractured instruments: a prospective clinical study, *J Endod* 38(4):442–444, 2012.
148. Ng Y, Gulabivala K: Evaluation of outcomes. In *Pathways of the pulp*, ed 11, St. Louis, MO, 2016, Elsevier, pp 474–531.
149. Ng YL, Aung TH, Alavi A, et al: Root and canal morphology of Burmese maxillary molars, *Int Endod J* 34(8):620–630, 2001.
150. Nicopoulou-Karayianni K, Bragger U, Patrikiou A, et al: Image processing for enhanced observer agreement in the evaluation of periapical bone changes, *Int Endod J* 35(7):615–622, 2002.
151. Nixdorf DR, Moana-Filho EJ, Law AS, et al: Frequency of persistent tooth pain after root canal therapy: a systematic review and meta-analysis, *J Endod* 36(2):224–230, 2010.
152. Oehlers FA: Dens invaginatus (dilated composite odontome). II. Associated posterior crown forms and pathogenesis, *Oral Surg Oral Med Oral Pathol* 10(12):1302–1316, 1957.
153. Ohrbach R, List T, Goulet JP, et al: Recommendations from the International Consensus Workshop: convergence on an orofacial pain taxonomy, *J Oral Rehabil* 37(10):807–812, 2010.
154. Orhan K, Aksoy U, Kalender A: Cone-beam computed tomographic evaluation of spontaneously healed root fracture, *J Endod* 36(9):1584–1587, 2010.
155. Ozer SY: Detection of vertical root fractures of different thicknesses in endodontically enlarged teeth by cone beam computed tomography versus digital radiography, *J Endod* 36(7):1245–1249, 2010.
156. Panitvisai P, Parunnit P, Sathorn C, et al: Impact of a retained instrument on treatment outcome: a systematic review and meta-analysis, *J Endod* 36(5):775–780, 2010.
157. Parashos P, Gordon I, Messer HH: Factors influencing defects of rotary nickel-titanium endodontic instruments after clinical use, *J Endod* 30(10):722–725, 2004.
158. Patel K, Mannocci F, Patel S: The assessment and management of external cervical resorption with periapical radiographs and cone-beam computed tomography: a clinical study, *J Endod* 42(10):1435–1440, 2016.
159. Patel S: New dimensions in endodontic imaging: Part 2. Cone beam computed tomography, *Int Endod J* 42(6):463–475, 2009.
160. Patel S: The use of cone beam computed tomography in the conservative management of dens invaginatus: a case report, *Int Endod J* 43(8):707–713, 2010.
161. Patel S, Brady E, Wilson R, et al: The detection of vertical root fractures in root filled teeth with periapical radiographs and CBCT scans, *Int Endod J* 46(12):1140–1452, 2013.
162. Patel S, Dawood A: The use of cone beam computed tomography in the management of external cervical resorption lesions, *Int Endod J* 40(9):730–737, 2007.
163. Patel S, Dawood A, Wilson R, et al: The detection and management of root resorption lesions using intraoral radiography and cone beam computed tomography - an in vivo investigation, *Int Endod J* 42(9):831–838, 2009.
164. Patel S, Durack C, Abella F, et al: Cone beam computed tomography in Endodontics - a review, *Int Endod J* 48(1):3–15, 2015.
165. Patel S, Ford TP: Is the resorption external or internal?, *Dent Update* 34(4):218–220, 222, 224–226, 229, 2007.
166. Patel S, Kanagasingam S, Pitt Ford T: External cervical resorption: a review, *J Endod* 35(5):616–625, 2009.
167. Patel S, Mavridou AM, Lambrechts P, et al: External cervical resorption-part 1: histopathology, distribution and presentation, *Int Endod J* 51(11):1205–223, 2018.
168. Peters E, Lau M: Histopathologic examination to confirm diagnosis of periapical lesions: a review, *J Can Dent Assoc* 69(9):598–600, 2003.
169. Peters OA, Peters CI, Schonenberger K, et al: ProTaper rotary root canal preparation: effects of canal anatomy on final shape analysed by micro CT, *Int Endod J* 36(2):86–92, 2003.
170. Peters OA, Schonenberger K, Laib A: Effects of four Ni-Ti preparation techniques on root canal geometry assessed by micro computed tomography, *Int Endod J* 34(3):221–230, 2001.
171. Pew Research Center: *US population projections: 2005-2050*. www.pewhispanic.org/2008/02/11/us-population-projections-2005-2050.
172. Piepenbring ME, Potter BJ, Weller RN, et al: Measurement of endodontic file lengths: a density profile plot analysis, *J Endod* 26(10):615–618, 2000.
173. Pigg M, List T, Petersson K, et al: Diagnostic yield of conventional radiographic and cone-beam computed tomographic images in patients with atypical odontalgia, *Int Endod J* 44(12):1092–1101, 2011.
174. Pineda F, Kuttler Y: Mesiodistal and buccolingual roentgenographic investigation of 7,275 root canals, *Oral Surg Oral Med Oral Pathol* 33(1):101–110, 1972.
175. Pogrel MA: Damage to the inferior alveolar nerve as the result of root canal therapy, *J Am Dent Assoc* 138(1):65–69, 2007.
176. Pope O, Sathorn C, Parashos P: A comparative investigation of cone-beam computed tomography and periapical radiography in the diagnosis of a healthy periapex, *J Endod* 40(3):360–365, 2014.
177. Renton T: Prevention of iatrogenic inferior alveolar nerve injuries in relation to dental procedures, *Dent Update* 37(6):350–352, 354–356, 358–360 passim, 2010.
178. Ricucci D, Langeland K: Apical limit of root canal instrumentation and obturation, part 2. A histological study, *Int Endod J* 31(6):394–409, 1998.
179. Rodrigues CD, Estrela C: Traumatic bone cyst suggestive of large apical periodontitis, *J Endod* 34(4):484–489, 2008.
180. Rodrigues CD, Villar-Neto MJ, Sobral AP, et al: Lymphangioma mimicking apical periodontitis, *J Endod* 37(1):91–96, 2011.
181. Rodriguez G, Patel S, Duran-Sindreu F, et al: Influence of cone-beam computed tomography on endodontic retreatment strategies among general dental practitioners and endodontists, *J Endod* 43(9):1433–1437, 2017.
182. Rud J, Omnell KA: Root fractures due to corrosion. Diagnostic aspects. *Scand J Dent Res* 78(5):397–403, 1970.
183. Ruddle C: Endodontic disinfection-tsunami irrigation, *Endod Prac* (Feb):7–15, 2008.
184. Ruddle CJ: Endodontic diagnosis, *Dent Today* 21(10):90–92, 94, 96–101; quiz 01, 78, 2002.
185. Samei E, Krupinski E: *Medical imaging: perception and techniques*, Cambridge, UK, 2010, Cambridge University Press.
186. Scannapieco FA: Position paper of The American Academy of Periodontology: periodontal disease as a potential risk factor for systemic diseases, *J Periodontol* 69(7):841–850, 1998.
187. Scarfe WC: Imaging of maxillofacial trauma: evolutions and emerging revolutions, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 100(2 Suppl):S75–S96, 2005.
188. Scarfe WC, Levin MD, Gane D, et al: Use of cone beam computed tomography in endodontics, *Int J Dent* 2009:634567, 2009.
189. Schilder H: Cleaning and shaping the root canal, *Dent Clin North Am* 18(2):269–296, 1974.
190. Schilder H: Canal debridement and disinfection. In Cohen S, Burns C, editors. *Pathways of the pulp*, ed 1, St. Louis, 1976, Mosby, p 111–113.
191. Scolozzi P, Lombardi T, Jaques B: Successful inferior alveolar nerve decompression for dysesthesia following endodontic treatment: report of 4 cases treated by mandibular sagittal osteotomy, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97(5):625–631, 2004.
192. Sert S, Bayirli GS: Evaluation of the root canal configurations of the mandibular and maxillary permanent teeth by gender in the Turkish population, *J Endod* 30(6):391–398, 2004.
193. Sharan A, Madjar D: Correlation between maxillary sinus floor topography and related root position of posterior teeth using panoramic and cross-sectional computed tomography imaging, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102(3):375–381, 2006.
194. Shemesh H, Cristescu RC, Wesselink PR, et al: The use of cone-beam computed tomography and digital periapical radiographs to diagnose root perforations, *J Endod* 37(4):513–516, 2011.
195. Shishvan HH, Ebrahimnejad H: A study on the ability of panoramic, CT, cone-beam CT, MRI and ultrasonography in detecting different foreign-bodies in the maxillofacial region (an in-vitro study), *Electron J Gen Med* 15(16):2018.
196. Siewerdsen JH, Jaffray DA: Cone-beam computed tomography with a flat-panel imager: effects of image lag, *Med Phys* 26(12): 2635–2647, 1999.

197. Simonton JD, Azevedo B, Schindler WG, et al: Age- and gender-related differences in the position of the inferior alveolar nerve by using cone beam computed tomography, *J Endod* 35(7):944–949, 2009.
198. Sjogren U, Hagglund B, Sundqvist G, et al: Factors affecting the long-term results of endodontic treatment, *J Endod* 16(10):498–504, 1990.
199. Sonoda M, Takano M, Miyahara J, et al: Computed radiography utilizing scanning laser stimulated luminescence, *Radiology* 148(3):833–838, 1983.
200. Spili P, Parashos P, Messer HH: The impact of instrument fracture on outcome of endodontic treatment, *J Endod* 31(12):845–850, 2005.
201. Stratemann SA, Huang JC, Maki K, et al: Comparison of cone beam computed tomography imaging with physical measures, *Dentomaxillofac Radiol* 37(2):80–93, 2008.
202. Swets J: *Signal detection theory and ROC analysis in psychology and diagnostics: collected papers*, Mahwah, NJ, 1996, Lawrence Erlbaum Associates.
203. Tahmasbi M, Jalali P, Nair MK, et al: Prevalence of middle mesial canals and isthmi in the mesial root of mandibular molars: an in vivo cone-beam computed tomographic study, *J Endod* 43(7): 1080–1083, 2017.
204. Tian YY, Guo B, Zhang R, et al: Root and canal morphology of maxillary first premolars in a Chinese subpopulation evaluated using cone-beam computed tomography, *Int Endod J* 45(11): 996–1003, 2012.
205. Torabinejad M: *Endodontics: principles and practice*, St Louis, 2009, Saunders.
206. Torabinejad M, Bahjri K: Essential elements of evidenced-based endodontics: steps involved in conducting clinical research, *J Endod* 31(8):563–569, 2005.
207. Toure B, Faye B, Kane AW, et al: Analysis of reasons for extraction of endodontically treated teeth: a prospective study, *J Endod* 37(11):1512–1515, 2011.
208. Tronstad L: Root resorption—etiology, terminology and clinical manifestations, *Endod Dent Traumatol* 4(6):241–252, 1988.
209. Trope M: Root resorption of dental and traumatic origin: classification based on etiology, *Pract Periodontics Aesthet Dent* 10(4):515–522, 1998.
210. Tsesis I, Rosen E, Schwartz-Arad D, et al: Retrospective evaluation of surgical endodontic treatment: traditional versus modern technique, *J Endod* 32(5):412–416, 2006.
211. Tsesis I, Rosen E, Tamse A, et al: Diagnosis of vertical root fractures in endodontically treated teeth based on clinical and radiographic indices: a systematic review, *J Endod* 36(9):1455–1458, 2010.
212. Tyndall DA, Ludlow JB, Platin E, et al: A comparison of Kodak Ektaspeed Plus film and the Siemens Sidexis digital imaging system for caries detection using receiver operating characteristic analysis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 85(1):113–118, 1998.
213. Tyndall DA, Price JB, Tetradis S, et al: Position statement of the American Academy of Oral and Maxillofacial Radiology on selection criteria for the use of radiology in dental implantology with emphasis on cone beam computed tomography, *Oral Surg Oral Med Oral Pathol Oral Radiol* 113(6):817–826, 2012.
214. Valmaseda-Castellon E, Berini-Aytes L, Gay-Escoda C: Inferior alveolar nerve damage after lower third molar surgical extraction: a prospective study of 1117 surgical extractions, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 92(4):377–383, 2001.
215. Vande Voorde HE BA: Estimating endodontic "working length" with paralleling radiographs, *Oral Surg Oral Med Oral Pathol.* 27:106–110, 1969.
216. Vaz de Souza D, Schirru E, Mannocci F, et al: External cervical resorption: a comparison of the diagnostic efficacy using 2 different cone-beam computed tomographic units and periapical radiographs, *J Endod* 43(1):121–125, 2017.
217. Versteeg CH, Sanderink GC, Lobach SR, et al: Reduction in size of digital images: does it lead to less detectability or loss of diagnostic information?, *Dentomaxillofac Radiol* 27(2):93–96, 1998.
218. Versteeg KH, Sanderink GC, van Ginkel FC, et al: Estimating distances on direct digital images and conventional radiographs, *J Am Dent Assoc* 128(4):439–443, 1997.
219. Vertucci F: Root canal morphology and its relationship to endodontic procedures, *Endod Topics* 10(3): 2005.
220. Vertucci FJ: Root canal morphology of mandibular premolars, *J Am Dent Assoc* 97(1):47–50, 1978.
221. Vertucci FJ: Root canal anatomy of the human permanent teeth, *Oral Surg Oral Med Oral Pathol* 58(5):589–599, 1984.
222. Vier-Pelisser FV, Pelisser A, Recuero LC, et al: Use of cone beam computed tomography in the diagnosis, planning and follow up of a type III dens invaginatus case, *Int Endod J* 45(2):198–208, 2012.
223. Wang P, Yan XB, Lui DG, et al: Detection of dental root fractures by using cone-beam computed tomography, *Dentomaxillofac Radiol* 40(5):290–298, 2011.
224. Weng XL, Yu SB, Zhao SL, et al: Root canal morphology of permanent maxillary teeth in the Han nationality in Chinese Guanzhong area: a new modified root canal staining technique, *J Endod* 35(5):651–656, 2009.
225. Wenzel A, Haiter-Neto F, Frydenberg M, et al: Variable-resolution cone-beam computerized tomography with enhancement filtration compared with intraoral photostimulable phosphor radiography in detection of transverse root fractures in an in vitro model, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108(6):939–945, 2009.
226. Westphalen VP, Gomes de Moraes I, Westphalen FH, et al: Conventional and digital radiographic methods in the detection of simulated external root resorptions: a comparative study, *Dentomaxillofac Radiol* 33(4):233–235, 2004.
227. White SC, Scarfe WC, Schulze RK, et al: The Image Gently in Dentistry campaign: promotion of responsible use of maxillofacial radiology in dentistry for children, *Oral Surg Oral Med Oral Pathol Oral Radiol* 118(3):257–261, 2014.
228. Wolner-Hanssen AB, von Arx T: Permanent teeth with horizontal root fractures after dental trauma, *A retrospective study. Schweiz Monatsschr Zahnmed* 120(3):200–212, 2010.
229. Wu MK, Dummer PM, Wesselink PR: Consequences of and strategies to deal with residual post-treatment root canal infection, *Int Endod J* 39(5):343–356, 2006.
230. Wu MK, Shemesh H, Wesselink PR: Limitations of previously published systematic reviews evaluating the outcome of endodontic treatment, *Int Endod J* 42(8):656–666, 2009.
231. Yoshioka T, Kobayashi C, Suda H, et al: An observation of the healing process of periapical lesions by digital subtraction radiography, *J Endod* 28(8):589–591, 2002.
232. Young GR: Contemporary management of lateral root perforation diagnosed with the aid of dental computed tomography, *Aust Endod J* 33(3):112–118, 2007.
233. Zadik Y, Sandler V, Bechor R, et al: Analysis of factors related to extraction of endodontically treated teeth, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 106(5):e31–e35, 2008.
234. Zahedi S, Mostafavi M, Lotfirikan N: Anatomic study of mandibular posterior teeth using cone-beam computed tomography for endodontic surgery, *J Endod* 44(5):738–743, 2018.
235. Zhang L, Wang T, Cao Y, et al: In vivo detection of subtle vertical root fracture in endodontically treated teeth by cone-beam computed tomography, *J Endod* 2019.
236. Zhang R, Wang H, Tian YY, et al: Use of cone-beam computed tomography to evaluate root and canal morphology of mandibular molars in Chinese individuals, *Int Endod J* 44(11):990–999, 2011.
237. Zhang Y, Zhang L, Zhu XR, et al: Reducing metal artifacts in cone-beam CT images by preprocessing projection data, *Int J Radiat Oncol Biol Phys* 67(3):924–932, 2007.
238. Zheng Q, Zhang L, Zhou X, et al: C-shaped root canal system in mandibular second molars in a Chinese population evaluated by cone-beam computed tomography, *Int Endod J* 44(9):857–862, 2011.
239. Zhou W, Han C, Li D, et al: Endodontic treatment of teeth induces retrograde peri-implantitis, *Clin Oral Implants Res* 20(12): 1326–1332, 2009.